



Amazon

CSE 326 : Information System Design
Group-2 (A2)

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Introduction

This report documents the design and implementation of an Amazon-like e-commerce platform for CSE 326: Information System Design. The project demonstrates comprehensive system analysis and full-stack development, covering product management, shopping cart, order placement, payment processing, and product reviews.

1.1 System Scope

The system supports customer product browsing with detailed information, shopping cart management, address-based checkout, secure payment processing via Stripe, order tracking, and product reviews. Multi-vendor support enables vendors to manage inventory and fulfill orders.

1.2 Key Features

- **Place Order:** Convert cart items to confirmed orders with address and payment selection
- **Checkout:** Complete purchase flow with order summary, address validation, and payment processing
- **Order Review:** Verified purchaser product reviews with aggregated ratings

BPMN Diagram

The presentation of the BPMN diagram was delivered by Anuron Maitro (2105037).

Modifications Suggested:

- Add a vendor pool to add,remove, and update products .
- Vendor Pool should also have the ability to manage orders (accept/reject) when the order is placed.

2.1 Full BPMN Diagram

2.1.1 Full BPMN(default size as page)

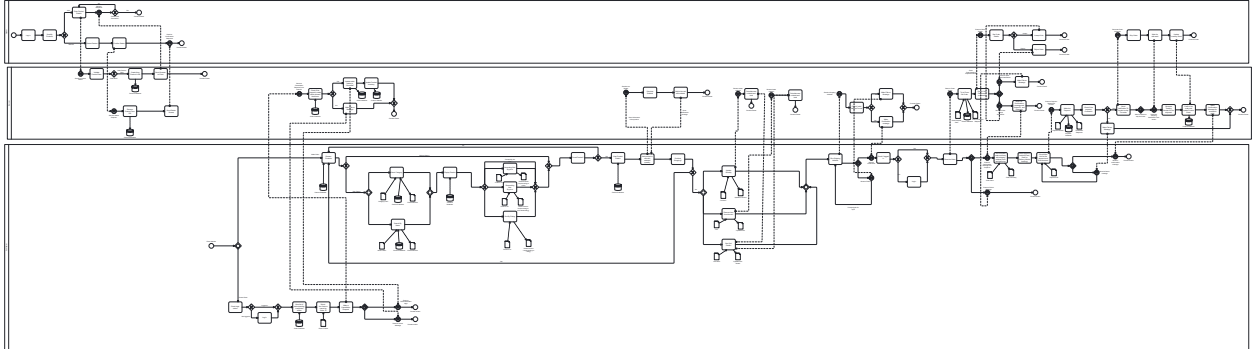
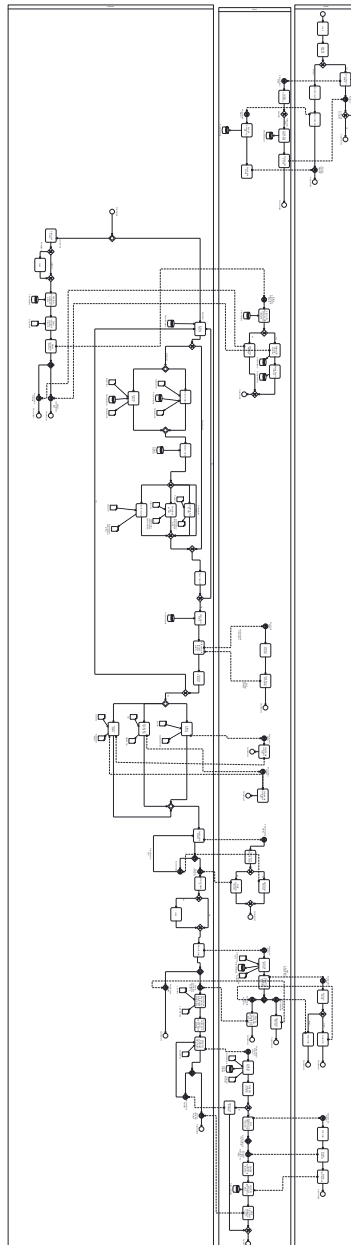


Figure 2.1: Full BPMN Diagram (Default Size as Page)

2.1.2 Full BPMN Diagram (Landscape)



2.2 BPMN Diagram (Server Pool)

2.2.1 BPMN Diagram (Server Pool Part 1)



Figure 2.2: BPMN Diagram (Server Pool Part 1)

2.2.2 BPMN Diagram (Server Pool Part 2)

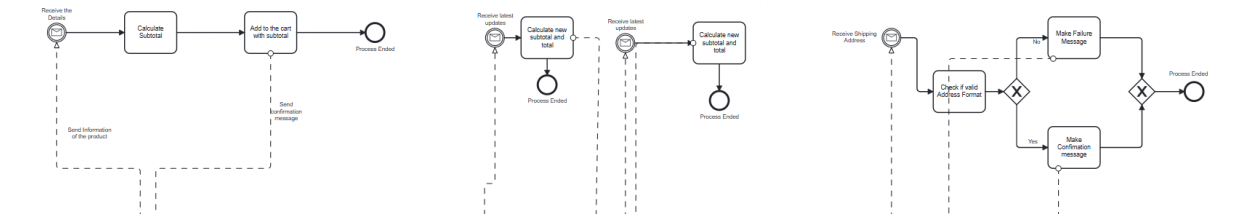


Figure 2.3: BPMN Diagram (Server Pool Part 2)

2.2.3 BPMN Diagram (Server Pool Part 3)

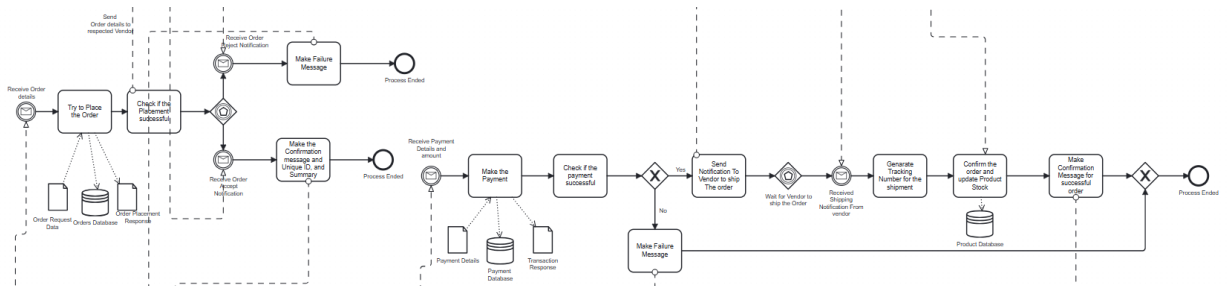


Figure 2.4: BPMN Diagram (Server Pool Part 3)

2.3 BPMN Diagram (Vendor Pool)

2.3.1 BPMN Diagram (Vendor Pool Part 1)

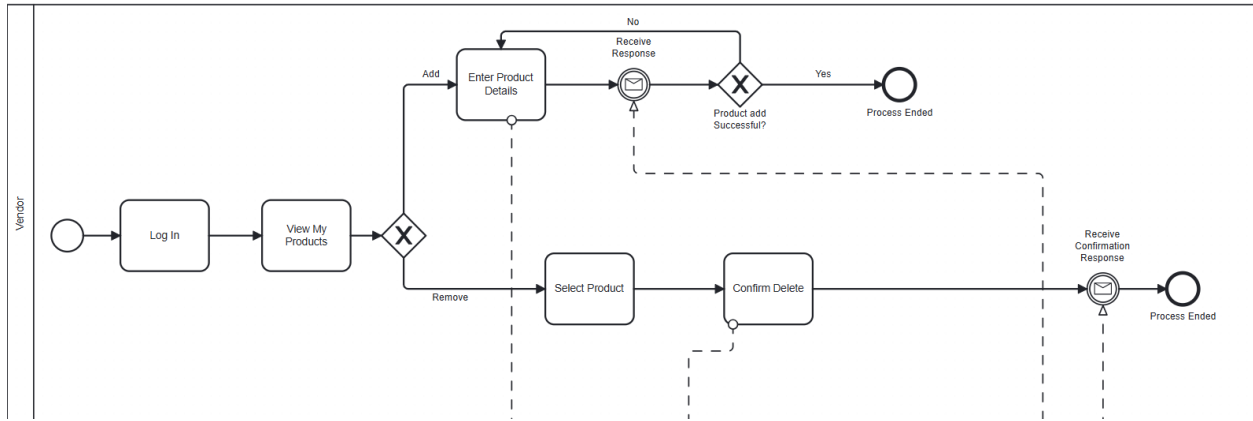


Figure 2.5: BPMN Diagram (Vendor Pool Part 1)

2.3.2 BPMN Diagram (Vendor Pool Part 2)

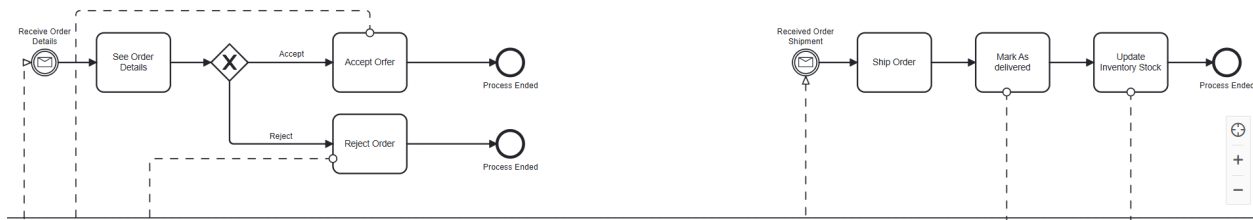


Figure 2.6: BPMN Diagram (Vendor Pool Part 2)

2.4 BPMN Diagram (Customer Pool)

2.4.1 BPMN Diagram (Customer Pool Part 1)

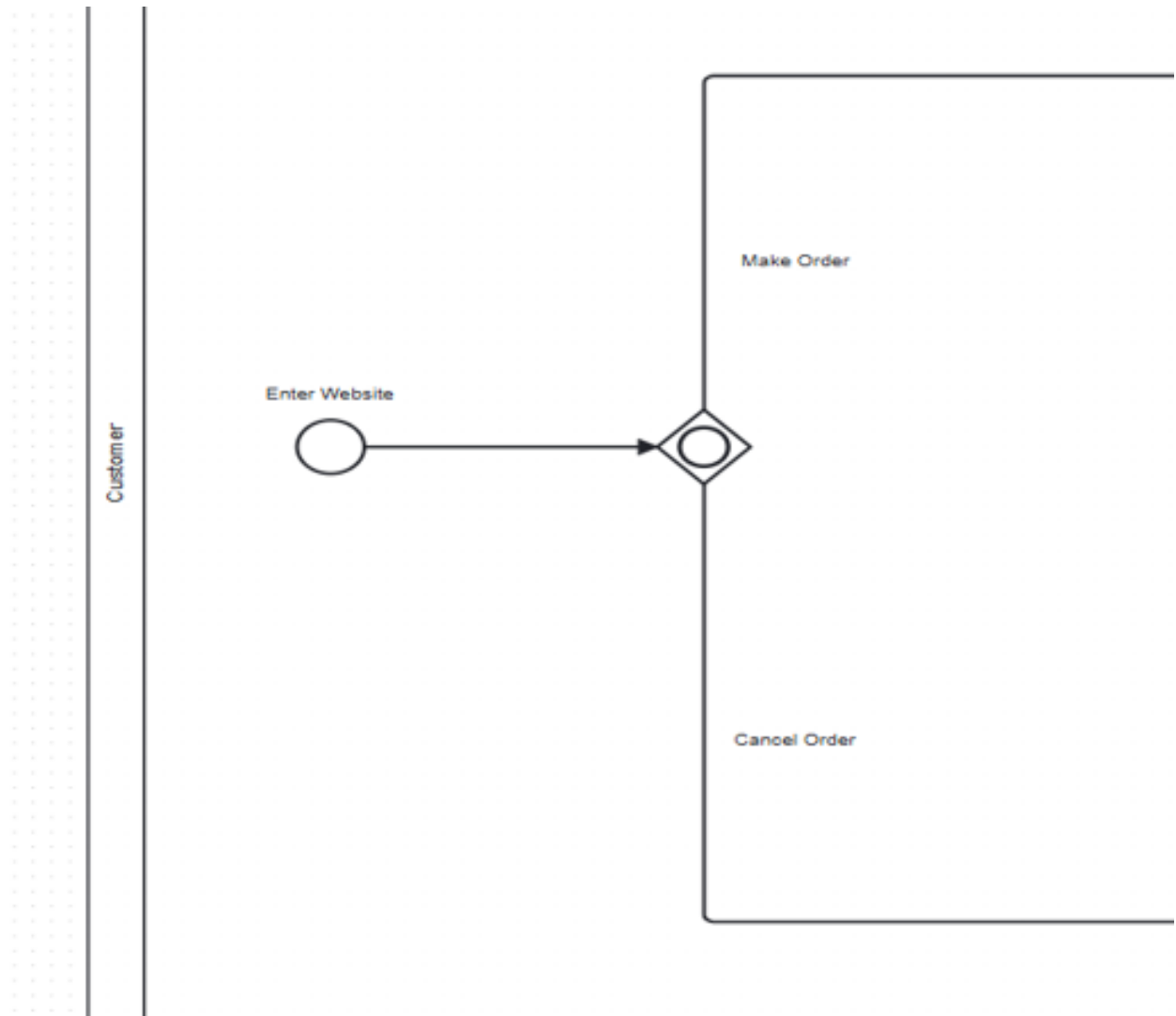


Figure 2.7: BPMN Diagram (Customer Pool Part 1)

2.4.2 BPMN Diagram (Customer Pool Part 2)

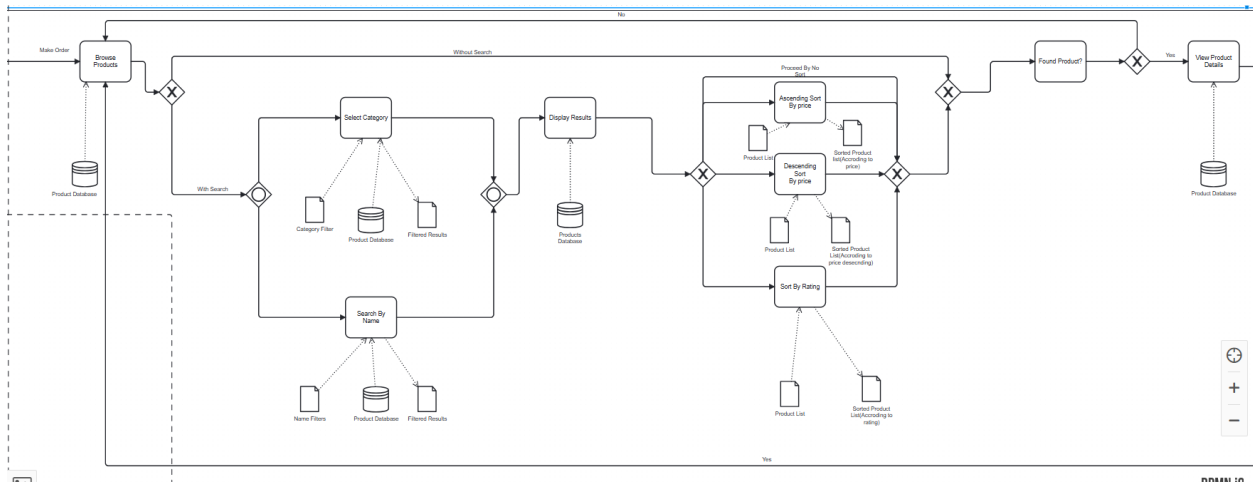


Figure 2.8: BPMN Diagram (Customer Pool Part 2)

2.4.3 BPMN Diagram (Customer Pool Part 3)

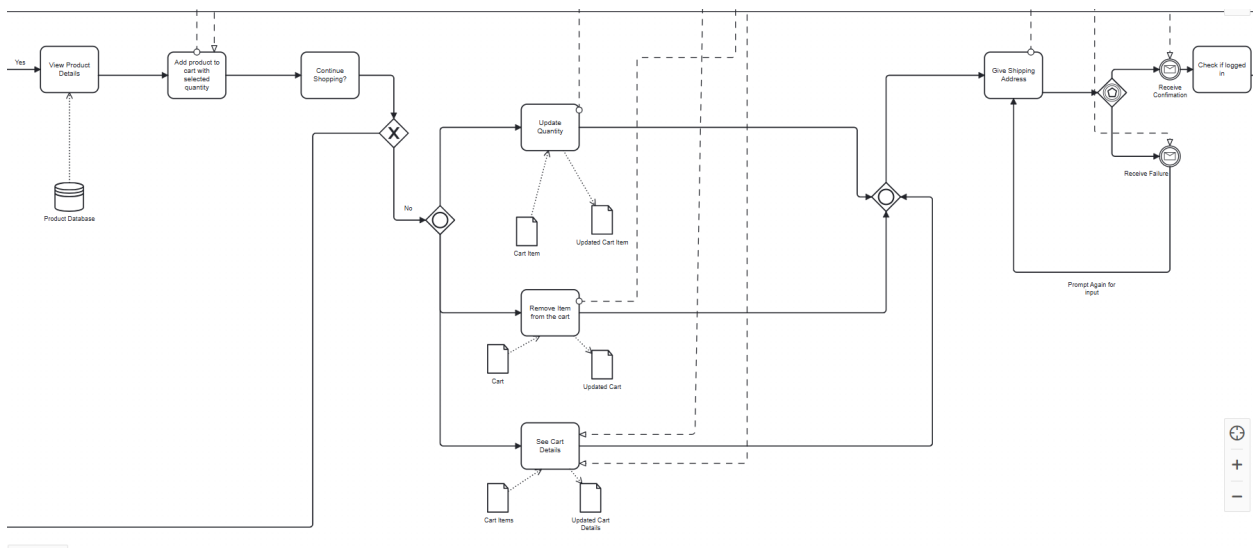


Figure 2.9: BPMN Diagram (Customer Pool Part 3)

2.4.4 BPMN Diagram (Customer Pool Part 4)

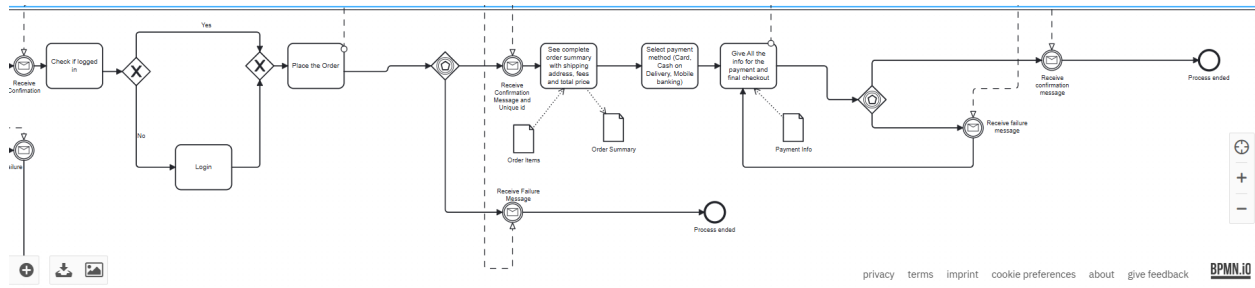


Figure 2.10: BPMN Diagram (Customer Pool Part 4)

2.4.5 BPMN Diagram (Customer Pool Part 5)

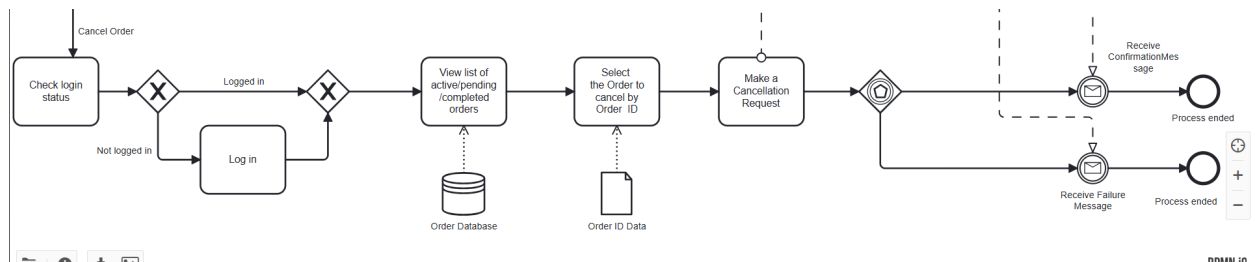


Figure 2.11: BPMN Diagram (Customer Pool Part 5)

2.5 Feature-wise BPMN Diagrams

2.5.1 Customer Pool: Starting the Order Process

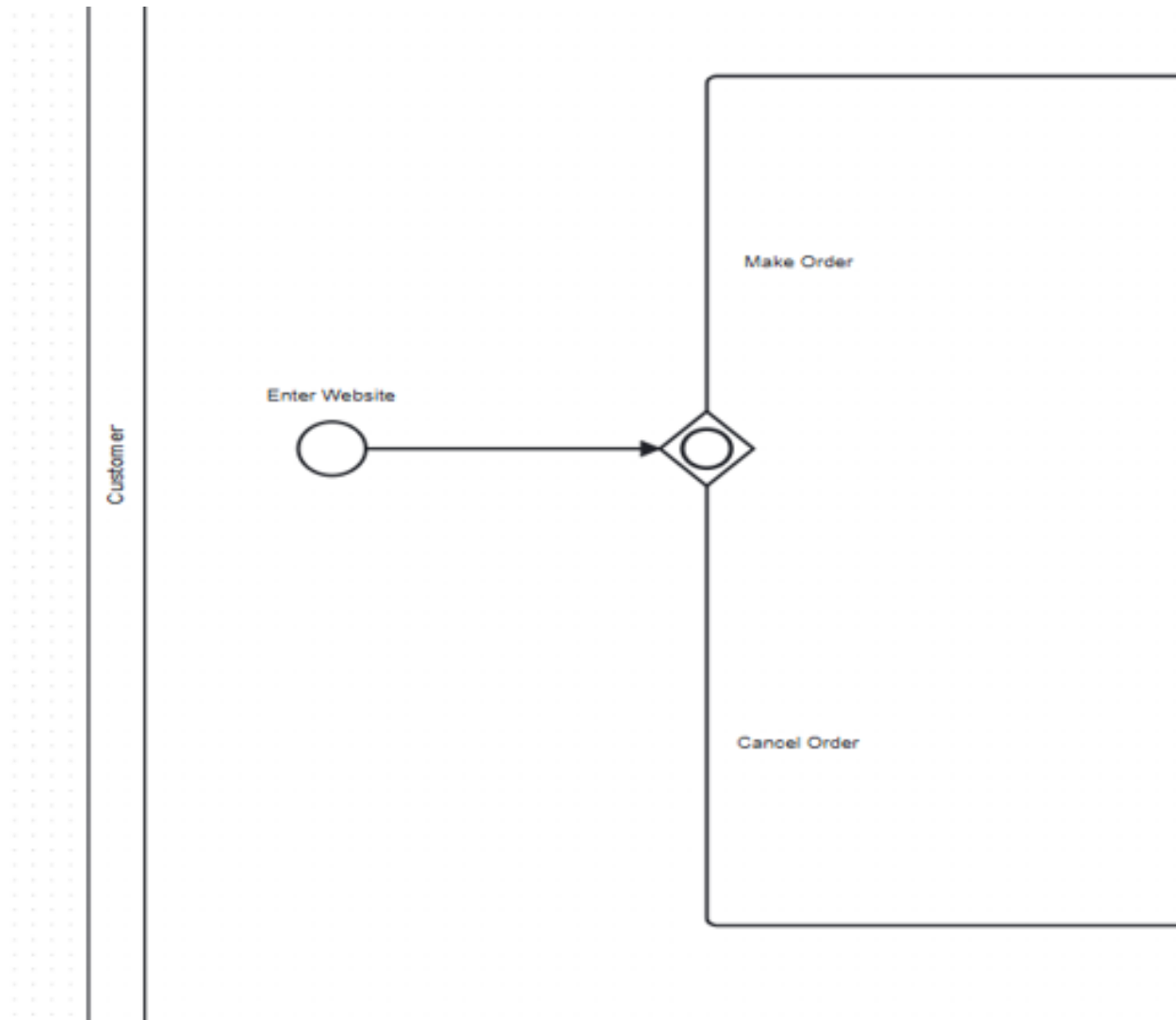


Figure 2.12: Customer Pool: Starting the Order Process

2.5.2 Customer Pool: Find the desired Product

Customer Pool: Find the desired Product Part 1

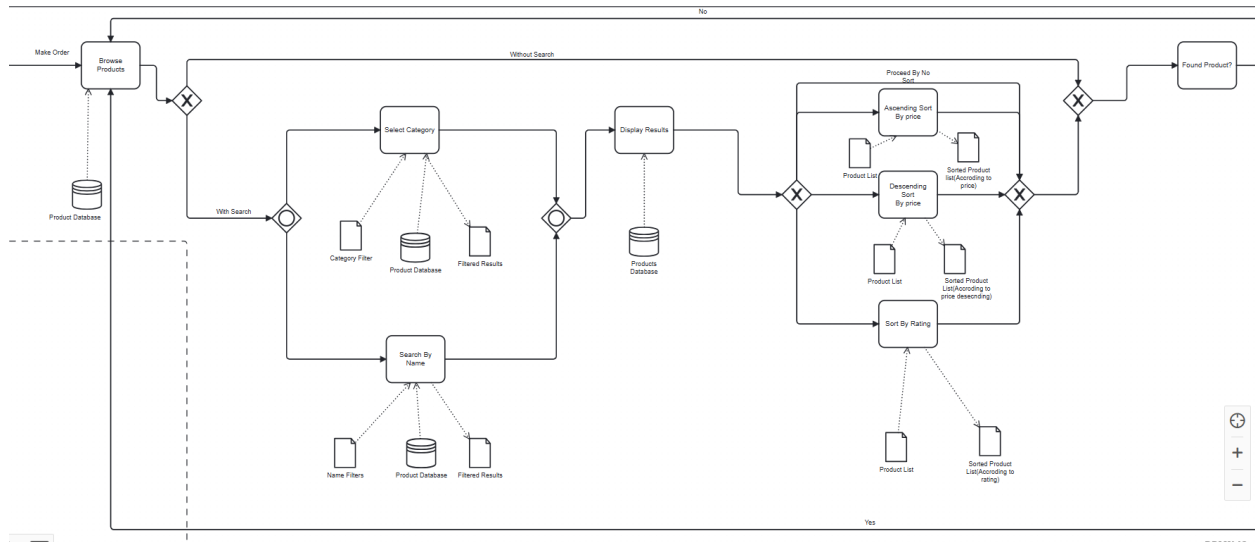


Figure 2.13: Customer Pool: Find the desired Product Part 1

Customer Pool: Find the desired Product Part 2

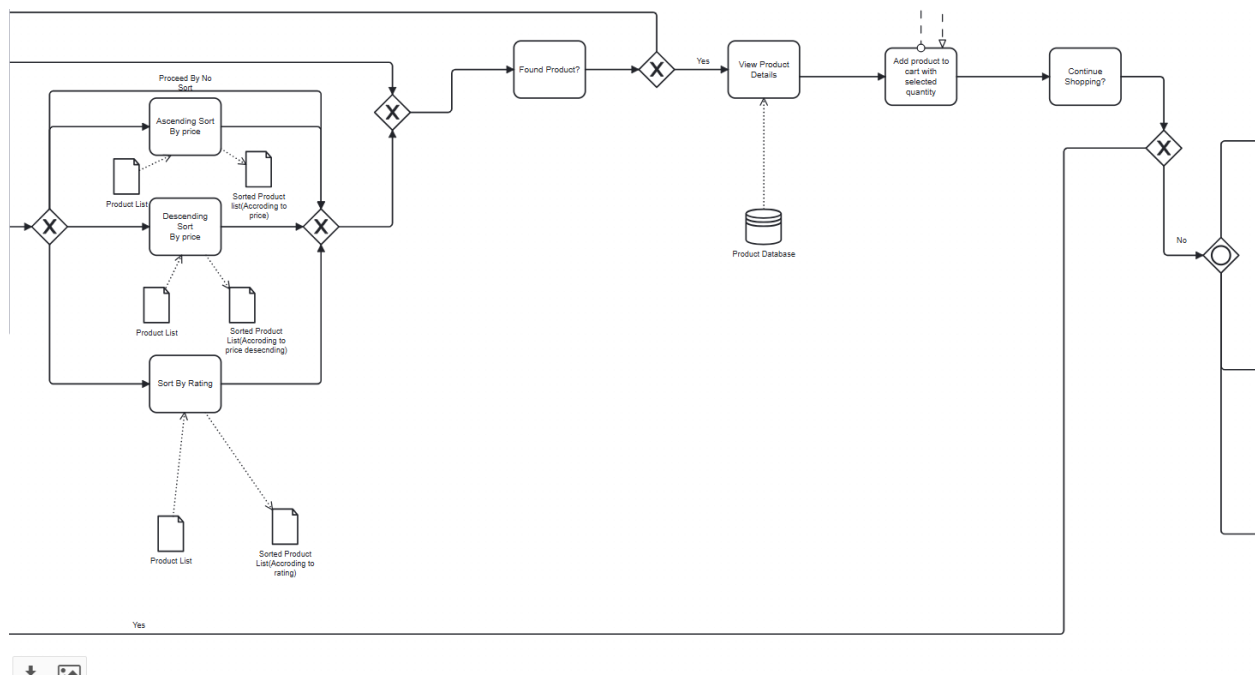


Figure 2.14: Customer Pool: Find the desired Product Part 2

2.5.3 Server Pool: Calculating Cart after Adding Product to Cart

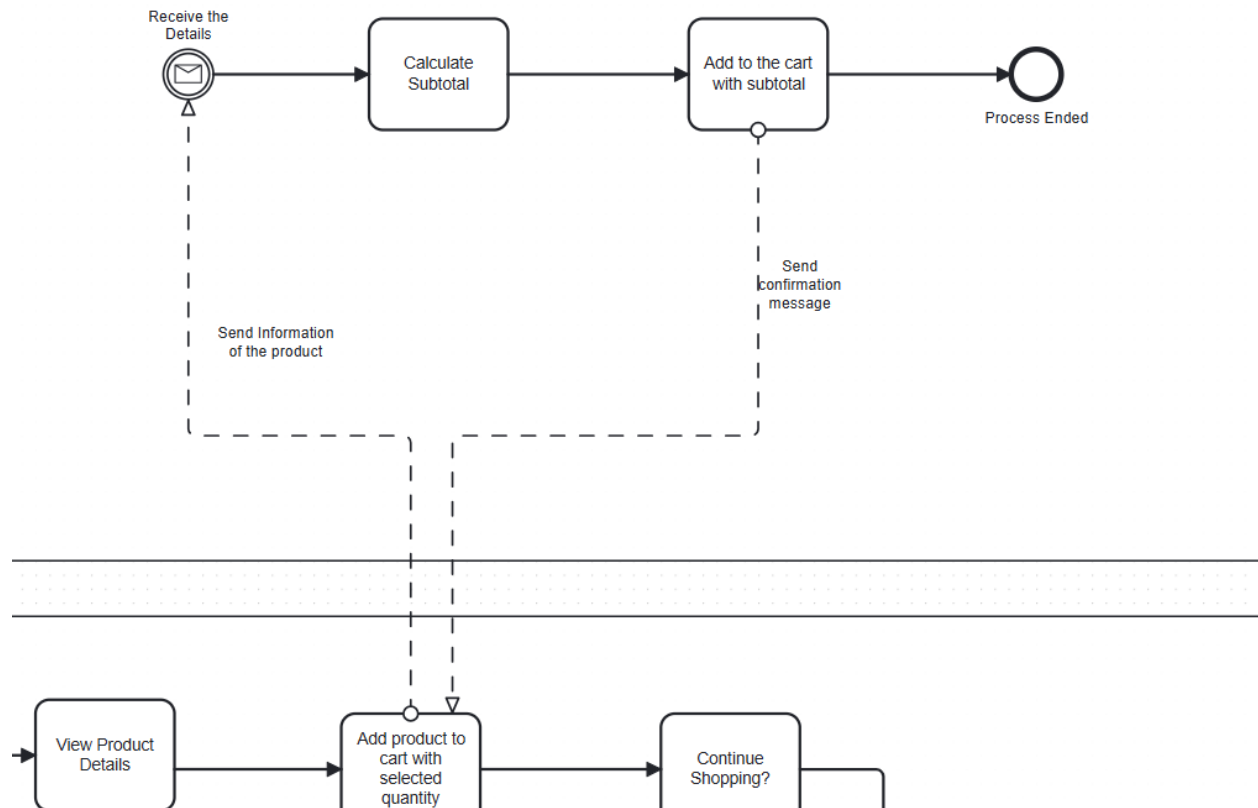


Figure 2.15: Server Pool: Calculating Cart after Adding Product to Cart

2.5.4 Customer Pool: Managing Cart

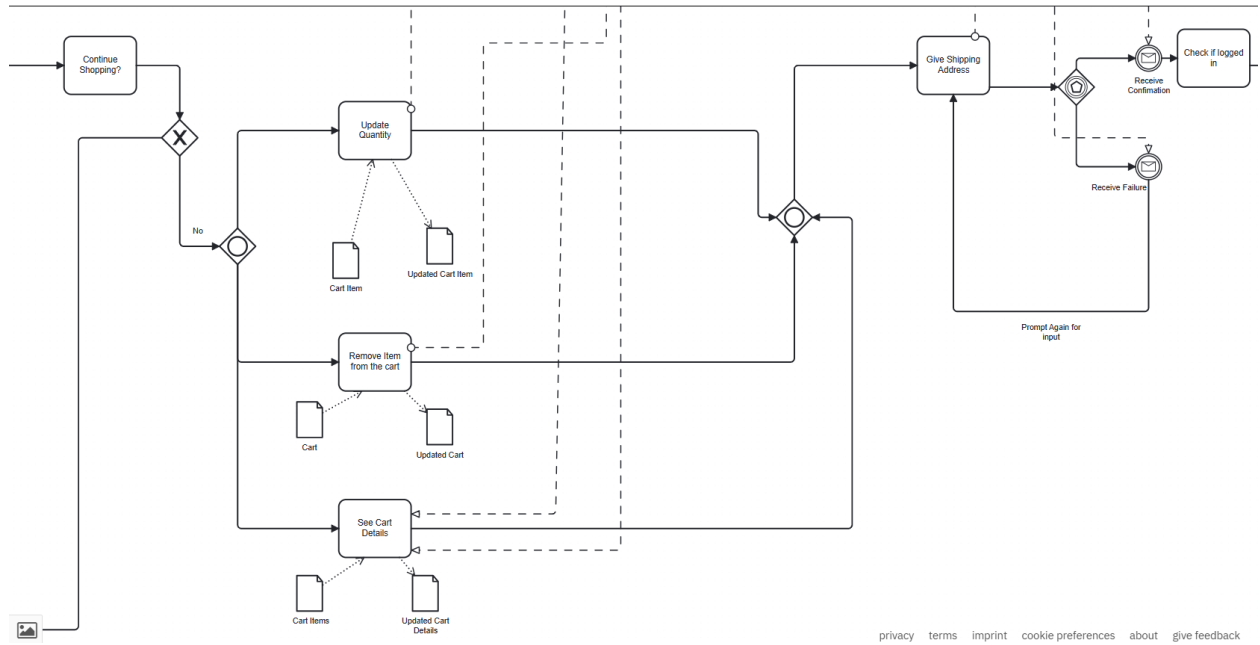


Figure 2.16: Customer Pool: Managing Cart

2.5.5 Server Pool: Updating Cart after Managing Cart

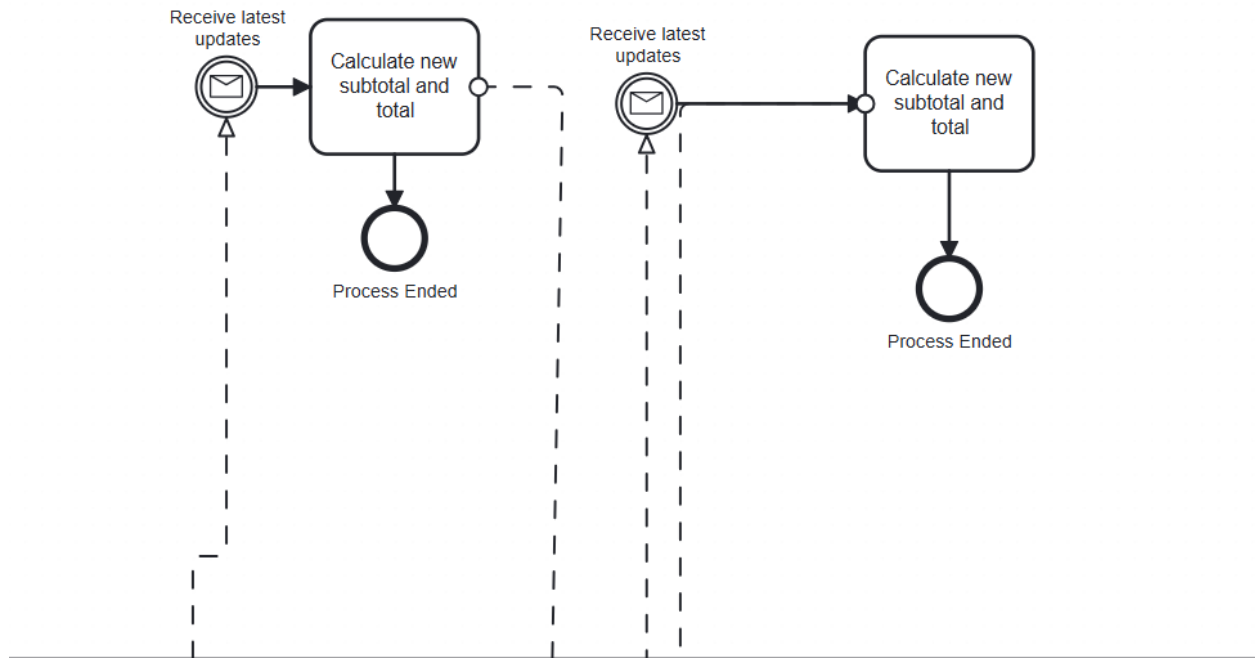


Figure 2.17: Server Pool: Updating Cart after Managing Cart

2.5.6 Customer Pool: Give Address and Place Order

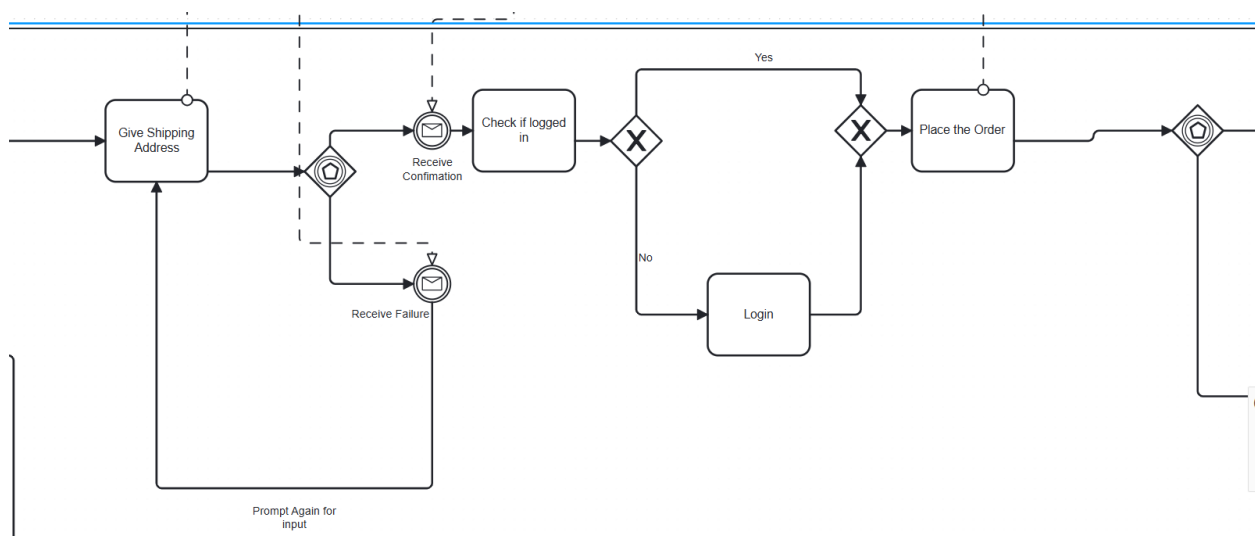


Figure 2.18: Customer Pool: Give Address and Place Order

2.5.7 Server Pool: Validate Address

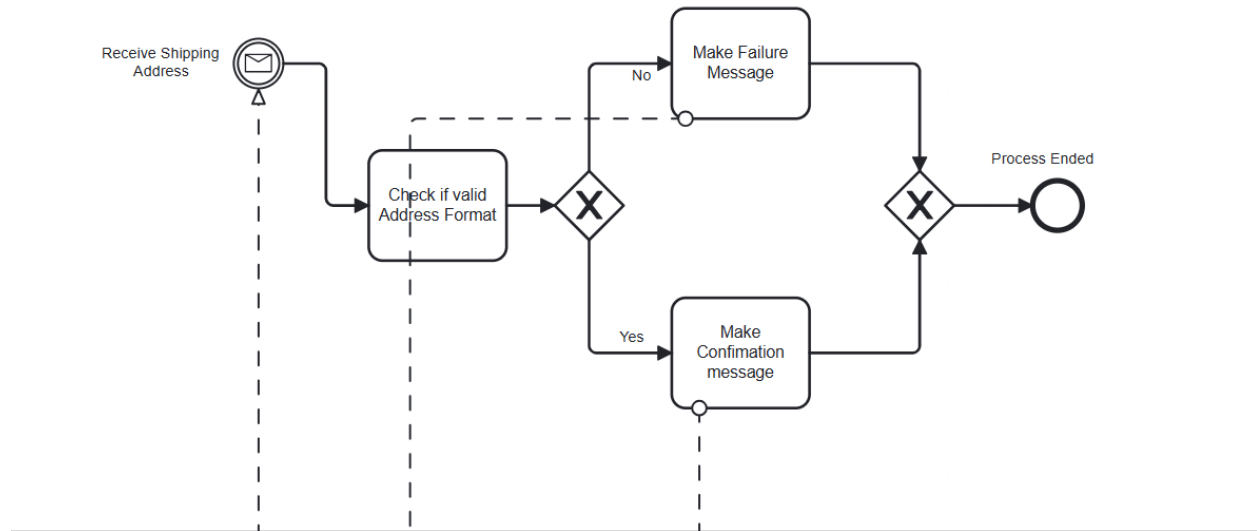


Figure 2.19: Server Pool: Validate Address

2.5.8 Server Pool: Process Placing Order (with taking help from Vendor Pool)

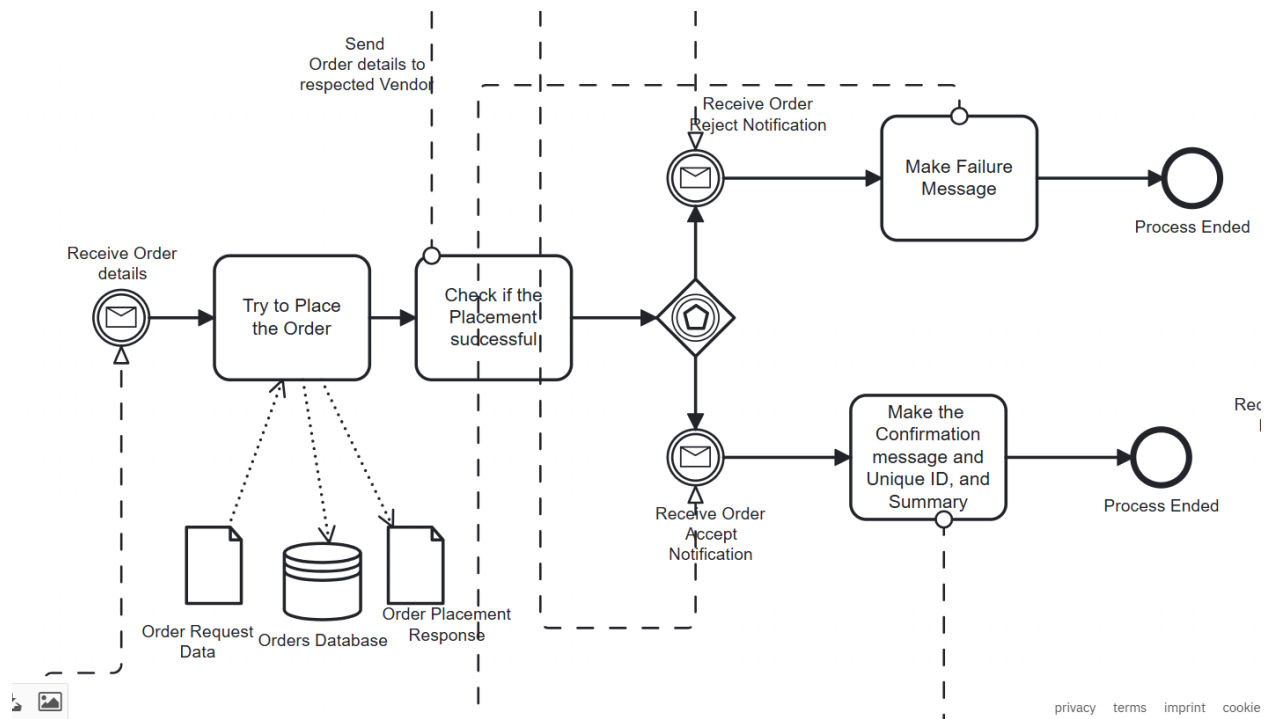


Figure 2.20: Server Pool: Process Placing Order (with taking help from Vendor Pool)

2.5.9 Vendor Pool: Accepting/Rejecting Order

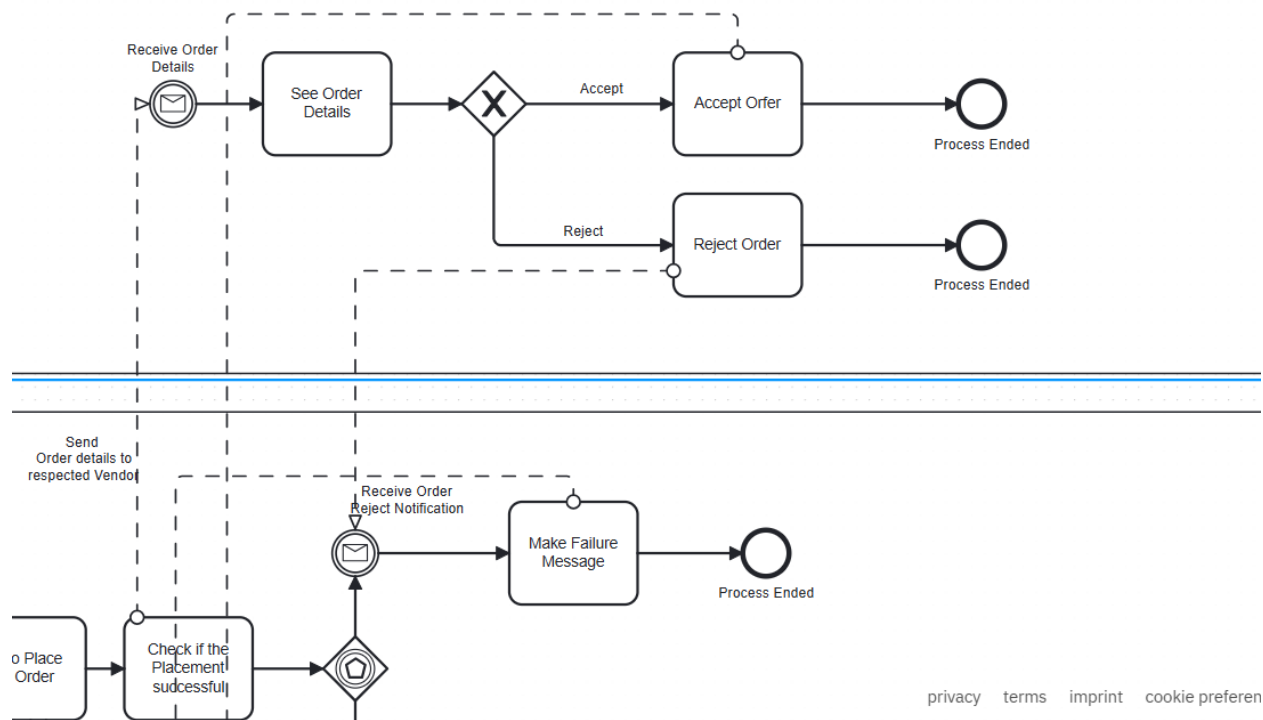


Figure 2.21: Vendor Pool: Accepting/Rejecting Order

2.5.10 Customer Pool :Payment Processing and Order Completion Workflow

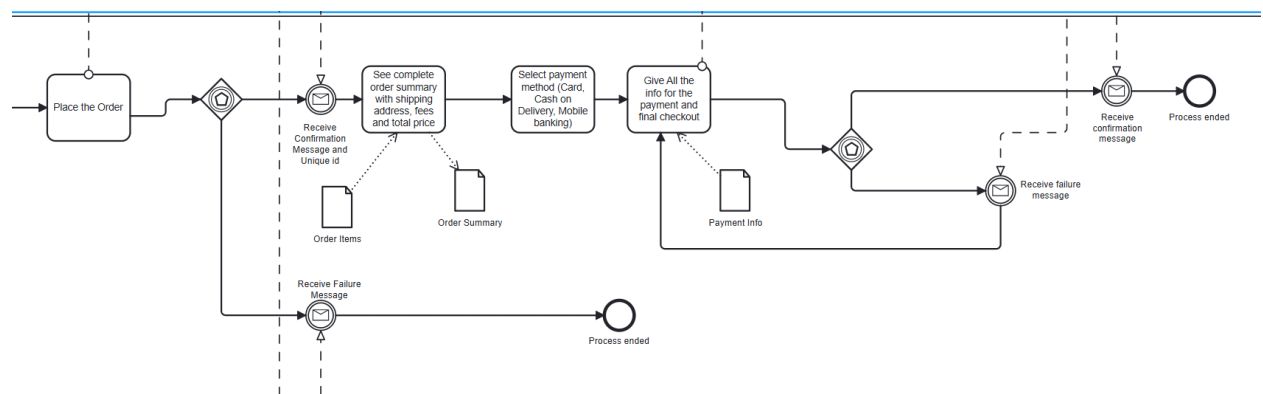


Figure 2.22: Customer Pool: Payment Processing and Order Completion Workflow

2.5.11 Server Pool: Handling Order Payment ,Shipping Information and Updating Stock

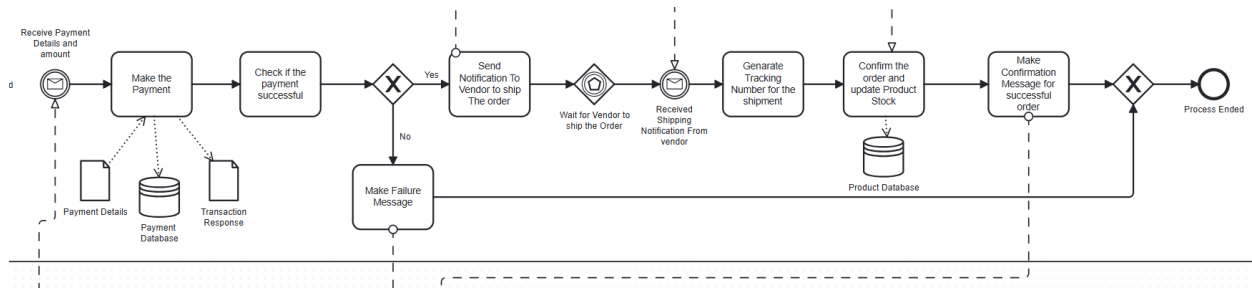


Figure 2.23: Server Pool: Handling Order Payment, Shipping Information and Updating Stock

2.5.12 Vendor Pool: Ship the Order and Update their Stock(Doing both communicating with server pool)

Here the Info of vendor of Shipping sent to Sever's "Received Shipping Notification From vendor" and Updating stock of vendor goes to Server's "Confirm the order and update Product Stock".

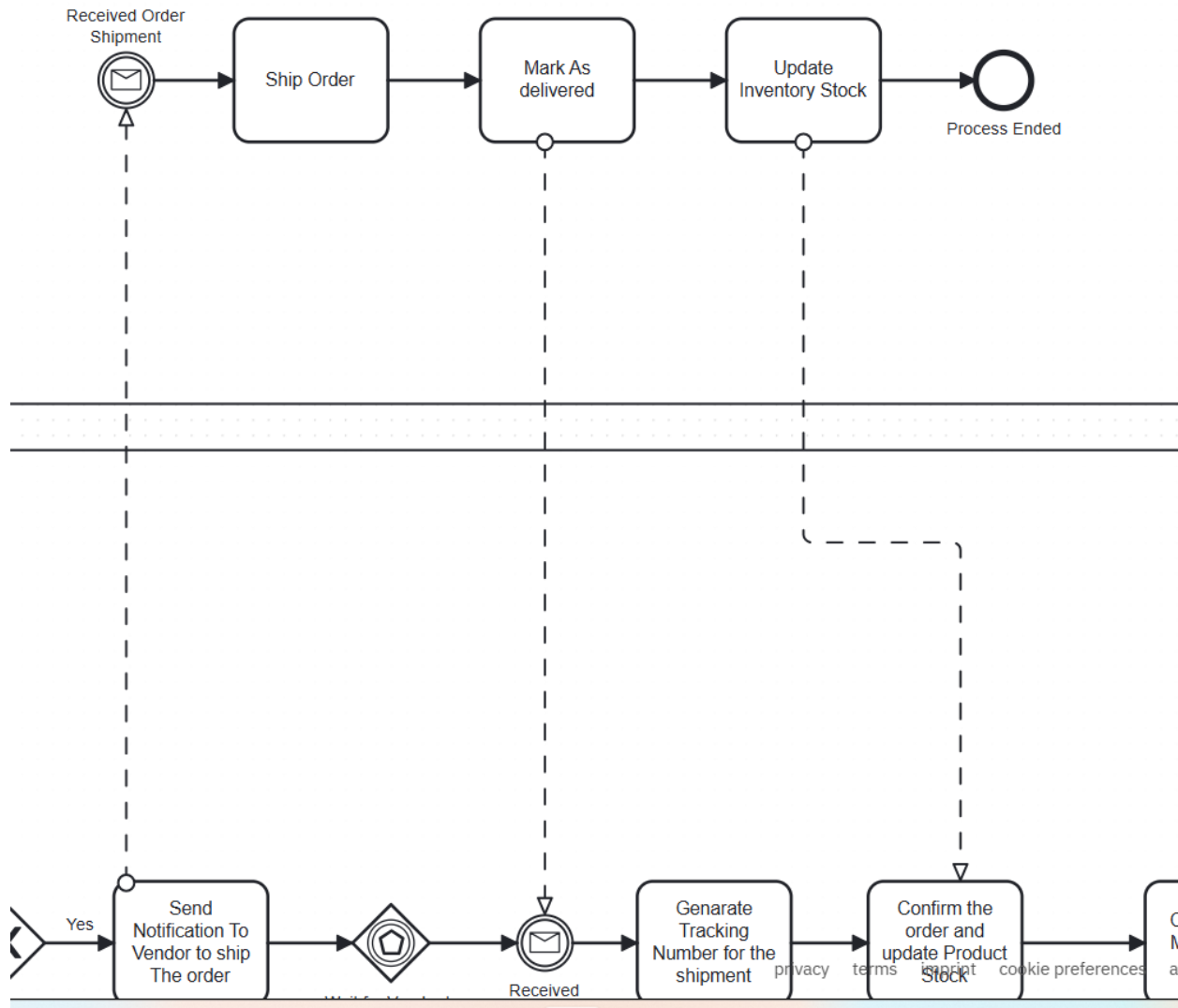


Figure 2.24: Vendor Pool: Ship the Order and Update their Stock (Doing both communicating with Server Pool)

2.5.13 Customer Pool: Order Cancellation Workflow

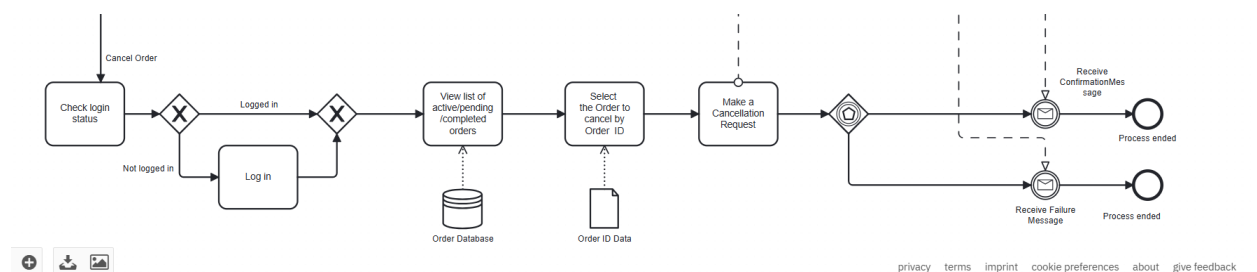


Figure 2.25: Customer Pool: Order Cancellation Workflow

2.5.14 Server Pool: Handling Order Cancellation

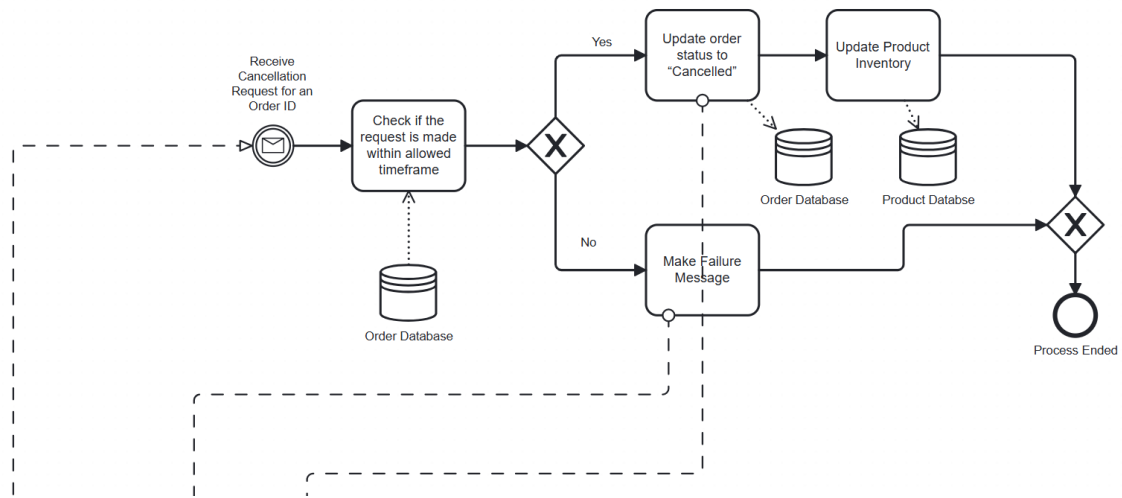


Figure 2.26: Server Pool: Handling Order Cancellation

2.5.15 Vendor Pool: Add or Remove Product (Separate Subprocess)

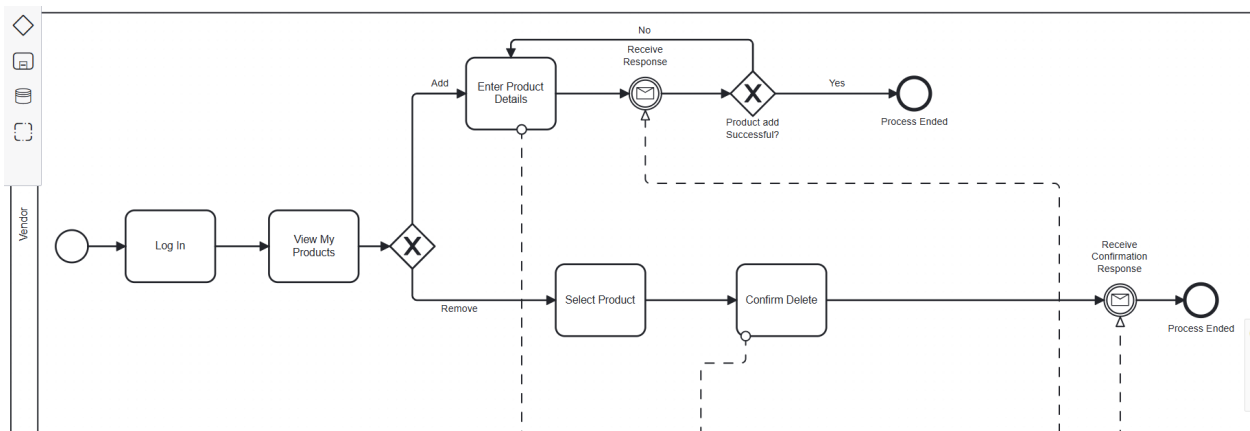


Figure 2.27: Vendor Pool: Add or Remove Product (Separate Subprocess)

2.5.16 Server Pool: Handling Adding or Removing Product by Vendor

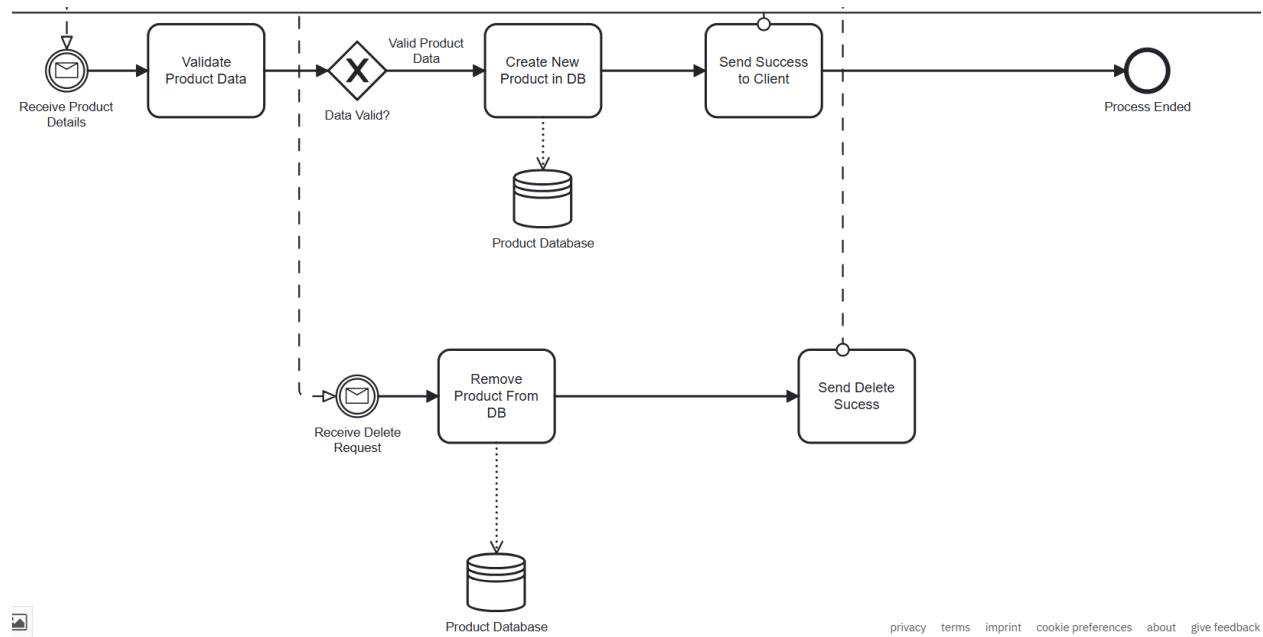


Figure 2.28: Server Pool: Handling Adding or Removing Product by Vendor

Use Case Diagram

The presentation of the use case diagram was delivered by Md. Asif Ali (2105038).

Modifications Suggested:

- Add internal server for order cancellation handling and other related tasks.

3.1 Full Use Case Diagram

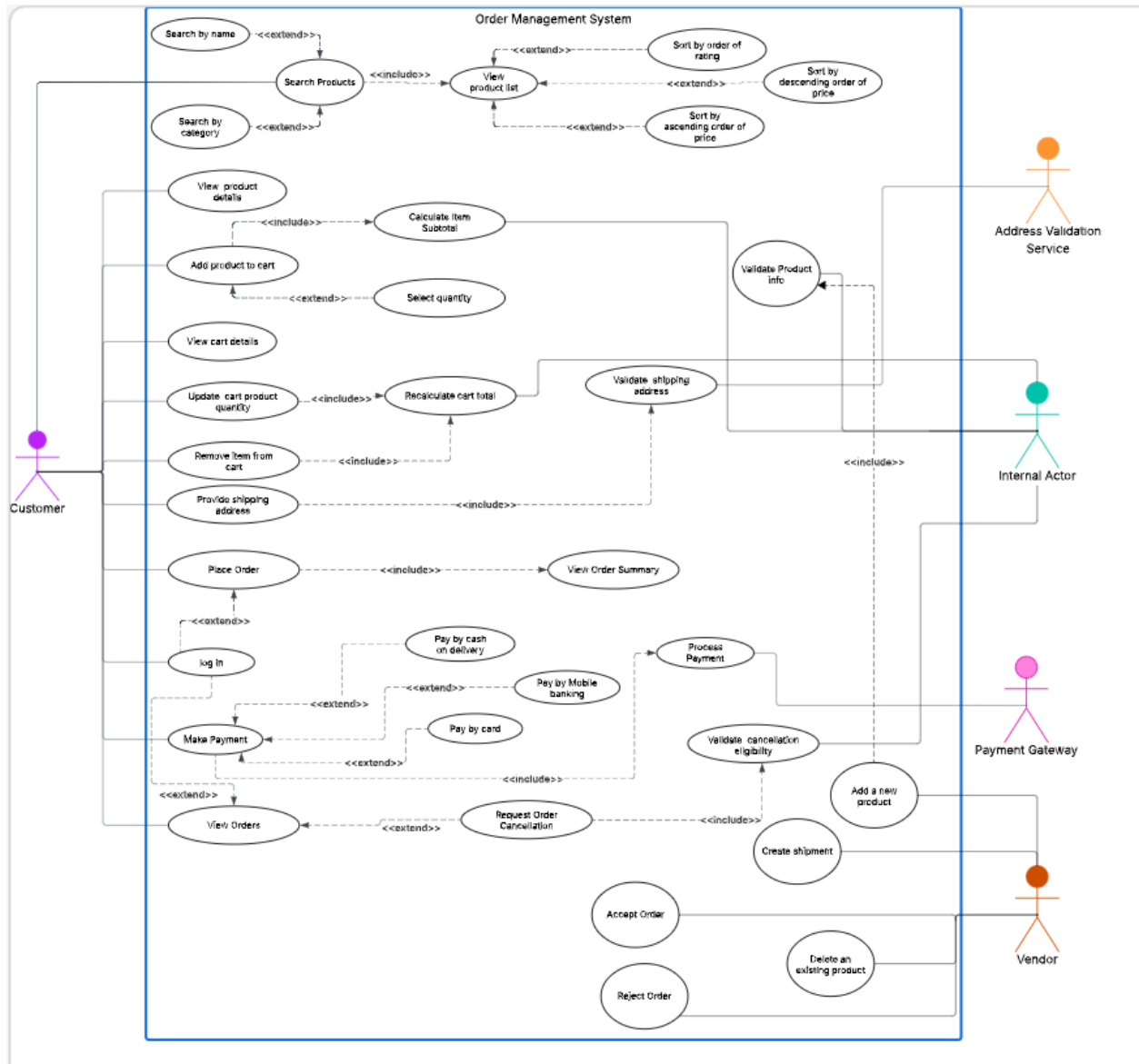


Figure 3.1: Full Use Case Diagram

3.2 Use Case Diagram (Part 1)

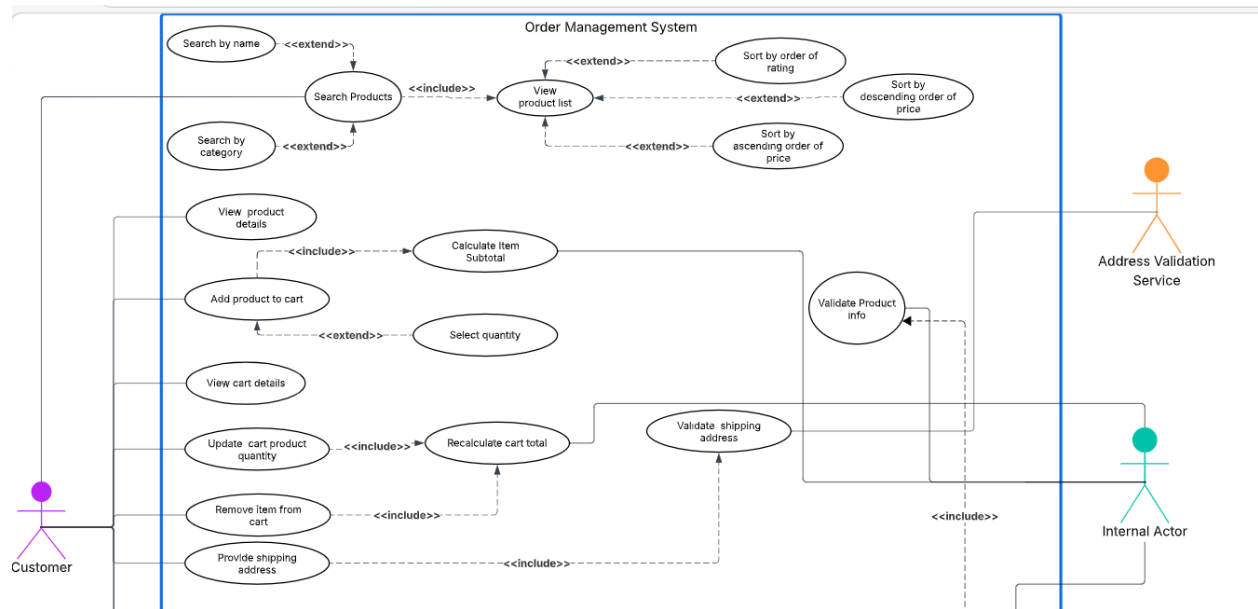


Figure 3.2: Use Case Diagram (Part 1)

3.3 Use Case Diagram (Part 2)

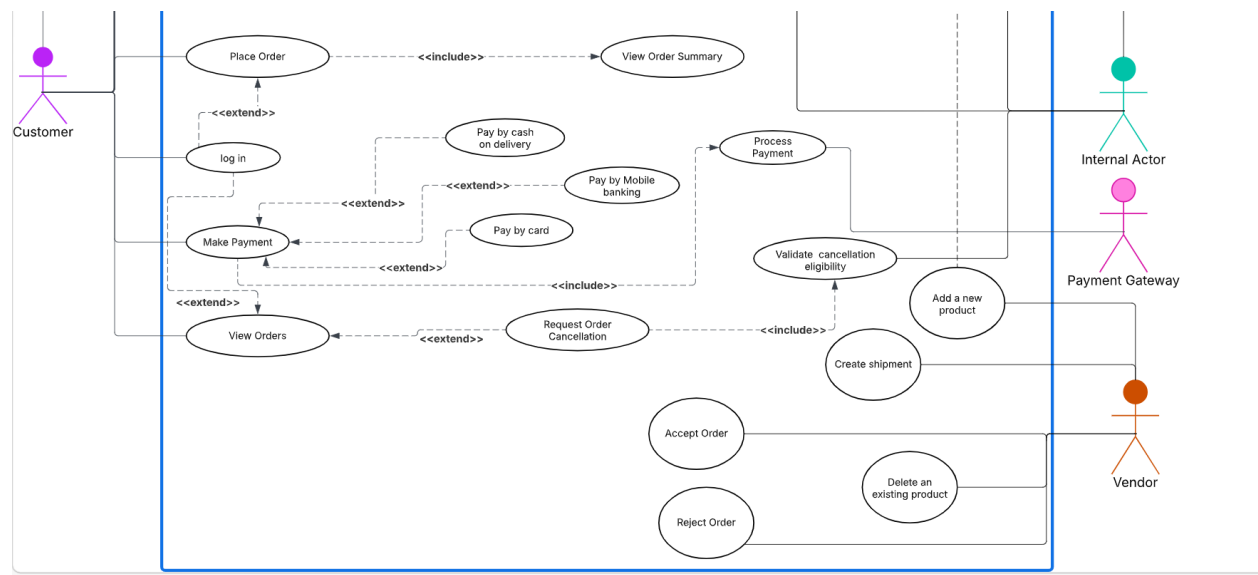


Figure 3.3: Use Case Diagram (Part 2)

Class

Diagram

The presentation of the class diagram was delivered by Md. Asif Ali (2105038).

Modifications Suggested:

- Add notification id and notification time attributes in the NotificationService class.

4.1 Full Class Diagram

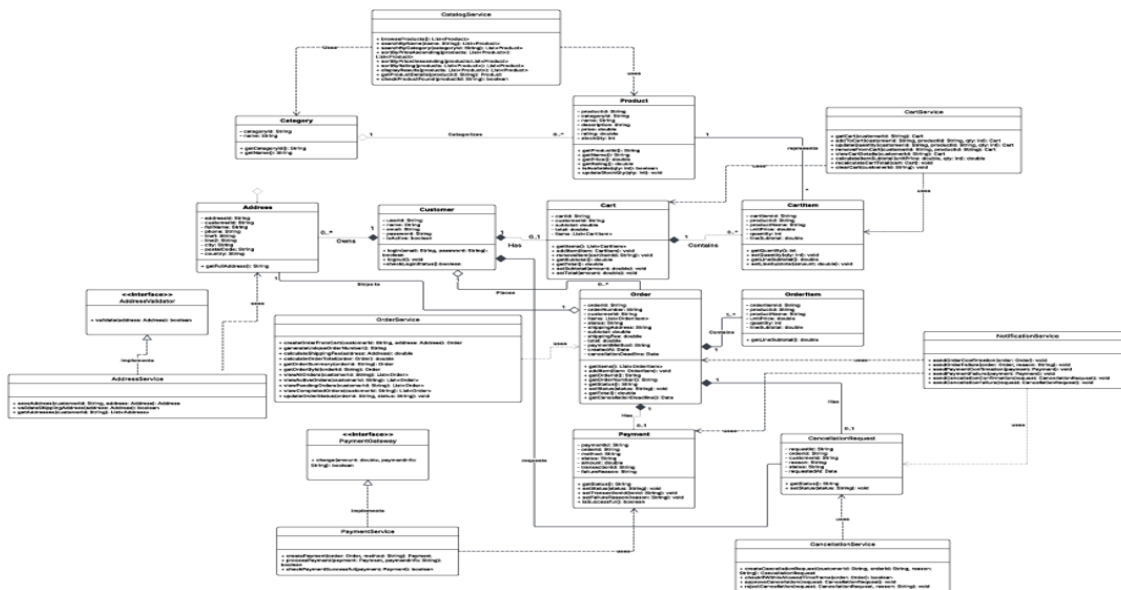


Figure 4.1: Full Class Diagram

4.2 Product Catalog

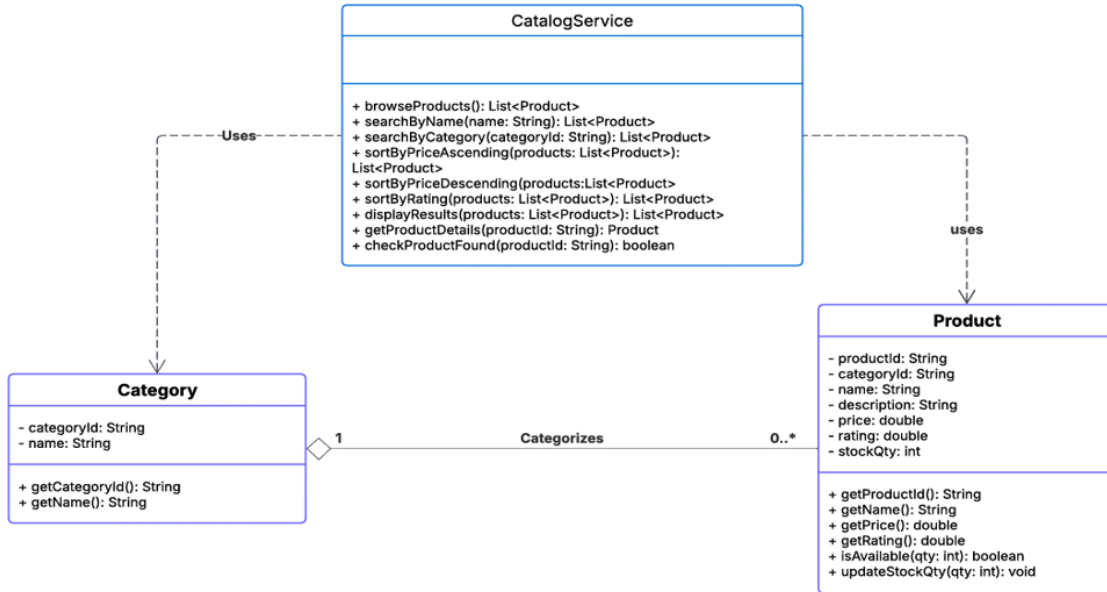


Figure 4.2: Product Catalog

4.3 Shopping Cart

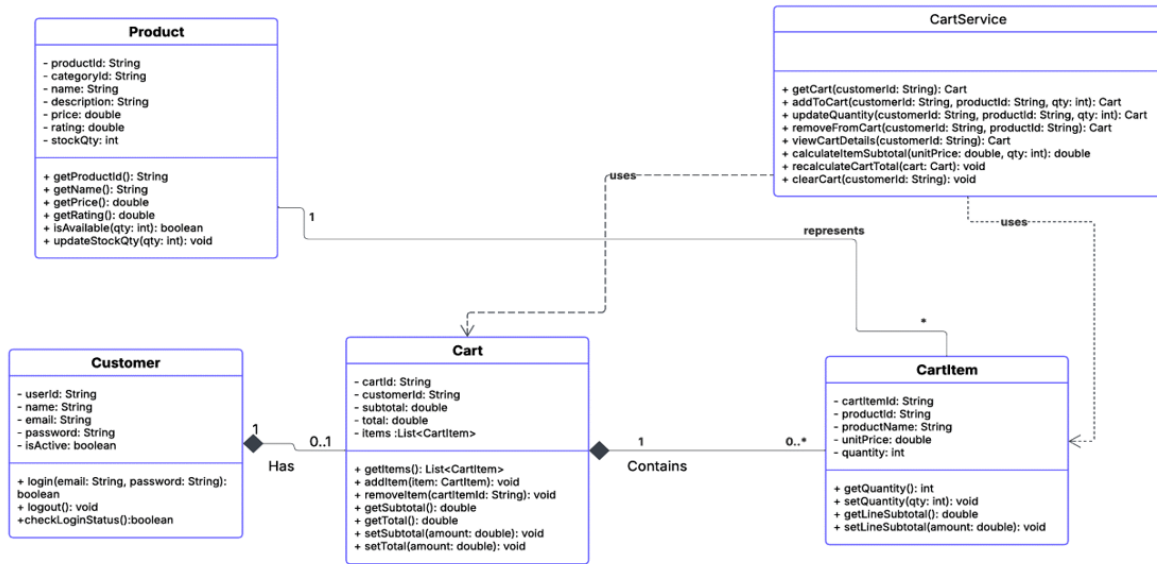


Figure 4.3: Shopping Cart

4.4 Order Creation

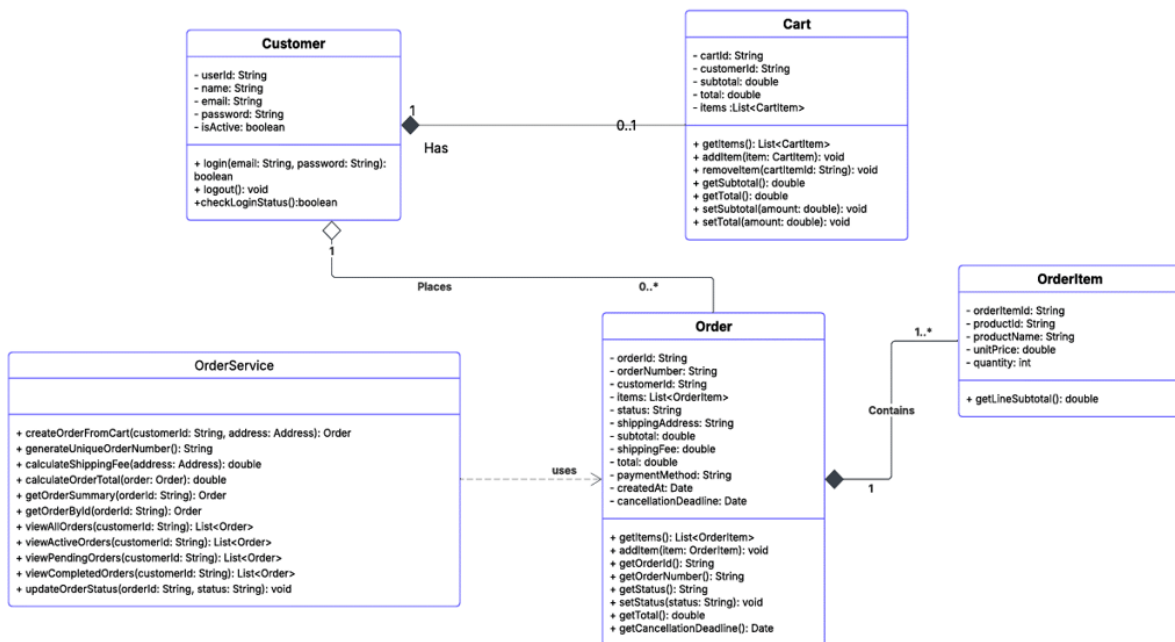


Figure 4.4: Order Creation

4.5 Shipping

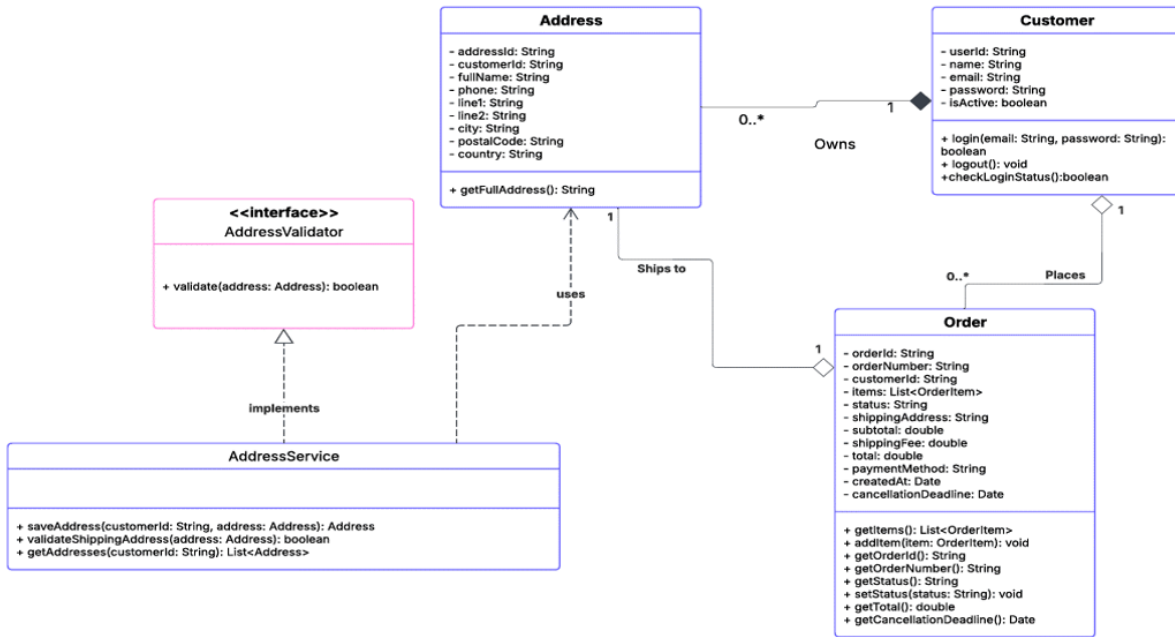


Figure 4.5: Shipping

4.6 Payment Processing

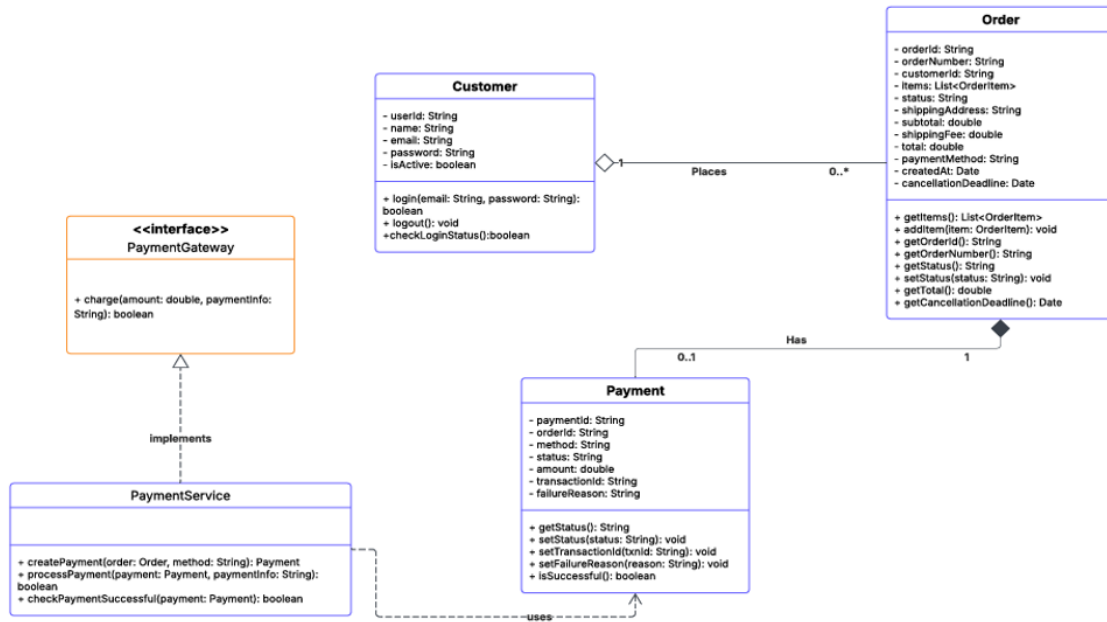


Figure 4.6: Payment Processing

4.7 Order Cancellation

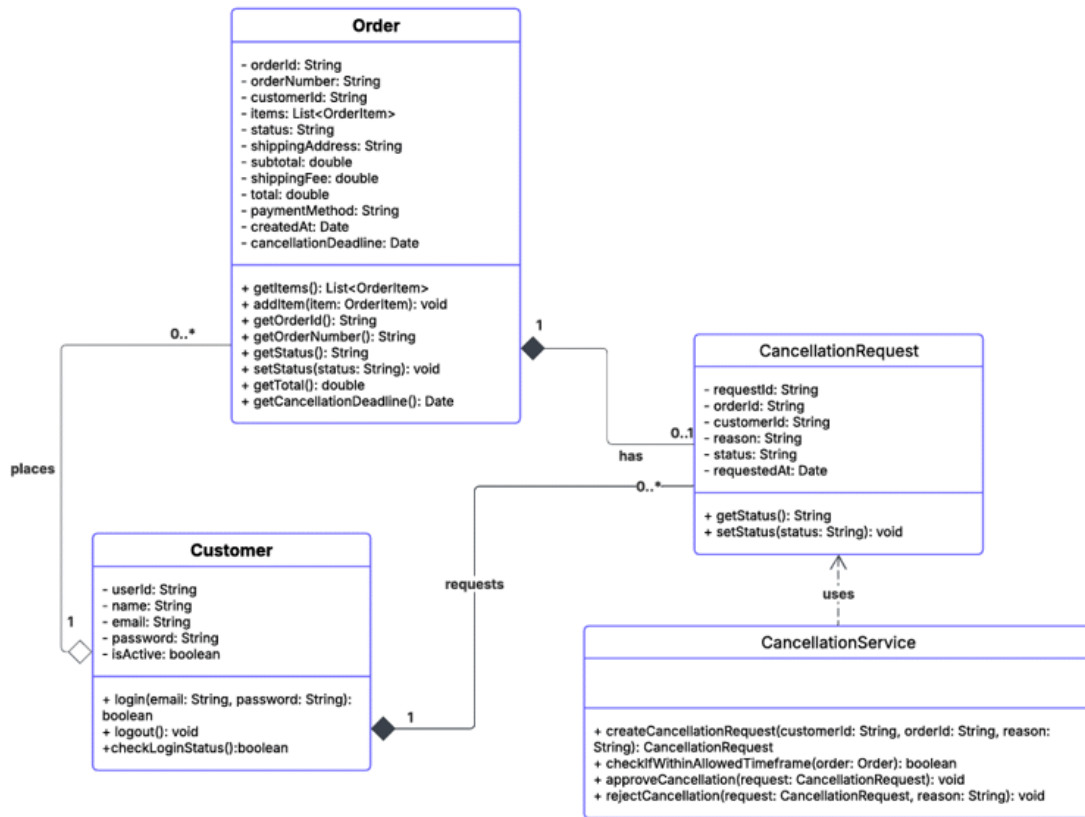


Figure 4.7: Order Cancellation

4.8 Notification Management

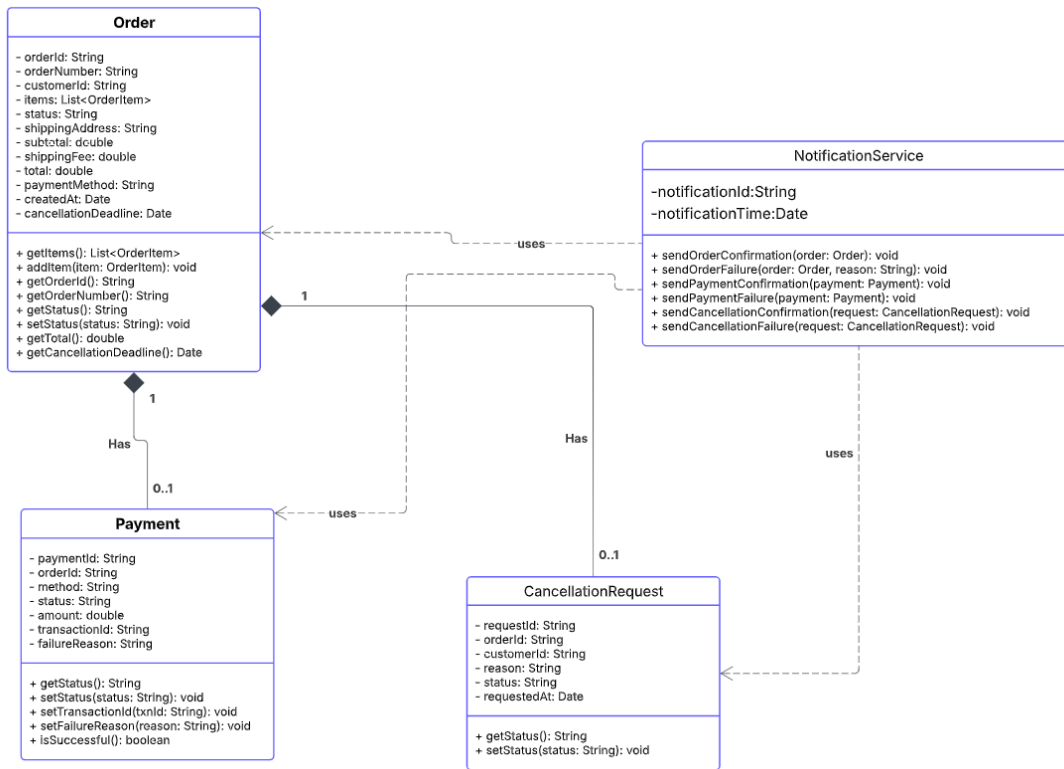


Figure 4.8: Notification Management

4.9 Vendor Side Product and Order Management

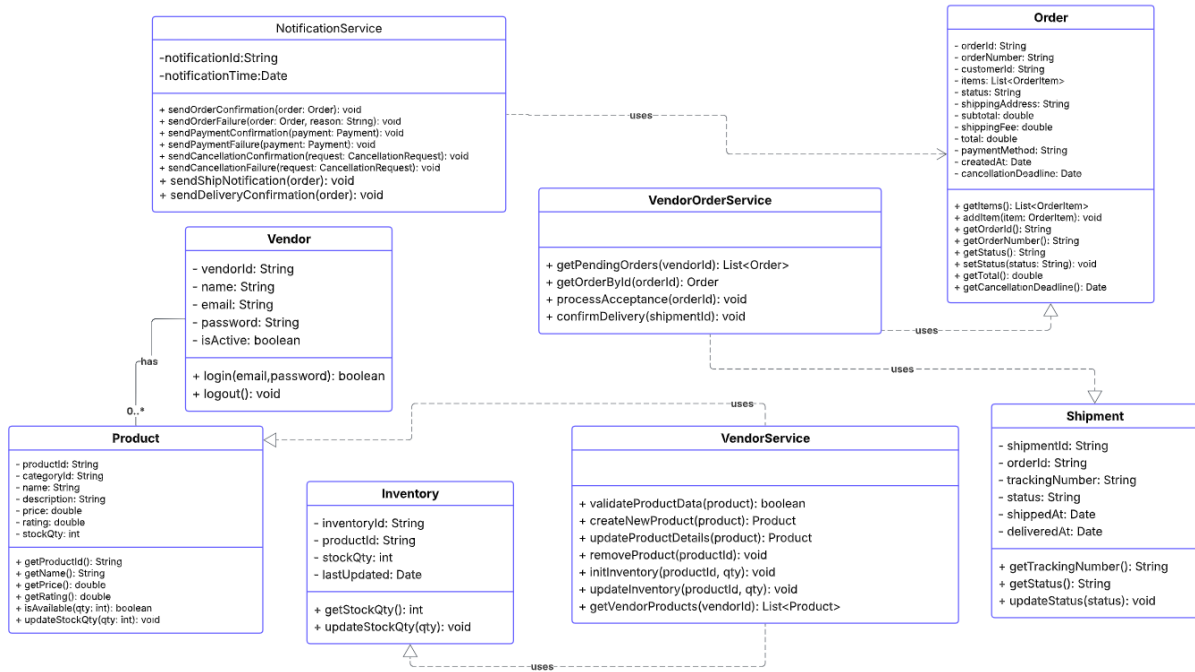


Figure 4.9: Vendor Side Product and Order Management

Mock

UI

The presentation of the Mock UI was delivered by Md. Mehedi Hasan (2105052).

Modifications Suggested:

- Add Review and Rating System for products.
- Add a wishlist page

5.1 Homepage

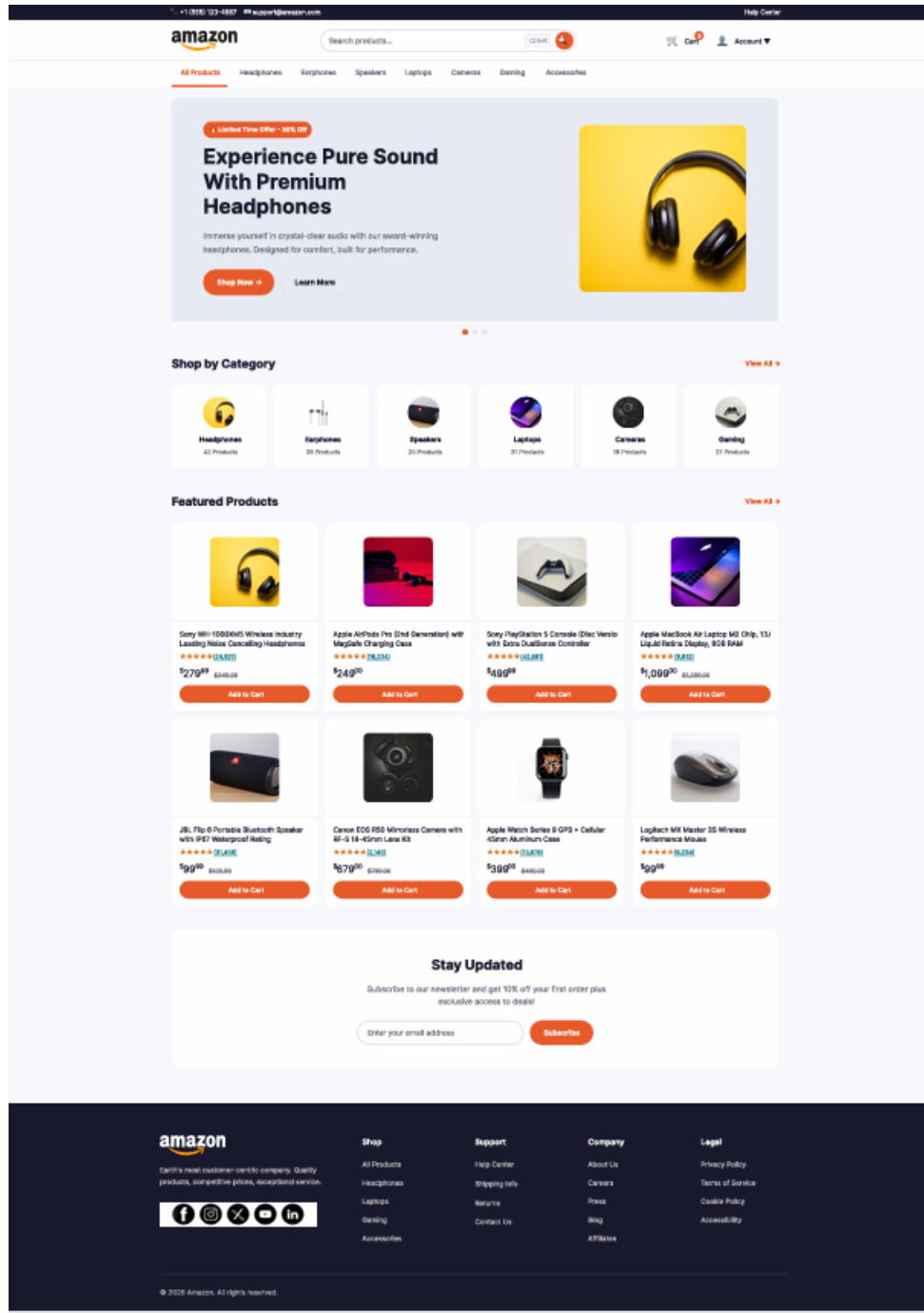


Figure 5.1: Homepage

5.2 View All Featured Products

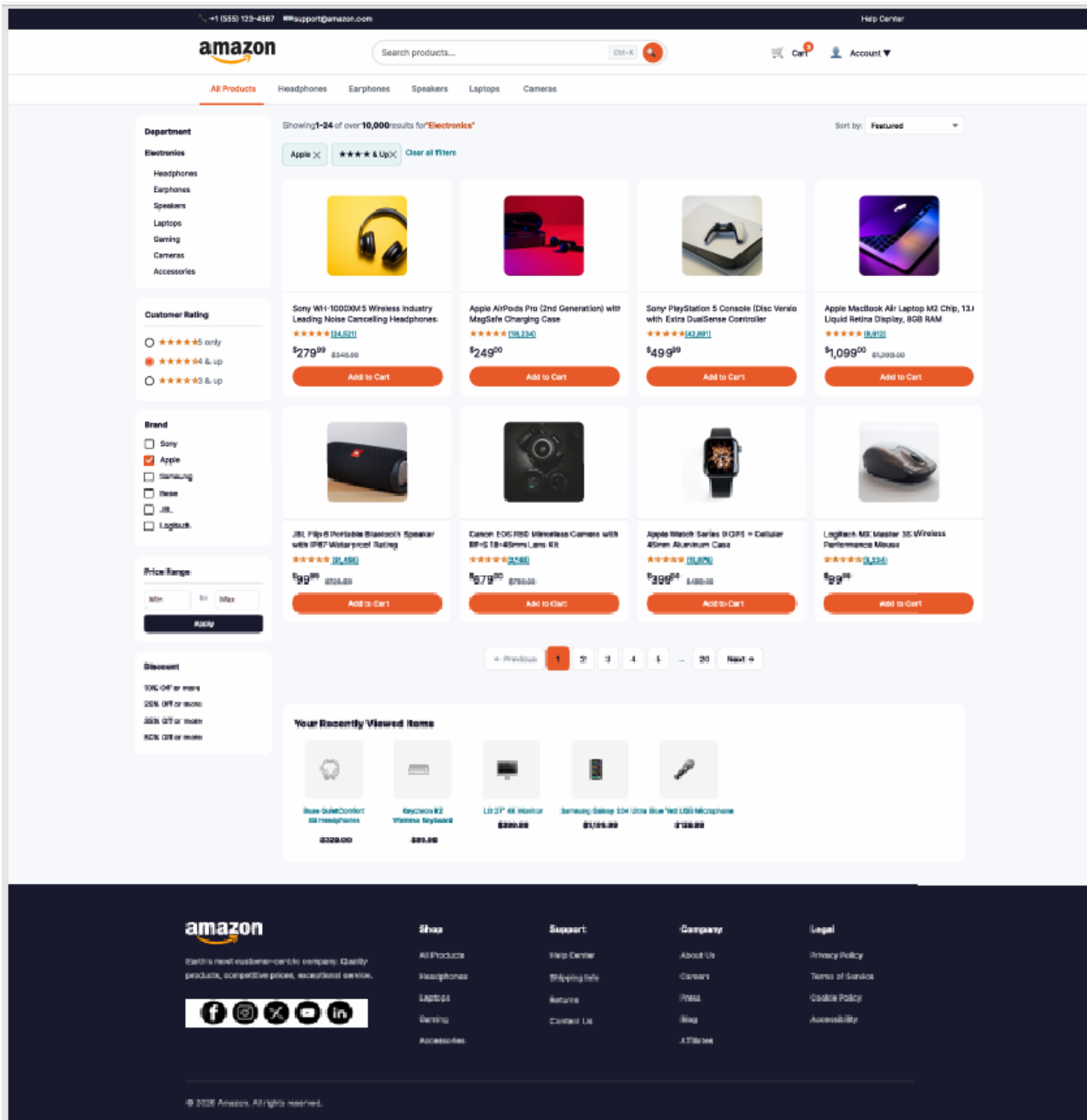


Figure 5.2: View All Featured Products

5.3 Search For Product Page

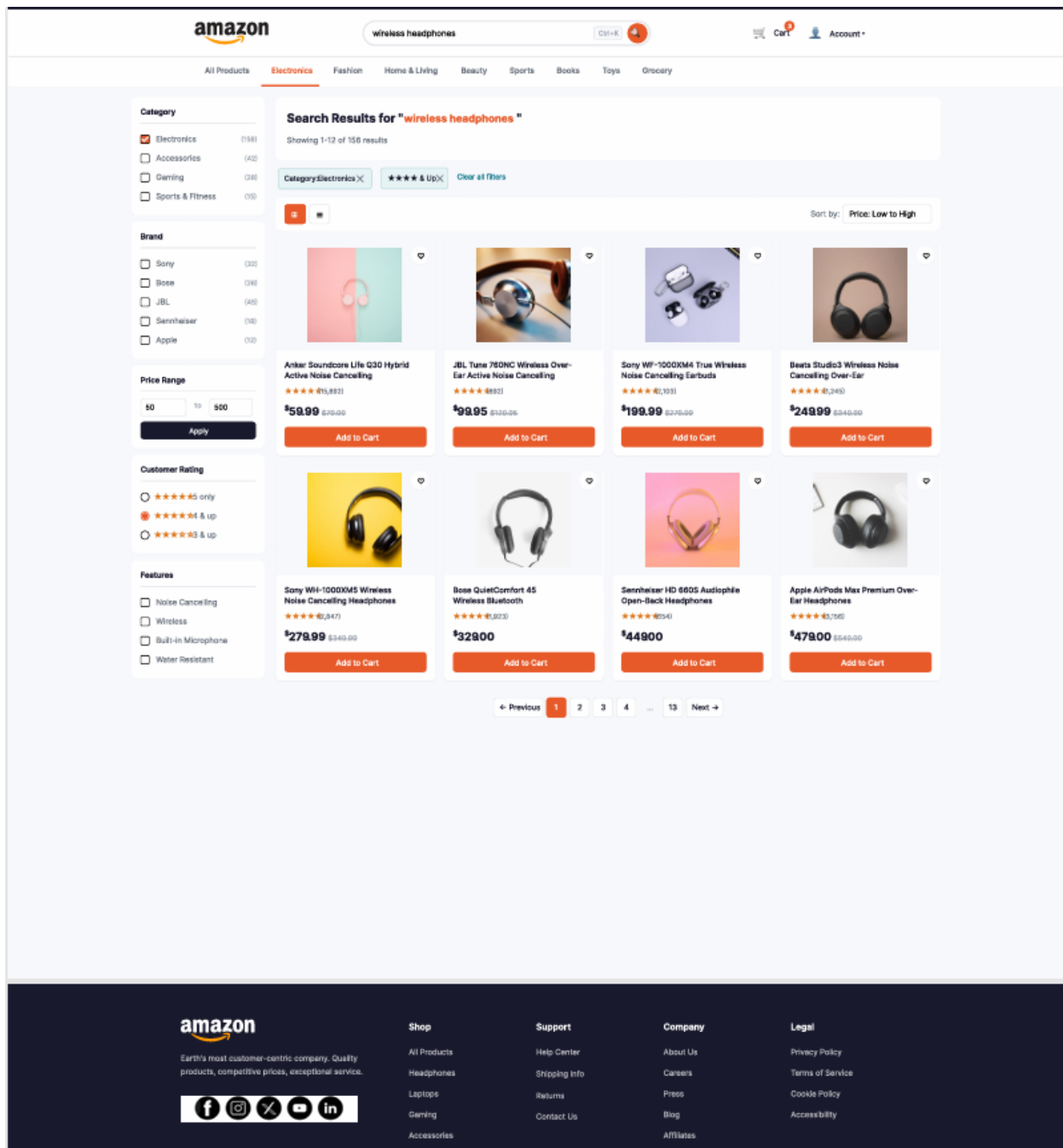


Figure 5.3: Search For Product Page

5.4 Product Details Page with related products

amazon

Search products, brands and more...

Cart

Account

All Products

Electronics

Fashion

Home & Living

Beauty

Sports

Books

Toys

Grocery

Home

Electronics

Headphones

Sony WH-1000XM5





Sony WH-1000XM5 Wireless Noise Cancelling Headphones

★★★★★ 4.8 2,847 reviews

\$279.99

~~\$349.00~~ Save \$70 (20% off)

Color:







Quantity:

-

1

+

In Stock

Add to Cart

Buy Now



✓ Industry-leading noise cancellation technology

✓ Up to 30 hours of battery life

✓ Premium sound quality with LDAC support

✓ Multipoint connection for seamless switching

✓ Speak-to-Chat technology automatically pauses music

 Free Delivery: Arrives by Jan 23

 Free Returns: 30-day return policy

 2-Year Warranty: Manufacturer warranty included

Description

Specifications

Reviews (2,847)

The Sony WH-1000XM5 headphones rewrite the rules for distraction-free listening. Two processors control 8 microphones for unprecedented noise cancellation. With Auto NC Optimizer, noise cancelling is automatically optimized based on your wearing conditions and environment.

Enjoy premium sound quality with our Integrated Processor V1 and supporting technology that enhances sound quality and reduces distortion. The lightweight design with soft fit leather ensures total comfort for extended listening sessions.

Up to 30 hours of battery life with quick charging (3 min charge for 3 hours of playback). Multi-point connection allows you to quickly switch between devices. Speak-to-Chat automatically pauses playback when you start talking.

Related Products

View All →



Bose QuietComfort 45 Wireless Headphones

★★★★★ (8,833)

\$329.00



Apple AirPods Max Premium Over-Ear Headphones

★★★★★ (1,194)

\$479.00



Sennheiser HD 600S Audiophile Headphones

★★★★★ (654)

\$449.00



Sony WF-1000XM4 True Wireless Earbuds

★★★★★ (2,533)

\$199.99

Figure 5.4: Product Details Page

44

5.5 Ratings and Reviews

The screenshot displays the Amazon product page for 'Amazon Basics Multipurpose Drawstring Trash Bags, Unscented, 30 Gallon, 50 Count (Previously Solimo)'. The product is priced at \$13.65. The page features a star rating of 4.0 based on 1 review, with a bar chart showing the distribution of ratings: 5 stars (0), 4 stars (1), 3 stars (0), 2 stars (0), and 1 star (0). A customer review from 'Md Maruf Hasan' dated March 18, 2020, is visible, stating 'very good product its perfect'. Below the reviews, there is a 'Related Products' section with eight items, each with its own rating and price. The footer contains links for 'Get to Know Us', 'Connect with Us', 'Make Money', and 'Let Us Help'.

Amazon | Deliver to: Your Location | Search products... | Md Maruf Hasan | Returns & Orders | Seller Dashboard | Cart

Home > Products > Household Cleaning Supplies > Amazon Basics Multipurpose Drawstring Trash Bags

HOUSEHOLD CLEANING SUPPLIES

Amazon Basics Multipurpose Drawstring Trash Bags, Unscented, 30 Gallon, 50 Count (Previously Solimo)

★★★★★ 4.0 (1 reviews)

40,000+ bought in the last month

\$13.65

Quantity: - 1 +

Add to Cart

Free Shipping | Secure Payment | 30-Day Returns

ASIN: B09SPL2MPZ | Category: Household Cleaning Supplies

Customer Reviews | Write a Review

4.0 | 5 star: 0 | 4 star: 1 | 3 star: 0 | 2 star: 0 | 1 star: 0

very good product its perfect
March 18, 2020

Related Products

Product Name	Rating	Price
HOUSEHOLD CLEANING SUPPLIES 360 by Whole Foods Market, Paper Towels 135 Sheet Jumbo Rolls 3...	★★★★★ 4.3 (0)	\$5.29
HOUSEHOLD CLEANING SUPPLIES Lysol Disinfectant Wipes Multi-Surface Antibacterial Cleaning Wipe...	★★★★★ 4.8 (0)	\$11.94
HOUSEHOLD CLEANING SUPPLIES Small Trash Bags 4-8 Gallon, Heavyduty 200 Count Biodegradable Trash Ba...	★★★★★ 4.4 (0)	\$14.99
HOUSEHOLD CLEANING SUPPLIES Bell ProGrade Contractor Trash Bags 55 Gallon 20 Bags w/Ties Heav...	★★★★★ 4.4 (0)	\$23.99
HOUSEHOLD CLEANING SUPPLIES The Honest Company Clean Conscious Wipes 99% Water...	★★★★★ 4.8 (0)	\$5.58
HOUSEHOLD CLEANING SUPPLIES Wall Map with 82 inches Long Handle, 360 Degree Rotating Microfiber Tr...	★★★★★ 4.3 (0)	\$24.99
HOUSEHOLD CLEANING SUPPLIES AIDEA Microfiber Cleaning Cloth 100Pack, Softer Highly Absorbent...	★★★★★ 4.7 (0)	\$30.05
HOUSEHOLD CLEANING SUPPLIES 3 in 1 Tile Tub Shower Scrubber/Cleaning Brush with 58"...	★★★★★ 4.2 (0)	\$15.99

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Make Money
Sell on Amazon
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Advertise

Let Us Help
Your Account
Returns & Orders
Help

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Figure 5.5: Ratings and Reviews

5.6 Wishlist

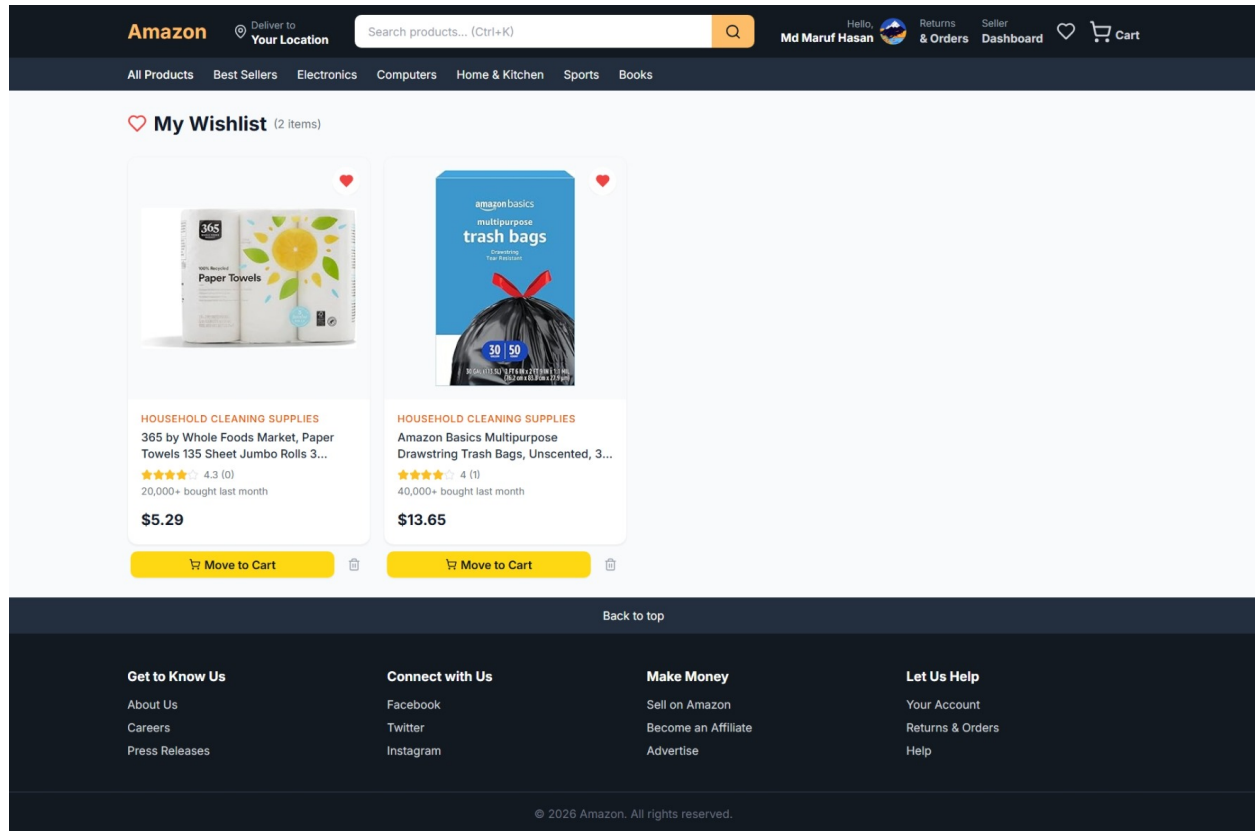


Figure 5.6: Wishlist

5.7 Cart Page

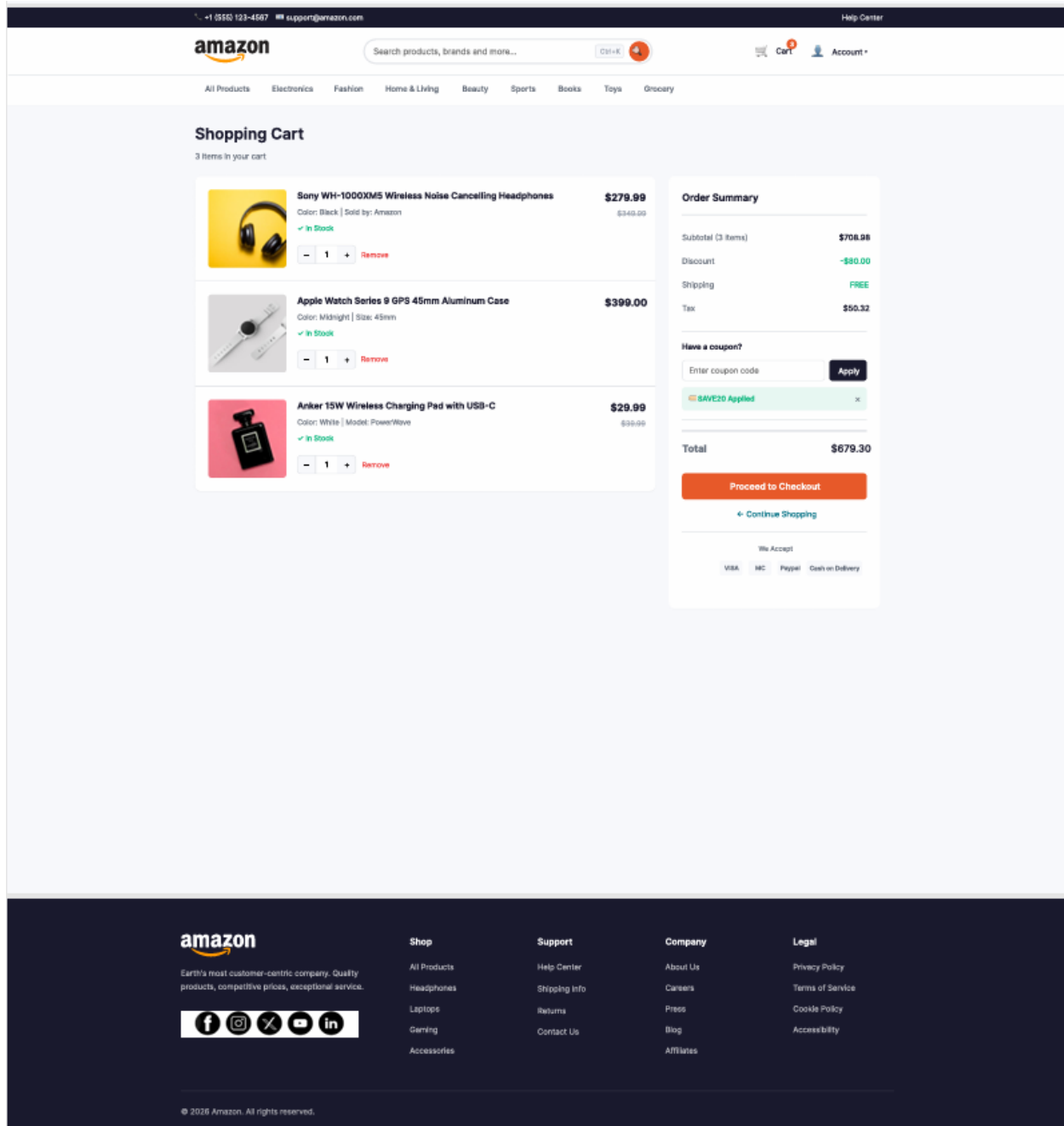


Figure 5.7: Cart Page

5.8 Selecting Shipping Address and View Order Summary Page

The screenshot displays the Amazon checkout process at the 'Shipping' step. The top navigation bar includes the Amazon logo, a search bar, and links for Cart (4 items) and Account. A progress indicator shows four steps: Cart (1), Shipping (2, active), Payment (3), and Confirmation (4).

Shipping Address

Select a saved address

HOME
John Doe
123 Main Street, Apt 4B
New York, NY 10001
United States
+1 (555) 123-4567

OFFICE
John Doe
456 Business Ave, Floor 12
New York, NY 10017
United States
+1 (555) 987-6543

+
Add New Address

Or enter a new address

Address Type

Home

Office

Other

First Name* John **Last Name*** Doe

Phone Number*
+1 (555) 123-4567

We'll send shipping updates to this number

Street Address*
House number, street name

Apartment, Suite, etc. (Optional)
Apt, Suite, Unit, Building, Floor

City* Enter city **State / Province*** New York

ZIP / Postal Code* Enter ZIP code **Country*** United States

Delivery instructions (Optional)
Add any special instructions for delivery...

☐ Save this address for future orders
☐ Set as default shipping address

[← Back to Cart](#) [Continue to Payment →](#)

Order Summary

Sony WH-1000XM5 Headphones Qty: 1	\$279.99
Apple Watch Series 9 Qty: 1	\$399.00
Anker Wireless Charger Qty: 1	\$29.99
Subtotal (3 items)	\$708.98
Discount (SAVE20)	-\$80.00
Shipping	FREE
Estimated Tax	\$50.32
Total	\$679.30

Your information is secure and encrypted

Figure 5.8: Selecting Shipping Address and View Order Summary Page

5.9 Select Payment Gateway Page

The screenshot displays the Amazon checkout page for selecting a payment gateway. The top navigation bar includes the Amazon logo, a search bar, and links to the Cart and Account. A progress indicator shows the current step is 'Payment'.

Payment Method

Shipping to: John Doe
123 Main Street, Apt 4B
New York, NY 10001, United States [Change](#)

Select payment method

- ☒ **Credit / Debit Card**
Pay securely with your card
- ☐ **PayPal**
You will be redirected to PayPal
- ☐ **Cash on Delivery**
Pay when your order arrives

Card Details

Your saved cards

- ☒ **VISA** **** * 4242 Expires 12/27

[Or use a new card](#)

Cardholder Name
Name on card

Card Number
1234 5678 9012 3456

Expiry Date
MM / YY

CVV
123

☐ Save this card for future purchases

[← Back](#) [Pay \\$679.30 →](#)

Order Summary

	Sony WH-1000XM5 Headphones Qty: 1	\$279.99
	Apple Watch Series 9 Qty: 1	\$399.00
	Anker Wireless Charger Qty: 1	\$29.99
Subtotal (3 items)		\$708.98
Discount (SAVE20)		-\$80.00
Shipping		FREE
Estimated Tax		\$50.32
Total		\$679.30

Footer

amazon
Earth's most customer-centric company. Quality products, competitive prices, exceptional service.

[Shop](#)
All Products
Headphones
Laptops
Gaming
Accessories

[Support](#)
Help Center
Shipping Info
Returns
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[Company](#)
About Us
Careers
Press
Blog
Affiliates

[Legal](#)
Privacy Policy
Terms of Service
Cookie Policy
Accessibility

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Figure 5.9: Select Payment Gateway Page

5.10 Order Confirmation Page after Placing Order

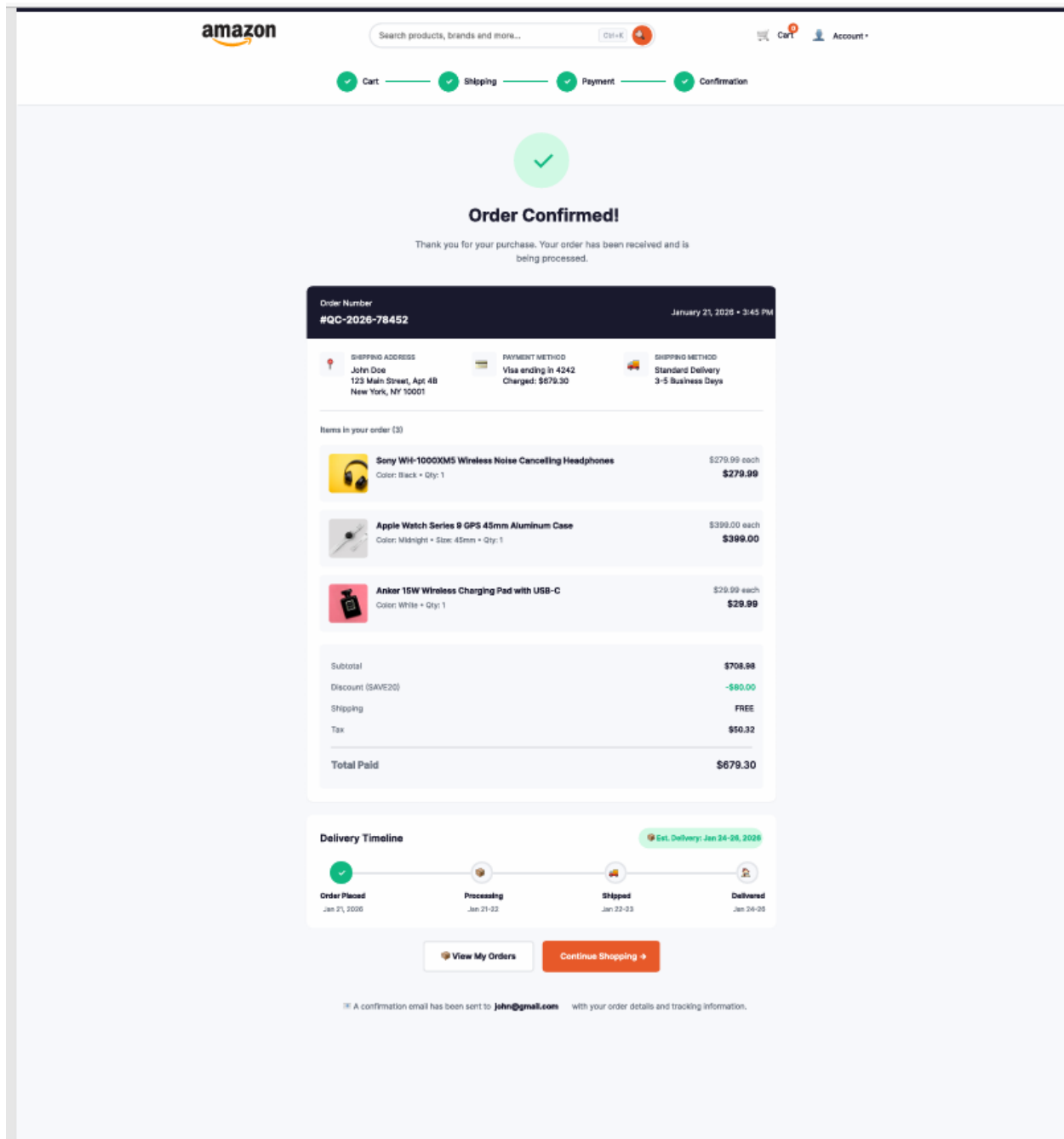


Figure 5.10: See Order Confirmation Page after Placing Order

5.11 View My Orders Page for Customer

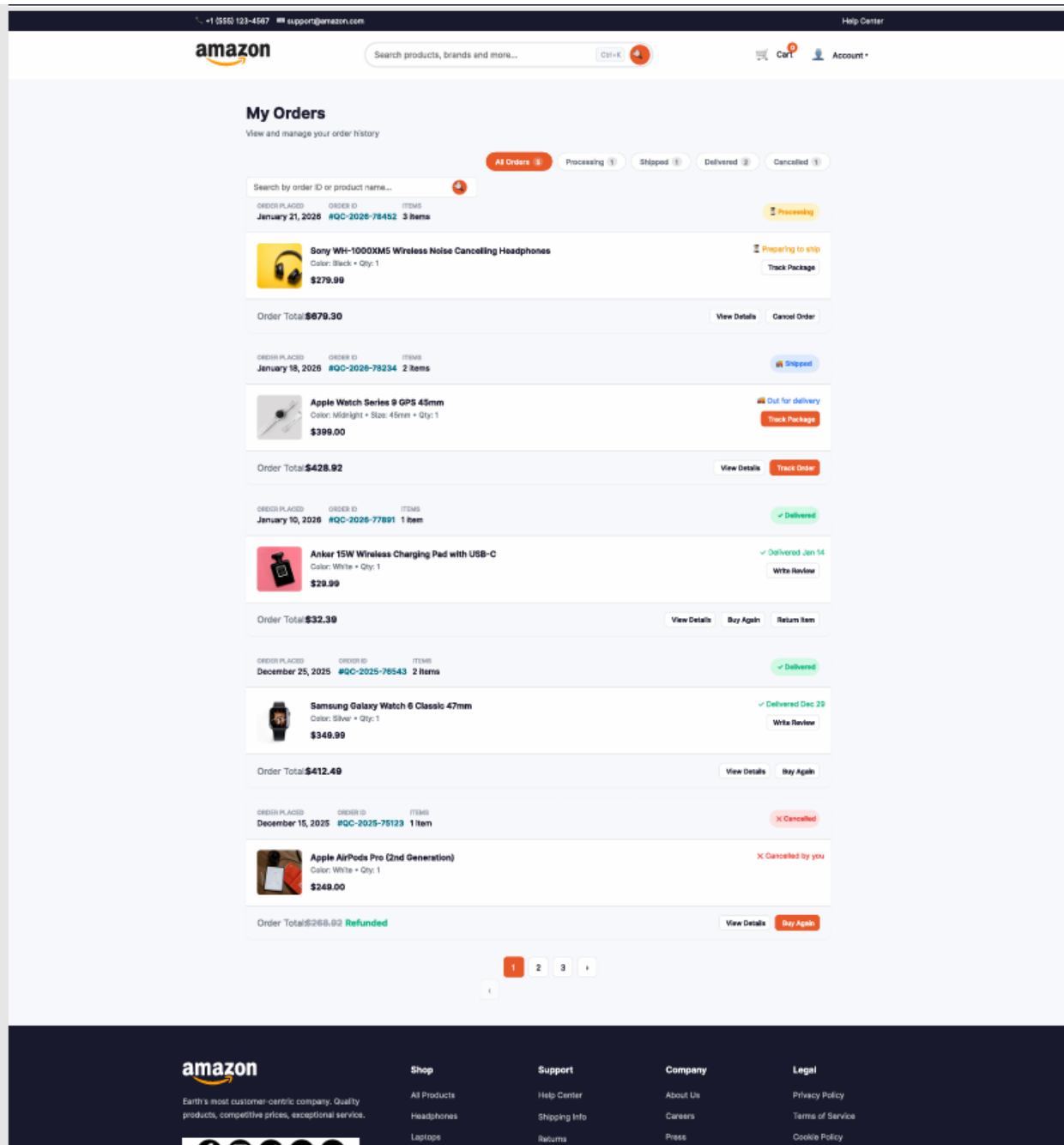


Figure 5.11: View My Orders Page for Customer

5.12 View My Profile Page for Customer

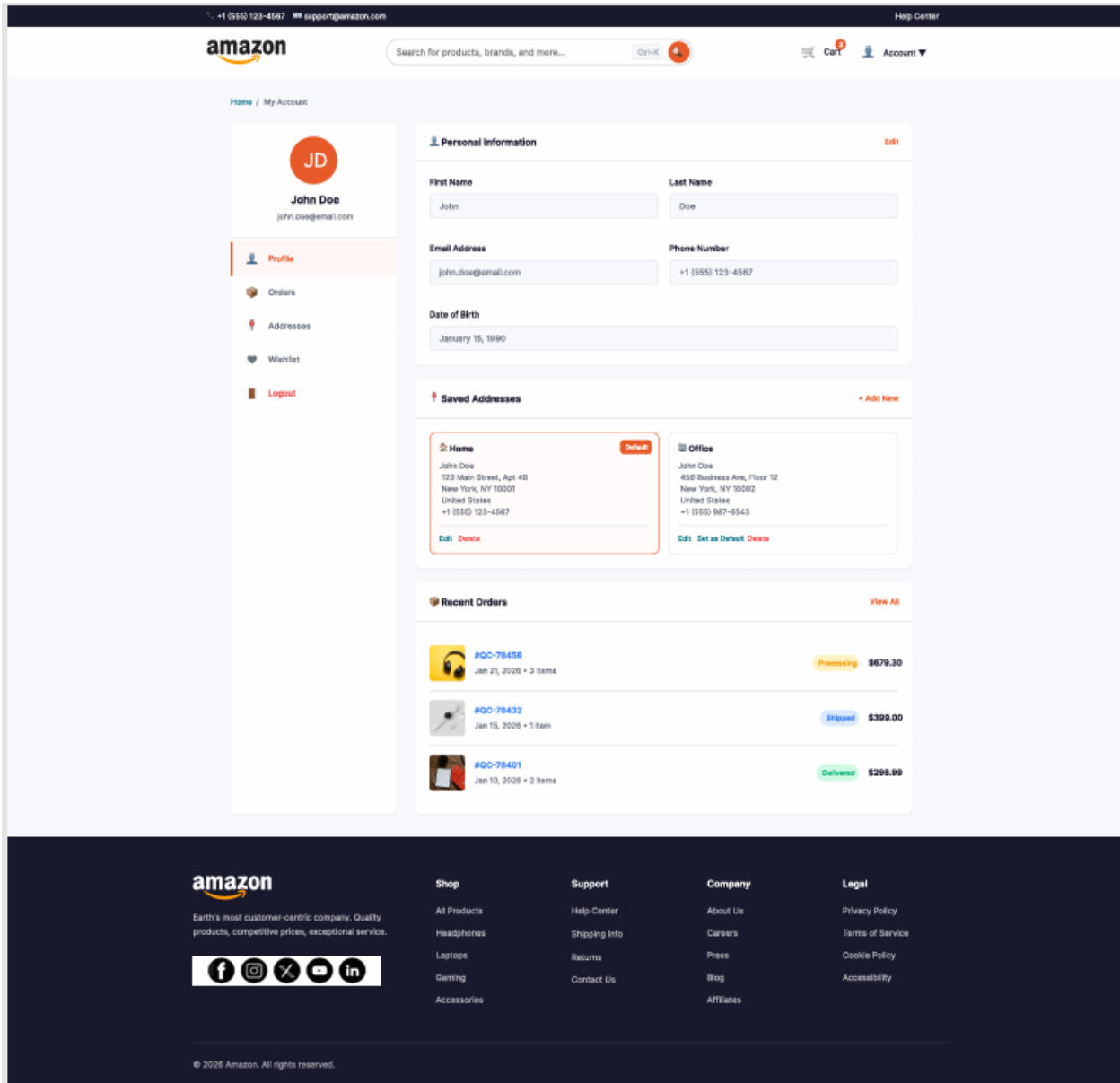


Figure 5.12: View My Profile Page for Customer

5.13 Vendor's Orders Overview Page

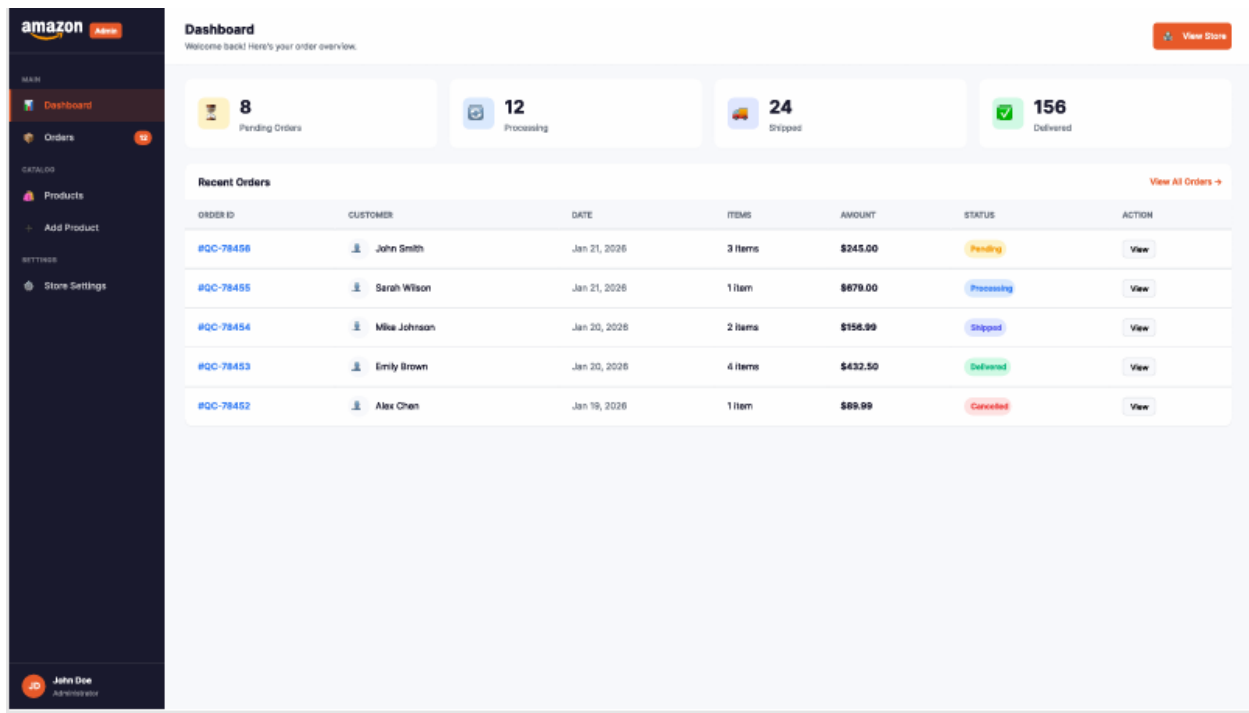


Figure 5.13: Vendor's Orders Overview Page

5.14 Vendor's Order Track and management Page

Orders
Manage and track customer orders

ALL Orders 100 Pending 12 Processing 24 Shipped 156 Delivered

Search by order ID or customer... 2026-01-01 to 2026-01-21

ORDER ID	CUSTOMER	ITEMS	DATE	AMOUNT	STATUS	UPDATE
#QC-78456	John Smith john.smith@gmail.com	Sony WH-1000XM5 +2 more items	Jan 21, 2026 10:45 AM	\$678.30	Pending	Pending
#QC-78455	Sarah Wilson sarah.w@gmail.com	Apple Watch Series 9	Jan 21, 2026 09:30 AM	\$399.00	Processing	Processing
#QC-78454	Mike Johnson mike.j@gmail.com	AirPods Pro 2nd Gen +1 more item	Jan 20, 2026 04:15 PM	\$298.99	Shipped	Shipped
#QC-78453	Emily Brown emily.b@gmail.com	Samsung Galaxy Watch 6	Jan 20, 2026 11:20 AM	\$299.00	Delivered	Delivered
#QC-78452	Alex Chen alex.chen@gmail.com	Anker Wireless Charger	Jan 19, 2026 02:45 PM	\$29.99	Cancelled	Cancelled
#QC-78451	Lisa Taylor lisa.t@gmail.com	Nike Air Max 270 +3 more items	Jan 19, 2026 10:00 AM	\$432.50	Delivered	Delivered

Showing 1-6 of 200 orders

1 2 3 ... 14

Figure 5.14: Vendor's Order Track and management Page

5.15 Add Product Page for Vendor

Add Product
Add a new product to your store

Basic Information

Product Name*
e.g., Sony WH-1000XM5 Wireless Headphones

Description*
Enter product description...

Category* **Brand**
Select category e.g., Sony, Apple, Nike

Product Images

Click to upload drag and drop
PNG, JPG or WEBP (max 5MB)

First image will be the main product image

Pricing

Regular Price* **Sale Price** **Cost Price**
\$ 0.00 \$ 0.00 \$ 0.00
For profit tracking

Inventory

SKU **Quantity***
e.g., WH-1000XM5-BLK 0

Stock Status

☒ In Stock ☐ Out of Stock

Settings

Published
Make product visible in store ☒

Featured Product
Show on homepage featured section ☐

Free Shipping
Offer free shipping for this product ☒

Figure 5.15: Add Product Page for Vendor

5.16 View and Manage My Products Page for Vendor

Products
Manage your product catalog

Search products... 4 All Categories All Status Sort by: Latest

☐	PRODUCT	CATEGORY	PRICE	STOCK	STATUS	ACTIONS
☐	 Sony WH-1000XM5 SKU: WH-1000XM5-BLK	Electronics	\$279.00 \$349.00	156	In Stock	 
☐	 Apple Watch Series 9 SKU: AW-SP-45MM	Electronics	\$399.00	82	In Stock	 
☐	 Anker Wireless Charger SKU: ANK-WC-15W	Electronics	\$29.99	8	Low Stock	 
☐	 Samsung Galaxy Watch 6 SKU: SGW-6-45MM	Electronics	\$299.00 \$349.00	0	Out of Stock	 
☐	 AirPods Pro 2nd Gen SKU: APP-PPD-2	Electronics	\$249.00	234	In Stock	 
☐	 Nike Air Max 270 SKU: NM-AM270-10	Fashion	\$150.00	45	In Stock	 

Showing 1-6 of 556 products 1 2 3 60

Figure 5.16: View and Manage My Products Page for Vendor

Collaboration

Diagram

The presentation of the Collaboration diagram was delivered by Himadri Gobinda Biswas
(2105047)

No modification was suggested for the collaboration diagram.

6.1 Homepage

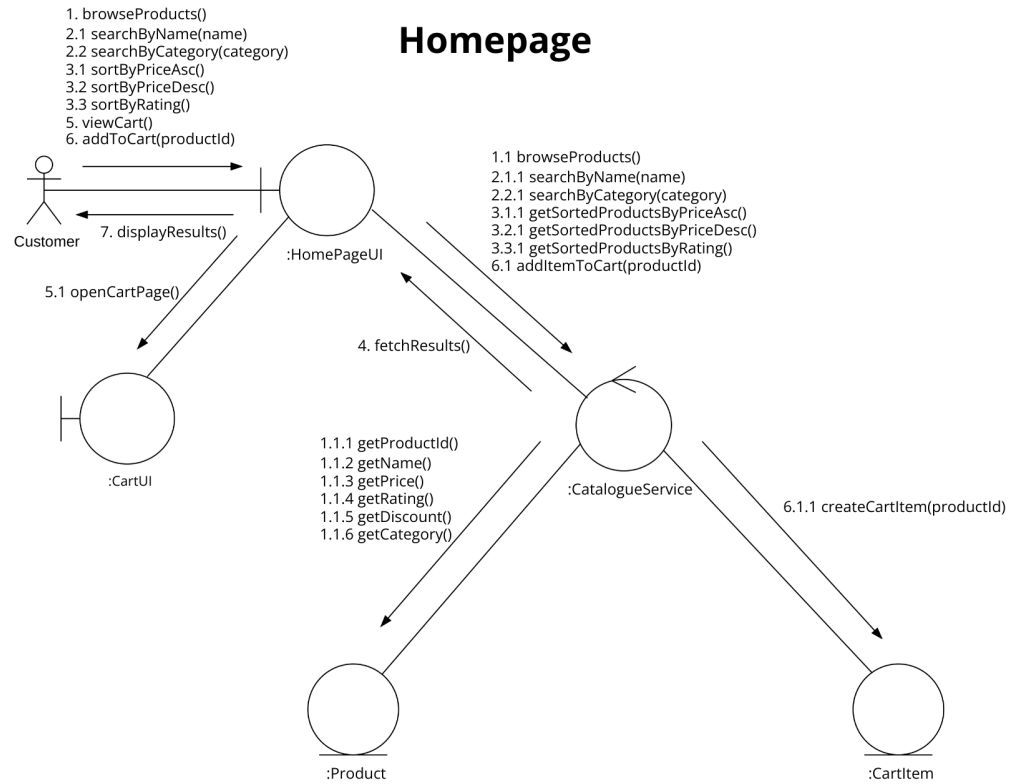


Figure 6.1: Homepage

6.2 Product Details

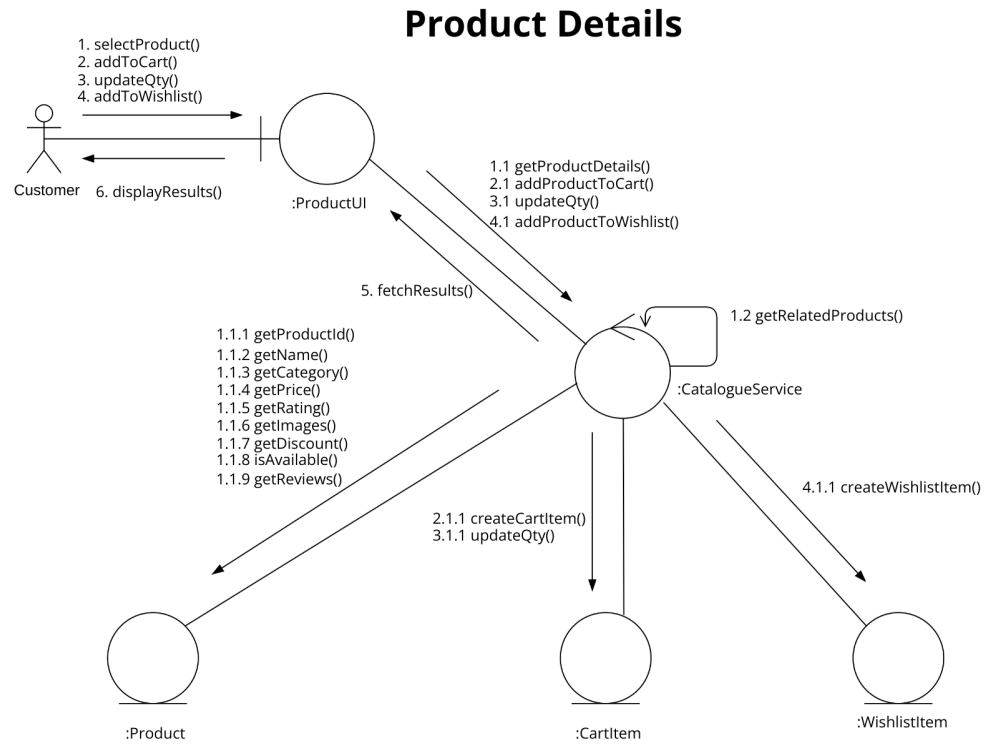


Figure 6.2: Product Details

6.3 Cart

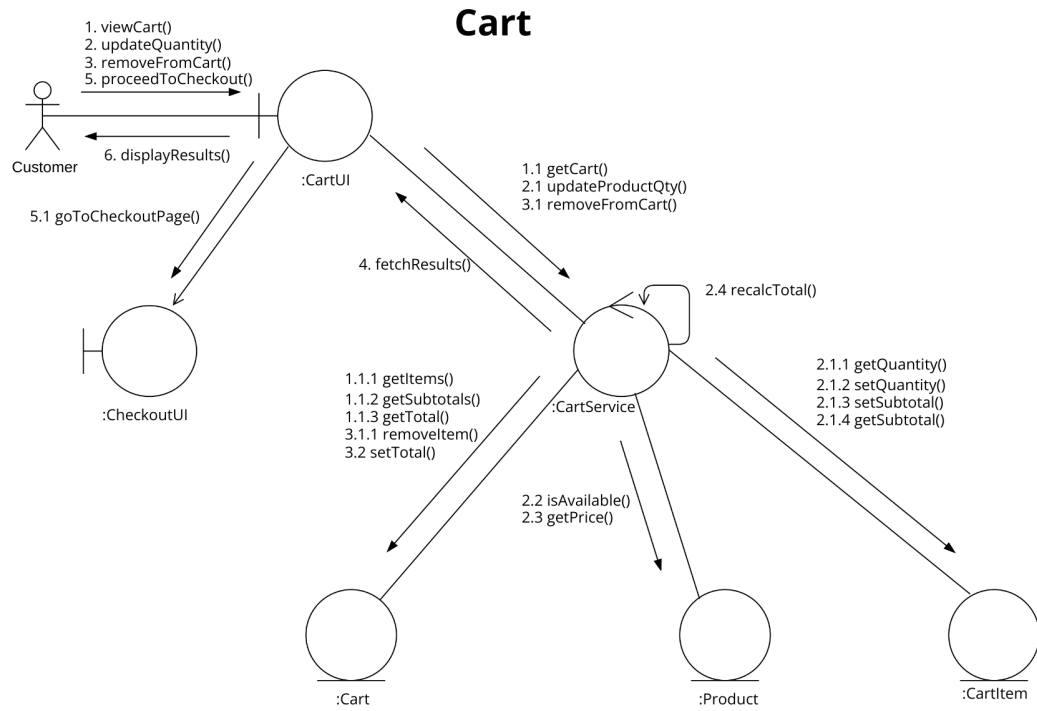


Figure 6.3: Cart

6.4 Checkout

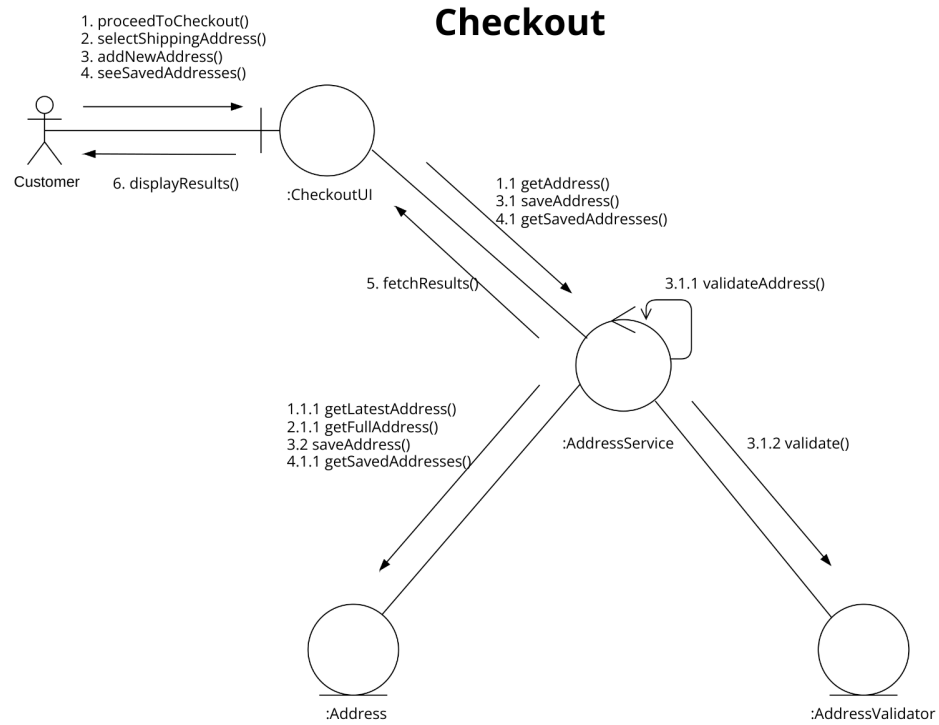


Figure 6.4: Checkout

6.5 Order-Summary

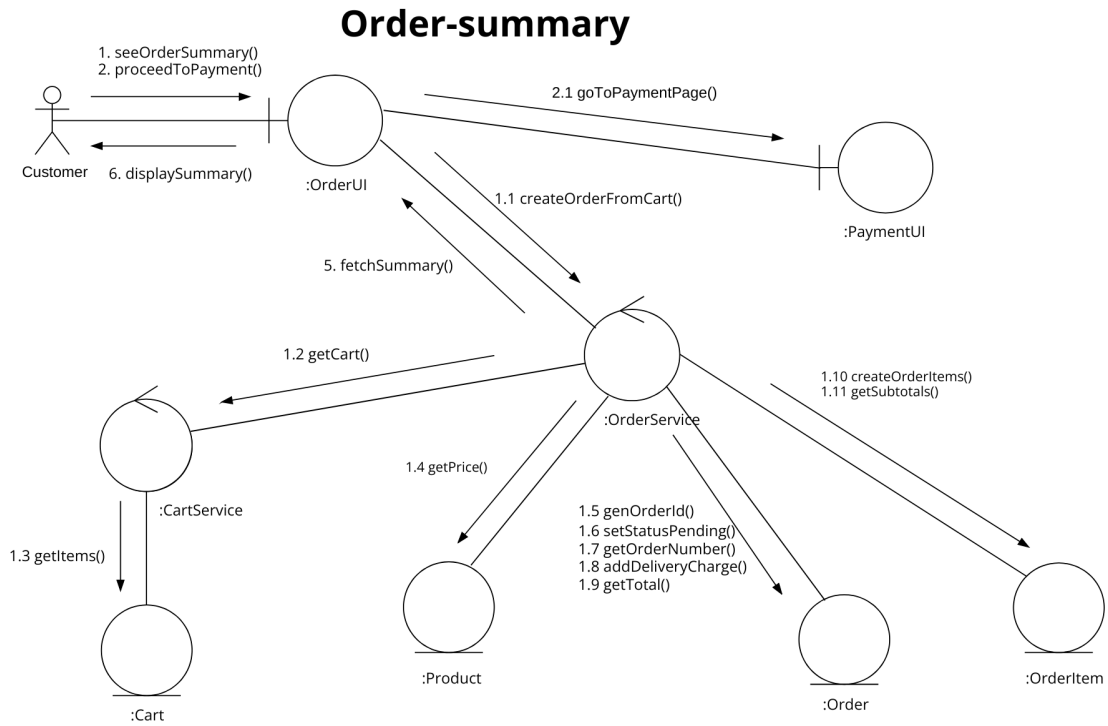


Figure 6.5: Order-Summary

6.6 Payment-Gateway

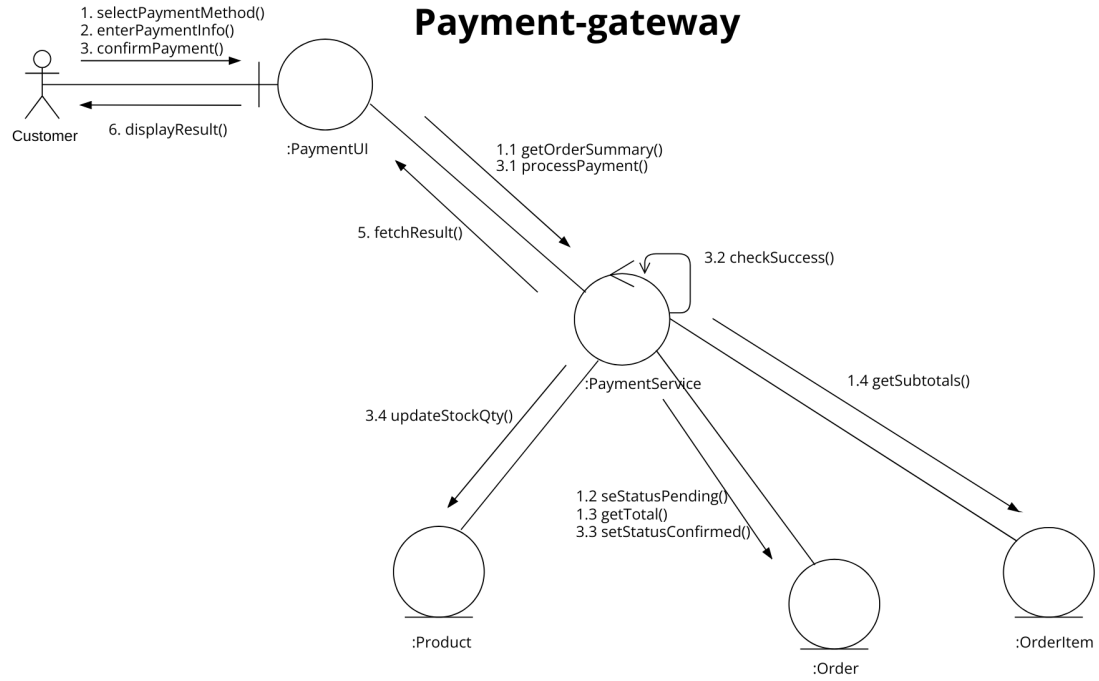


Figure 6.6: Payment-Gateway

6.7 View-Orders(Customer)

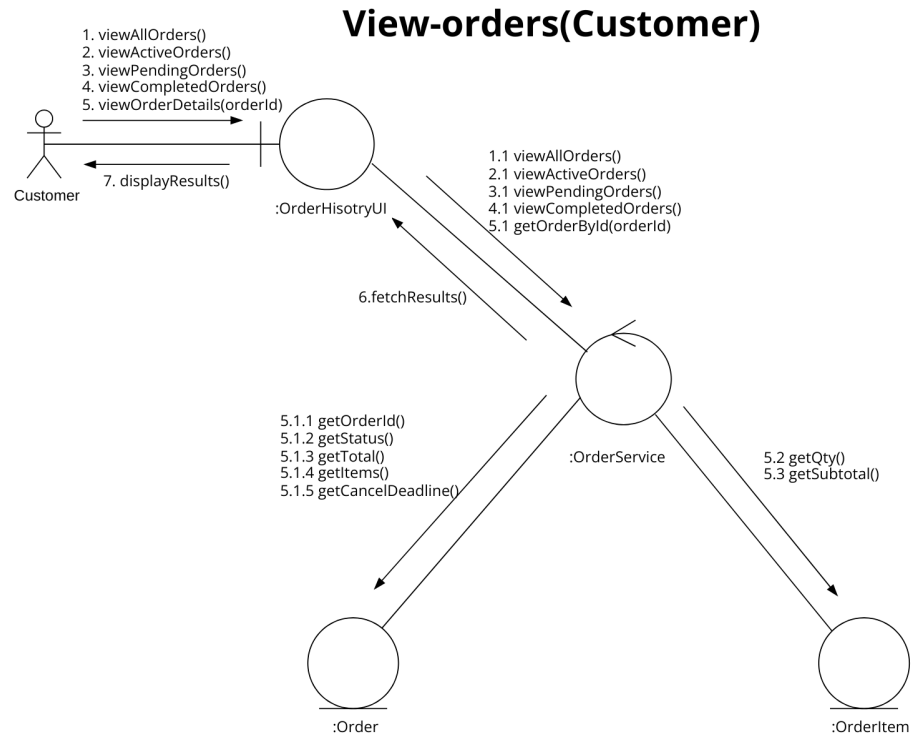


Figure 6.7: View-Orders(Customer)

6.8 Cancel Order

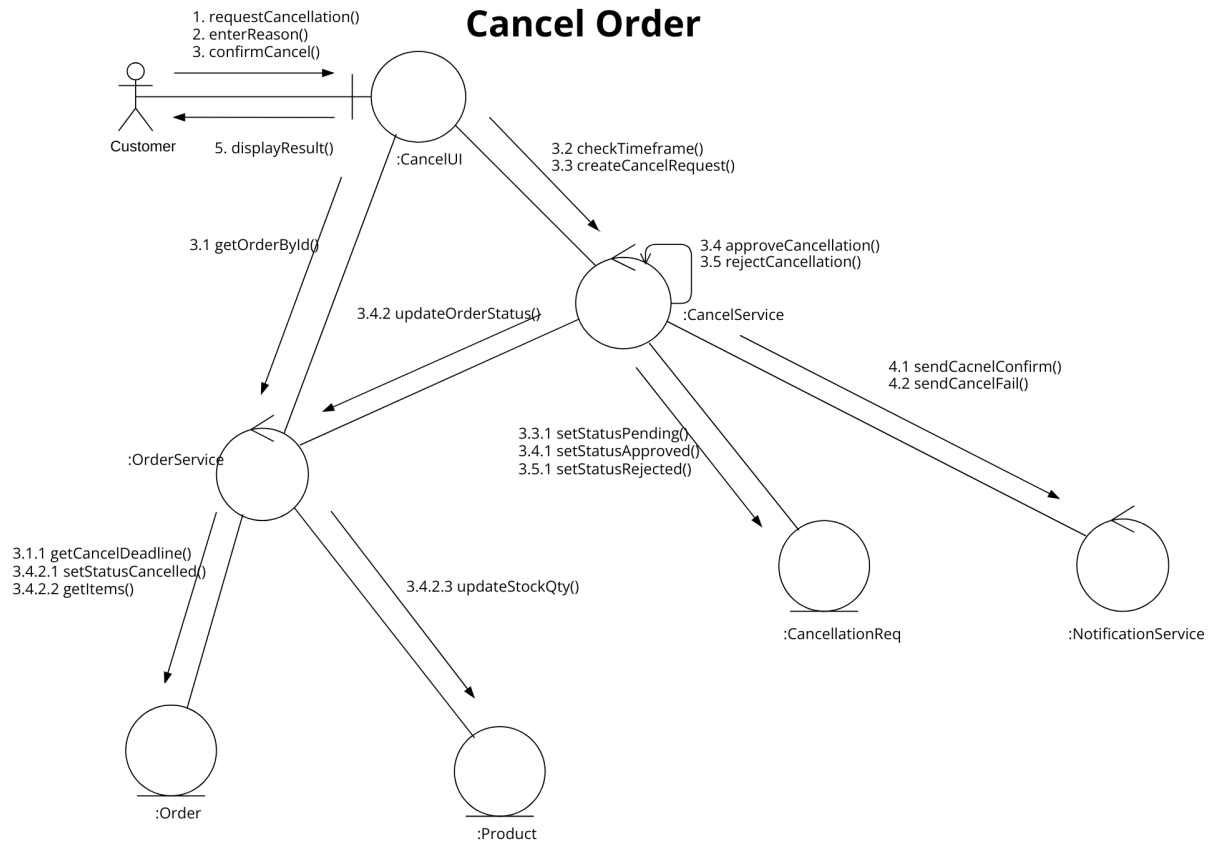


Figure 6.8: Cancel Order

6.9 Vendor-Products

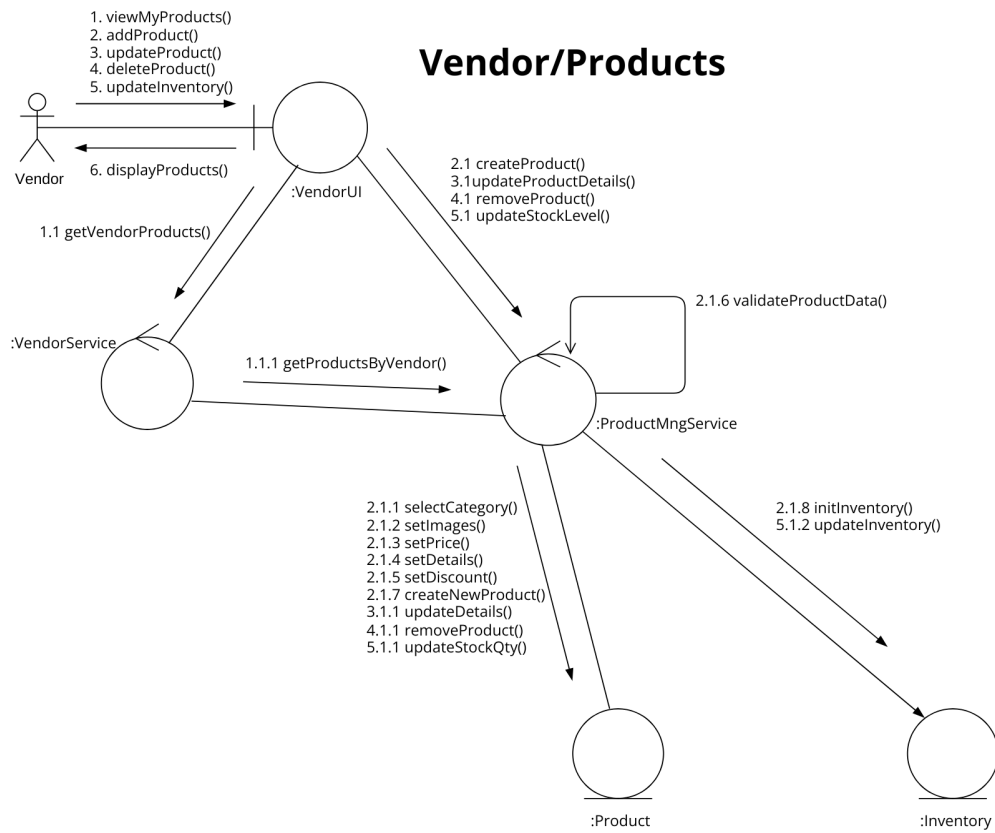


Figure 6.9: Vendor-Products

6.10 Vendor-Orders

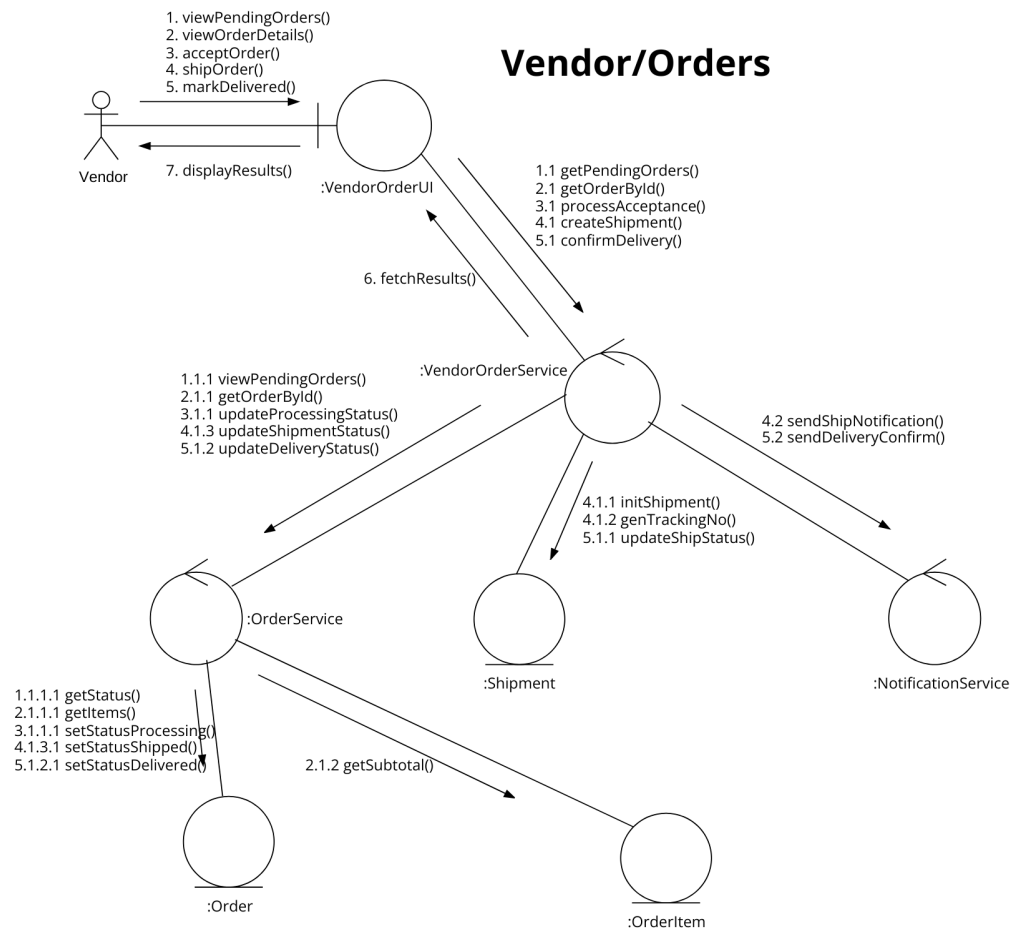


Figure 6.10: Vendor-Orders

6.11 Review Product

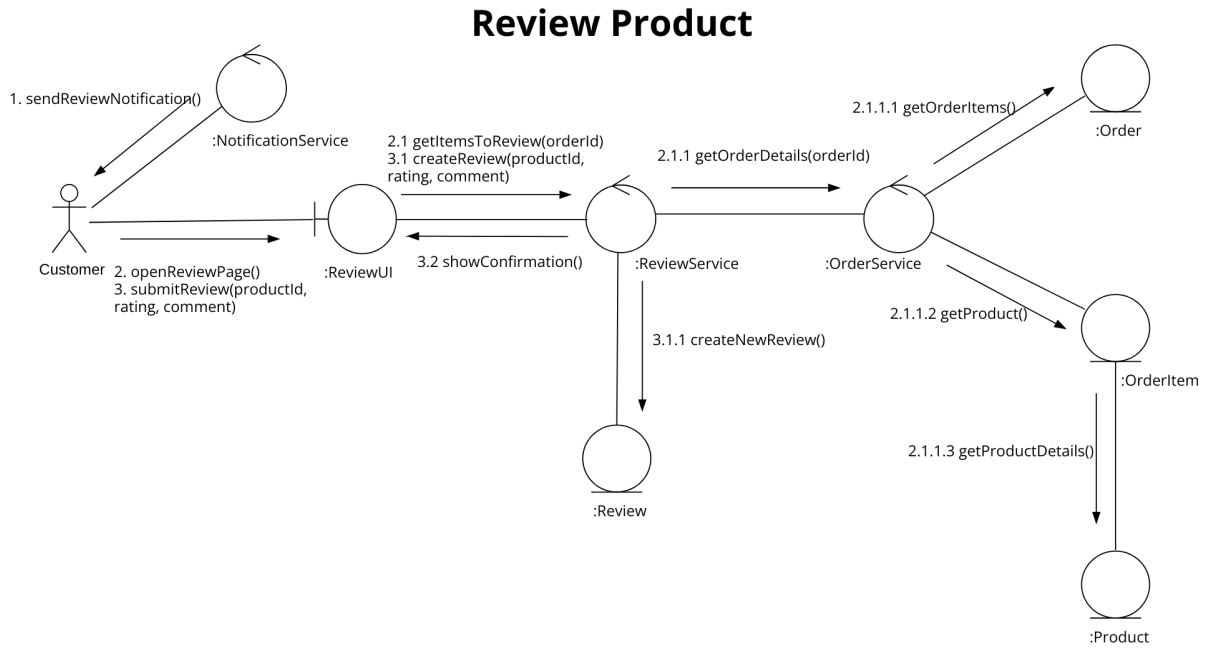


Figure 6.11: Review Product

6.12 Customer Authentication

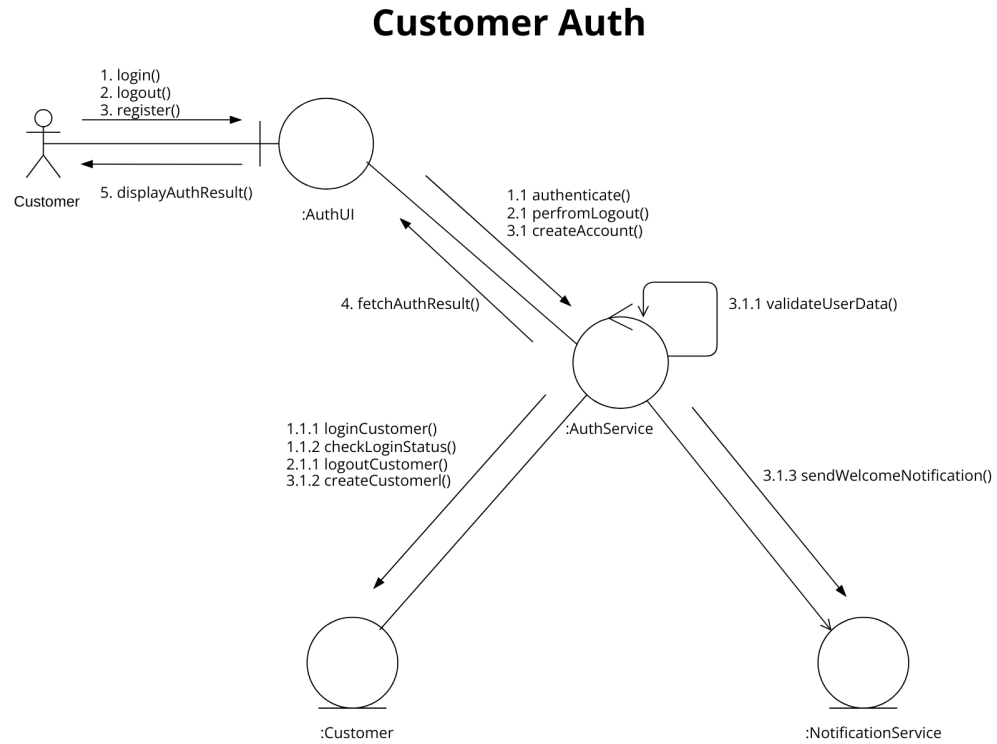


Figure 6.12: Customer Authentication

Sequence

Diagram

The presentation of the sequence diagrams was delivered by Md. Maruf Hasan Rubab (2105042).

Modifications Suggested:

- In the "Checkout-Address Selection" sequence diagram, when trying to save a new address, instead of using an address validator service, the address service should suggest similar addresses from the address database to the user typed address. The user can then confirm address from the suggestions or save a new address if no matches are found. So, address validation part was removed from the diagram.

7.1 Homepage - Browse & Search Products

7.1.1 Homepage - Browse & Search Products (Part 1)

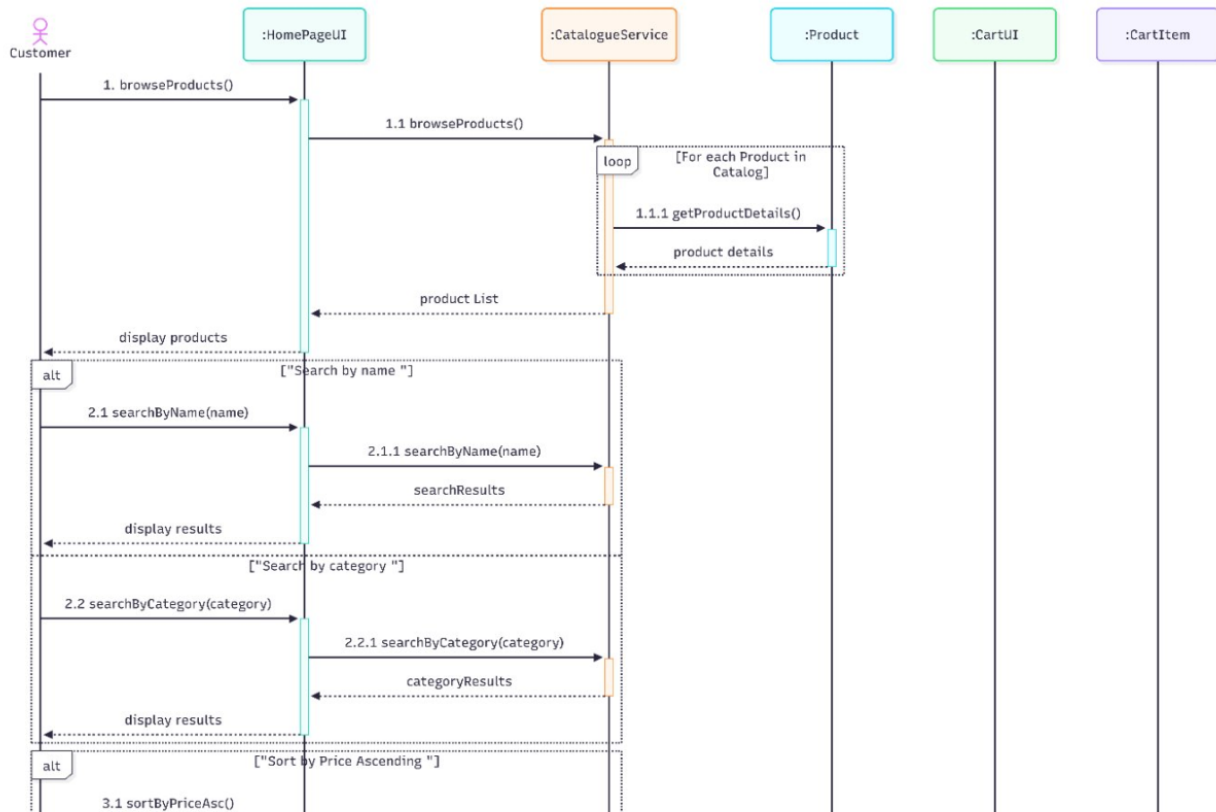


Figure 7.1: Homepage - Browse & Search Products (Part 1)

7.1.2 Homepage - Browse & Search Products (Part 2)

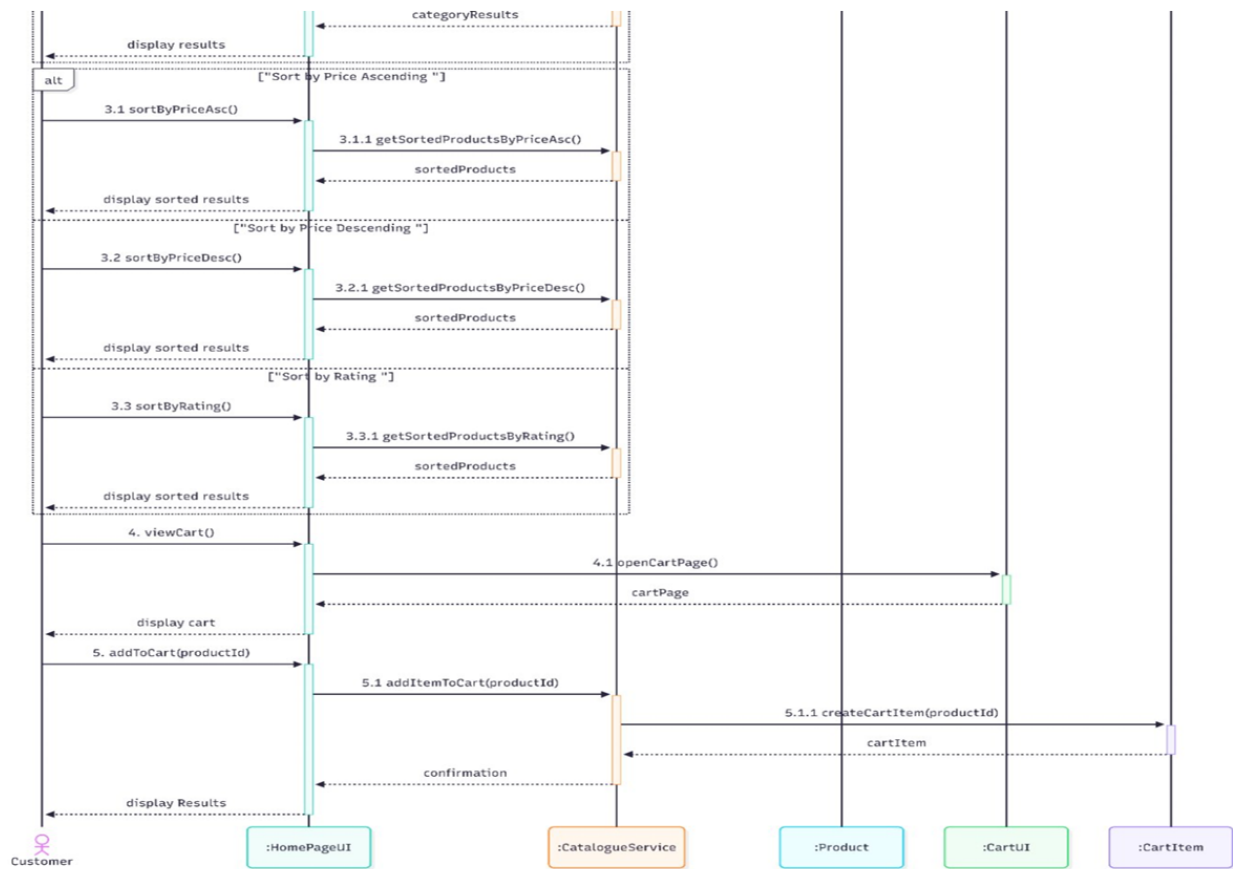


Figure 7.2: Homepage - Browse & Search Products (Part 2)

7.2 Product Details

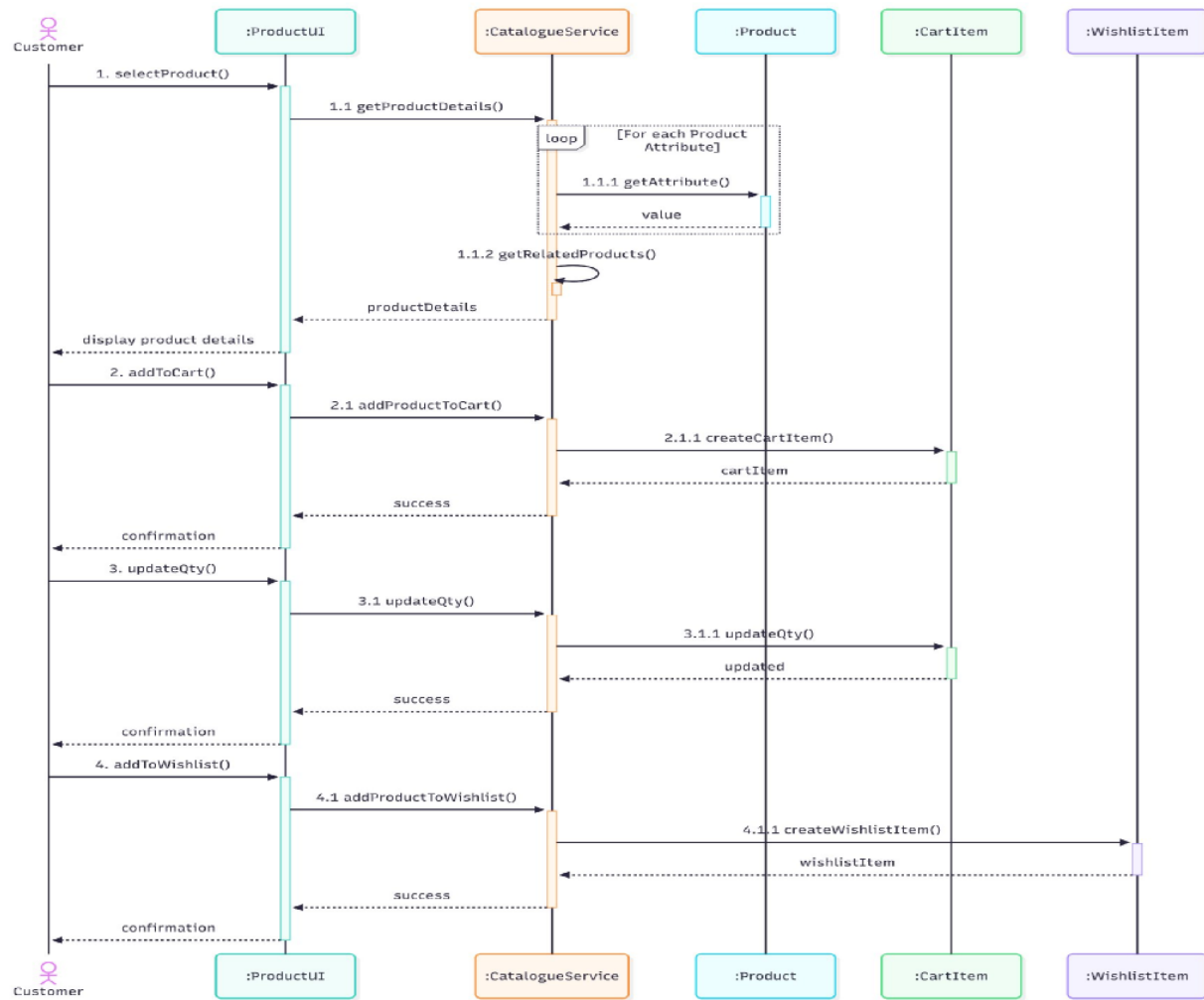


Figure 7.3: Product Details

7.3 Shopping Cart

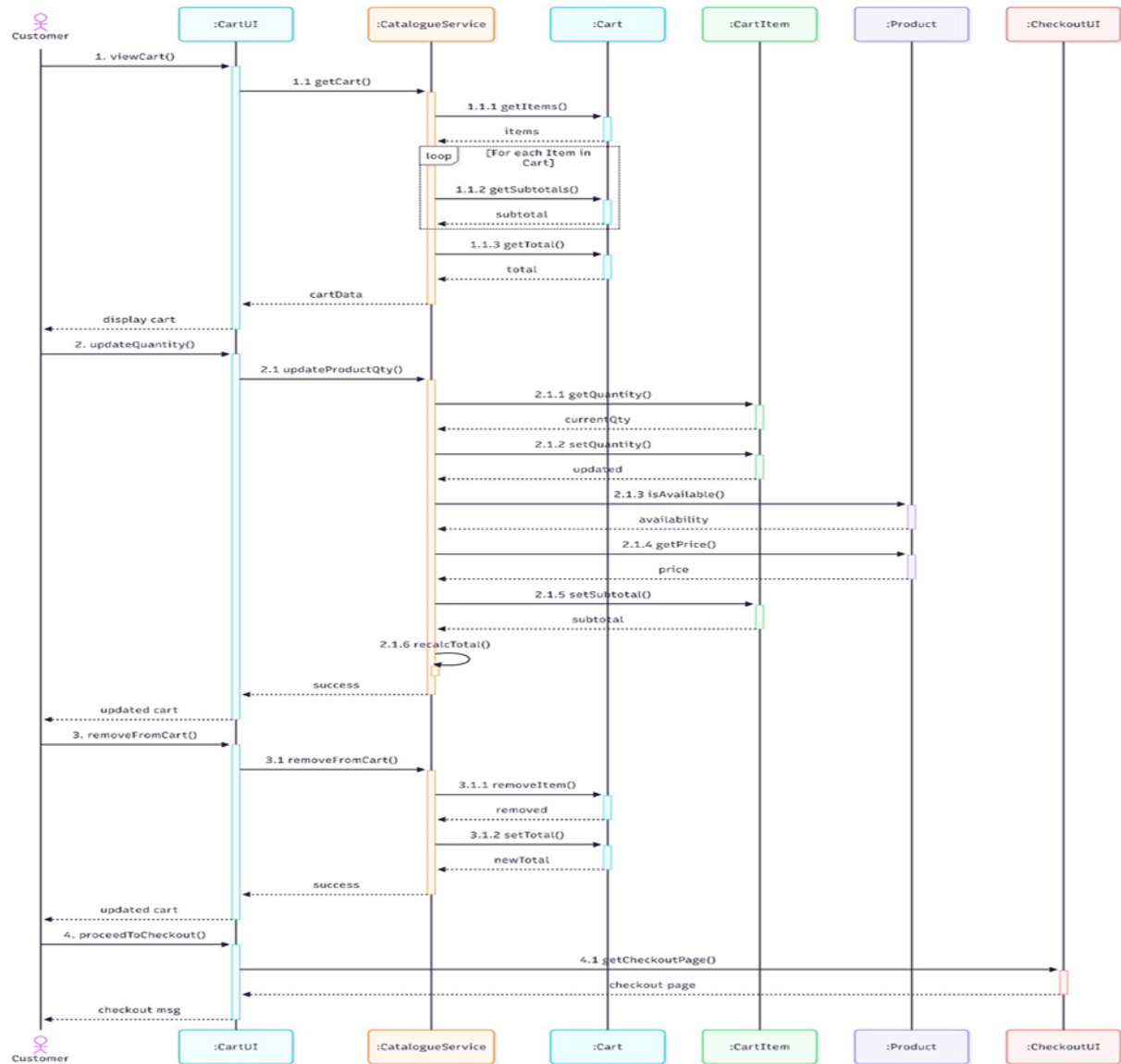


Figure 7.4: Shopping Cart

7.4 Checkout - Address Selection

7.4.1 Checkout - Address Selection (Part 1)

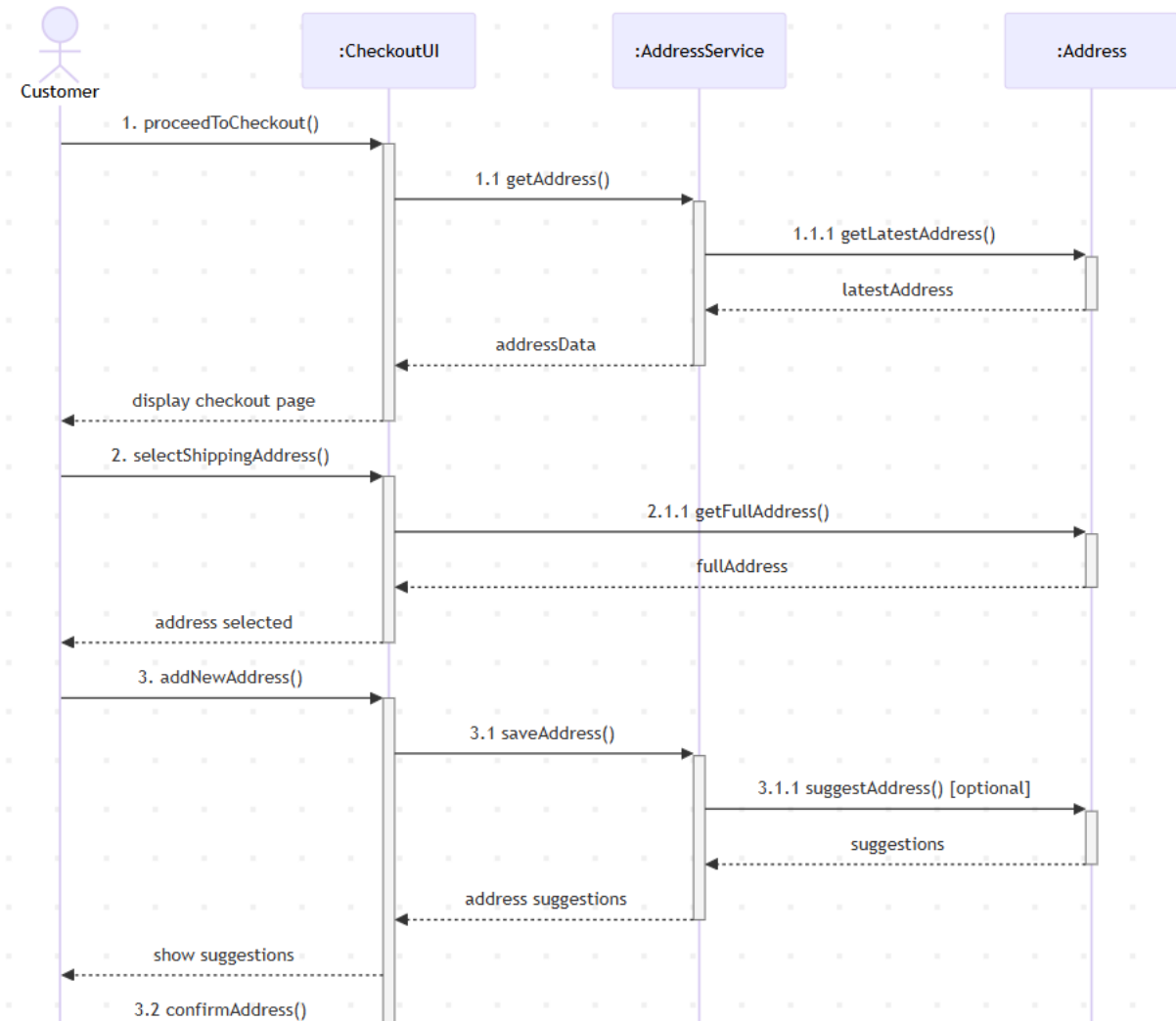


Figure 7.5: Checkout - Address Selection (Part 1)

7.4.2 Checkout - Address Selection (Part 2)

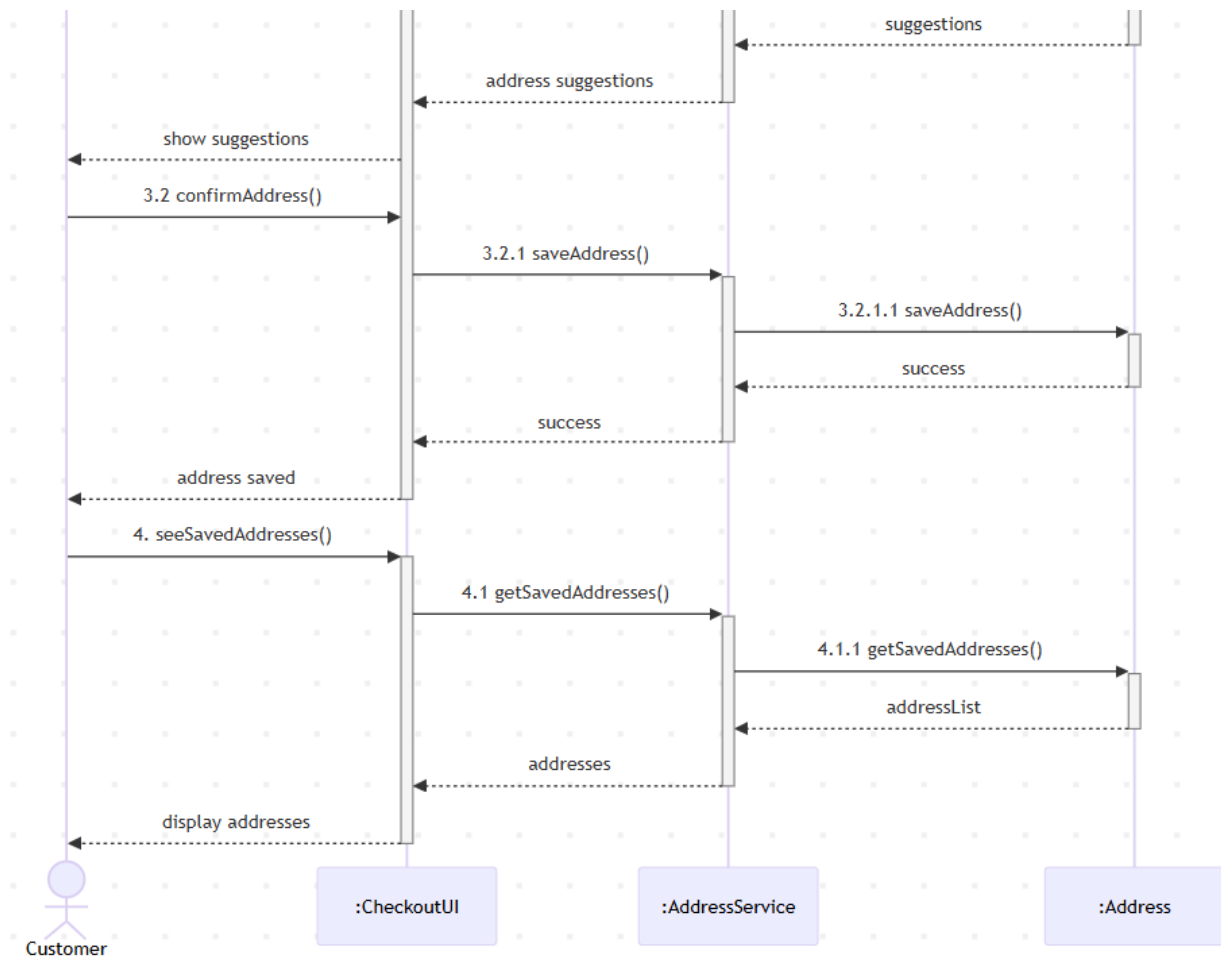


Figure 7.6: Checkout - Address Selection (Part 2)

7.5 Order Summary

7.5.1 Order Summary (Part 1)

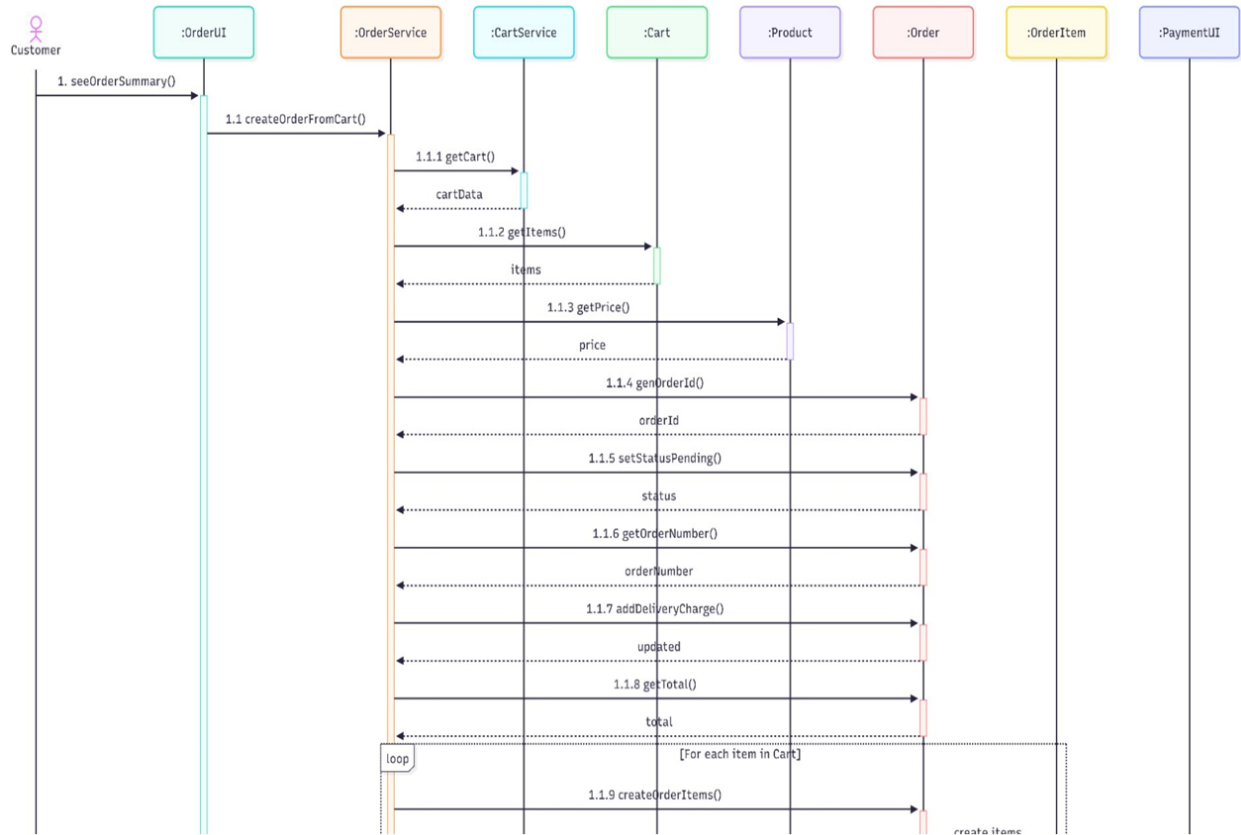


Figure 7.7: Order Summary (Part 1)

7.5.2 Order Summary (Part 2)

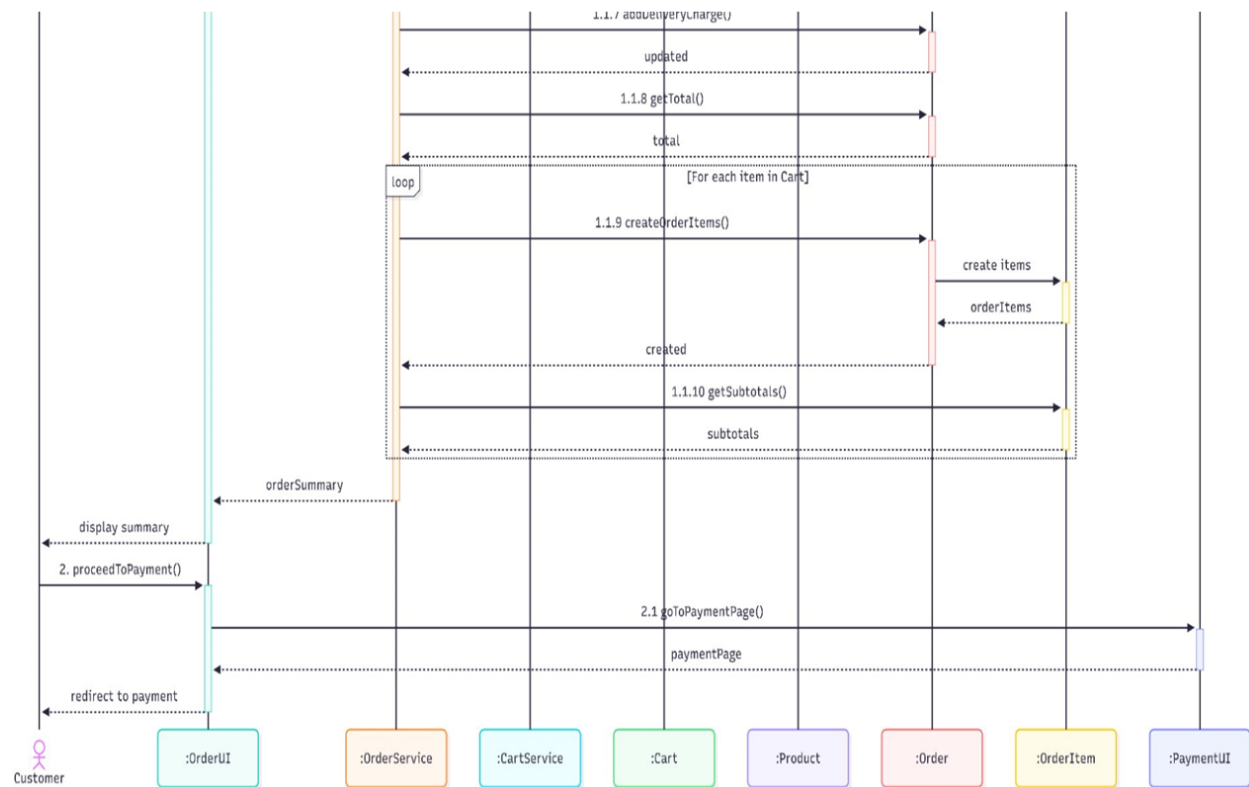


Figure 7.8: Order Summary (Part 2)

7.6 Payment Gateway

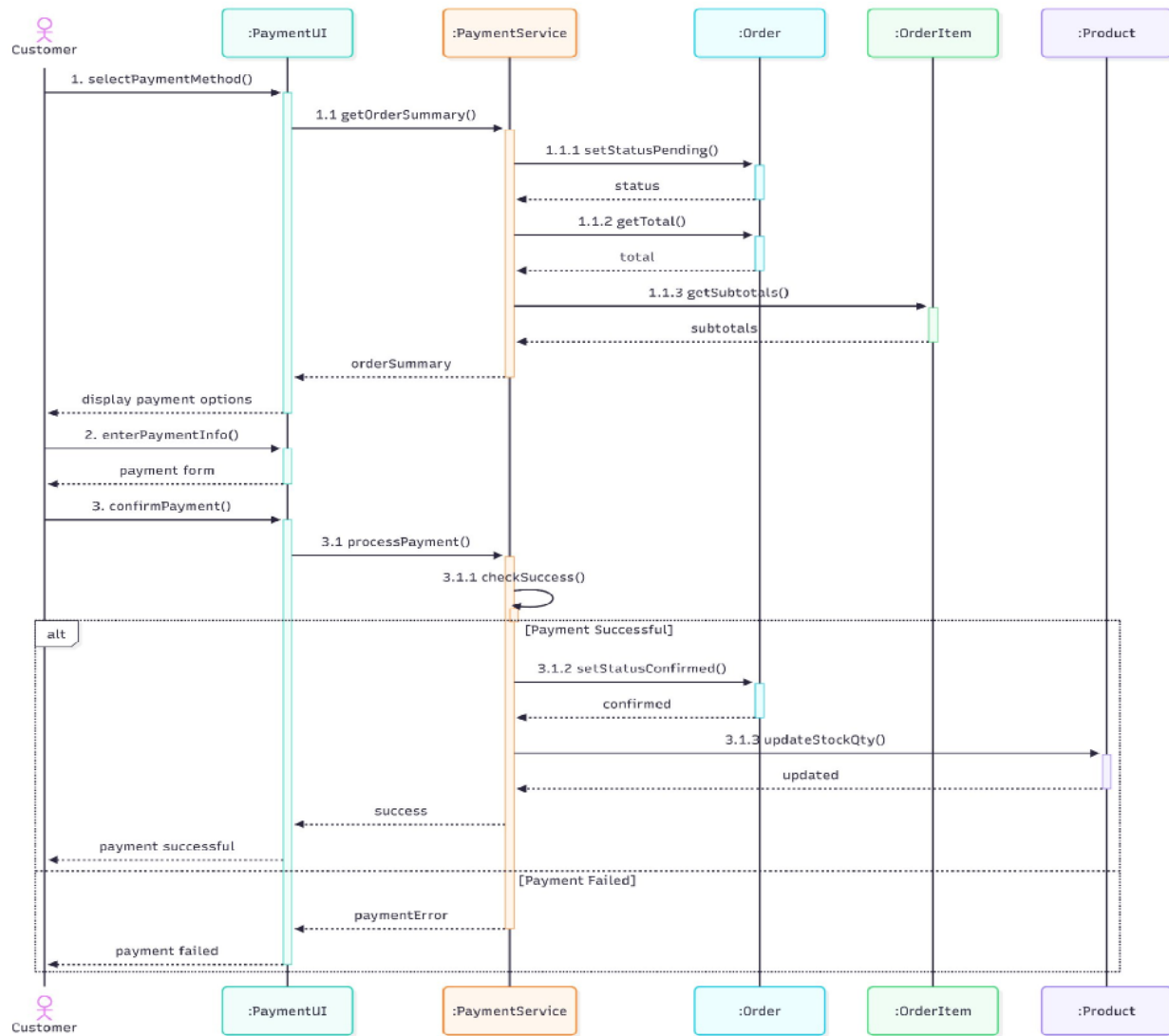


Figure 7.9: Payment Gateway

7.7 View Orders (Customer)

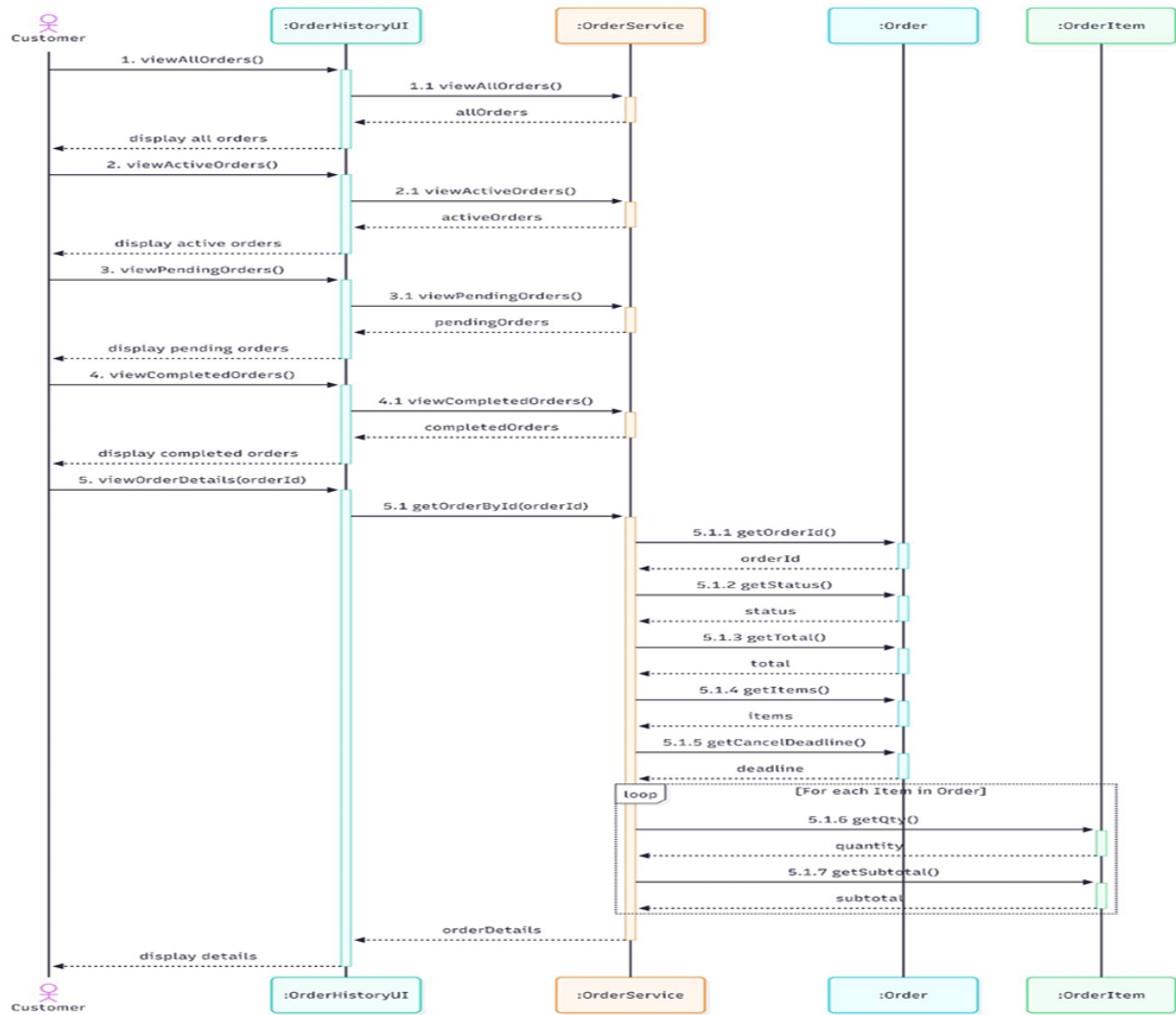


Figure 7.10: View Orders (Customer)

7.8 Cancel Order

7.8.1 Cancel Order (Part 1)

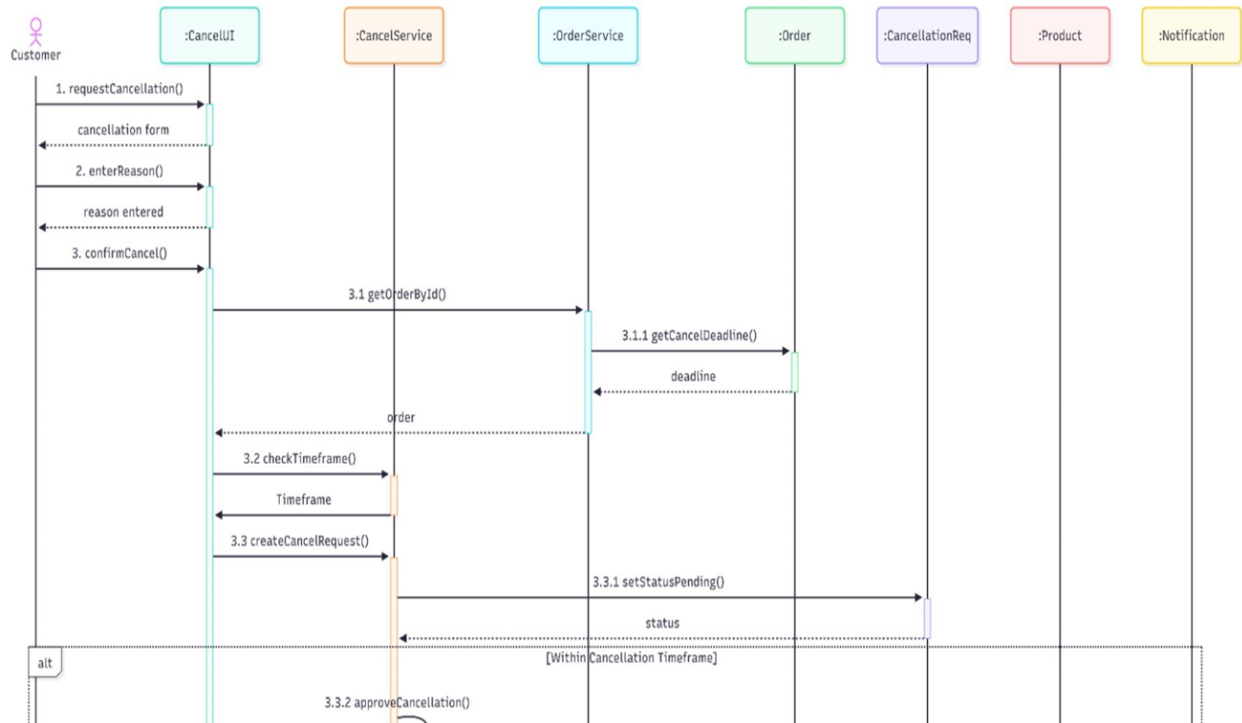


Figure 7.11: Cancel Order (Part 1)

7.8.2 Cancel Order (Part 2)

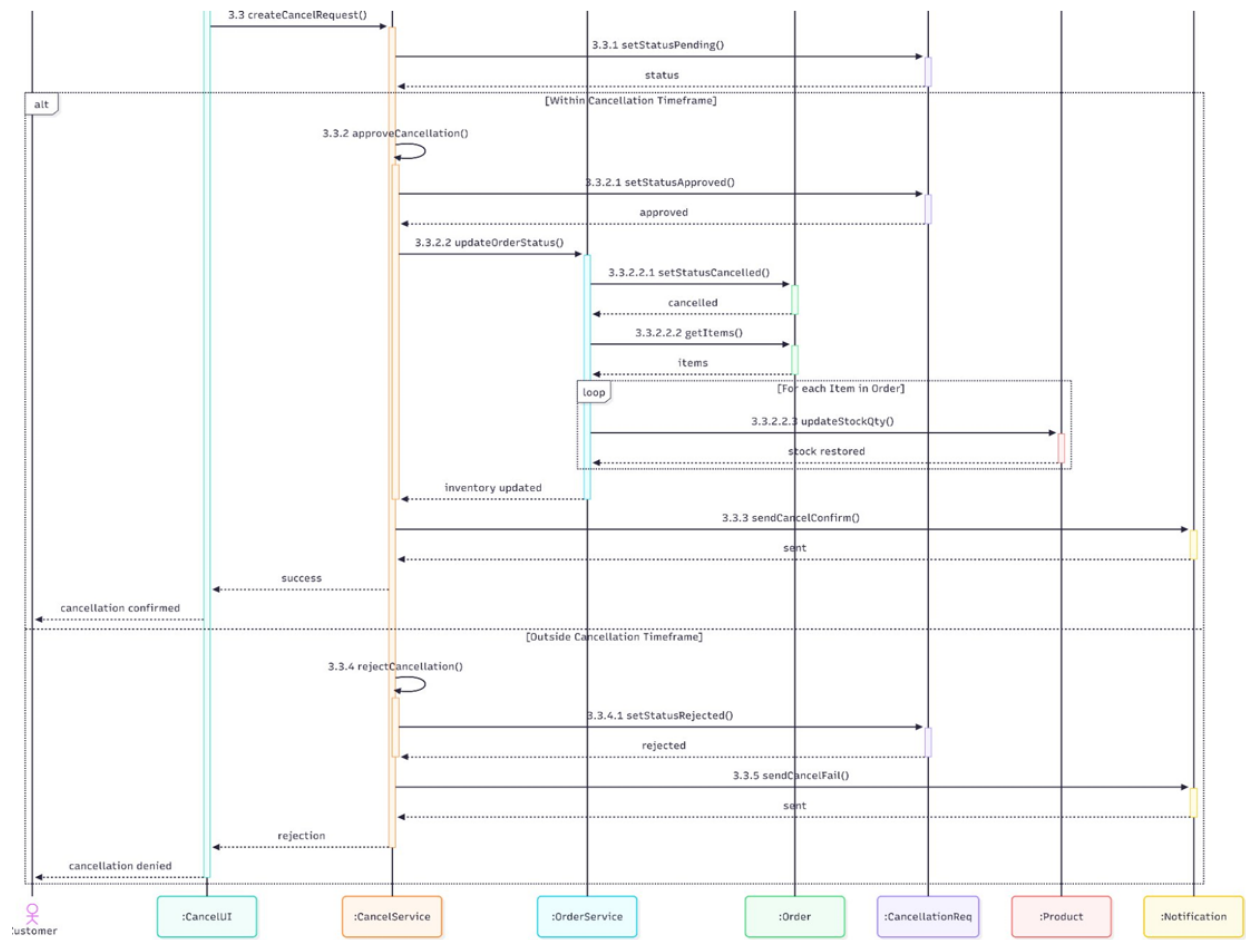


Figure 7.12: Cancel Order (Part 2)

7.9 Vendor Products Management

7.9.1 Vendor Products Management (Part 1)

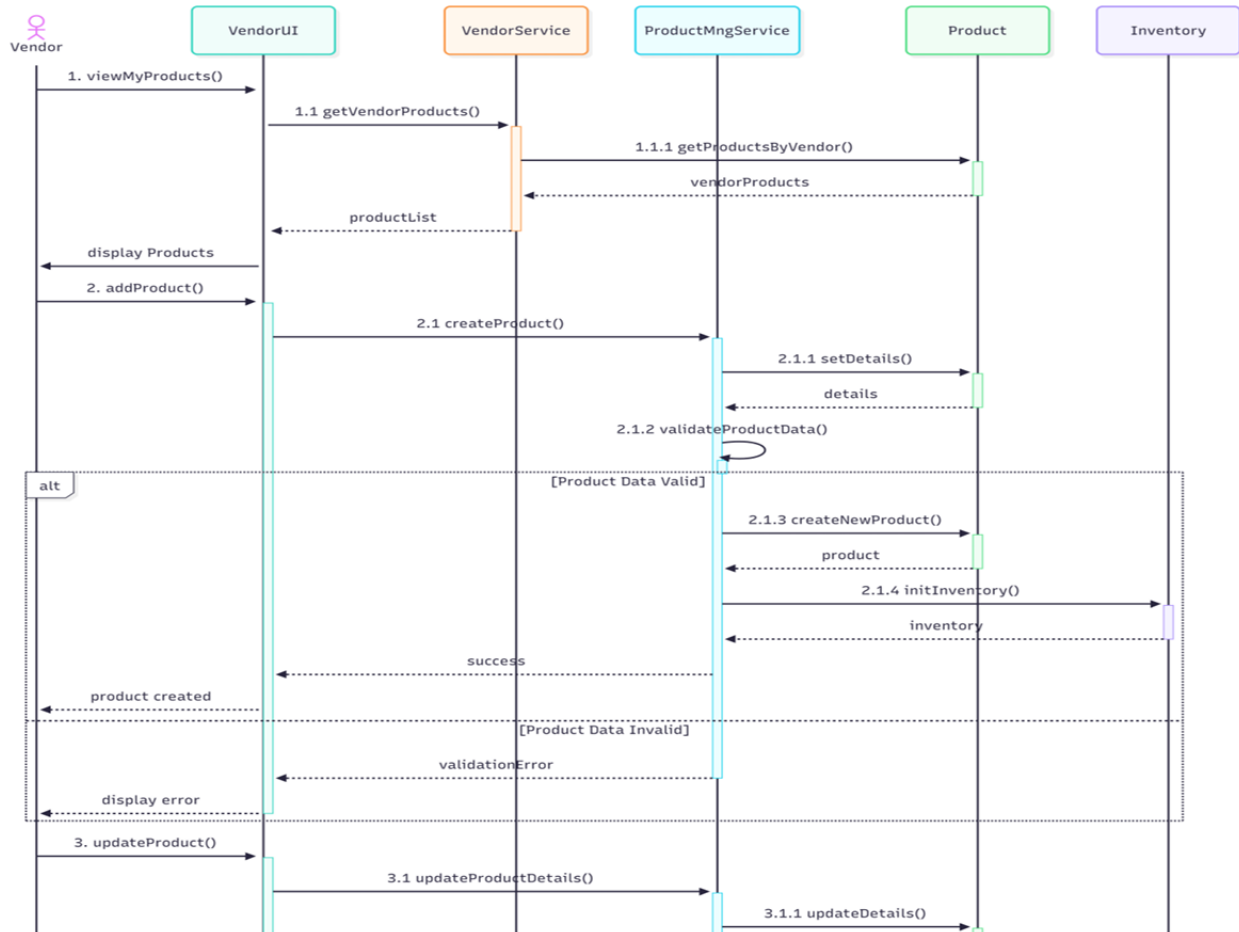


Figure 7.13: Vendor Products Management (Part 1)

7.9.2 Vendor Products Management (Part 2)

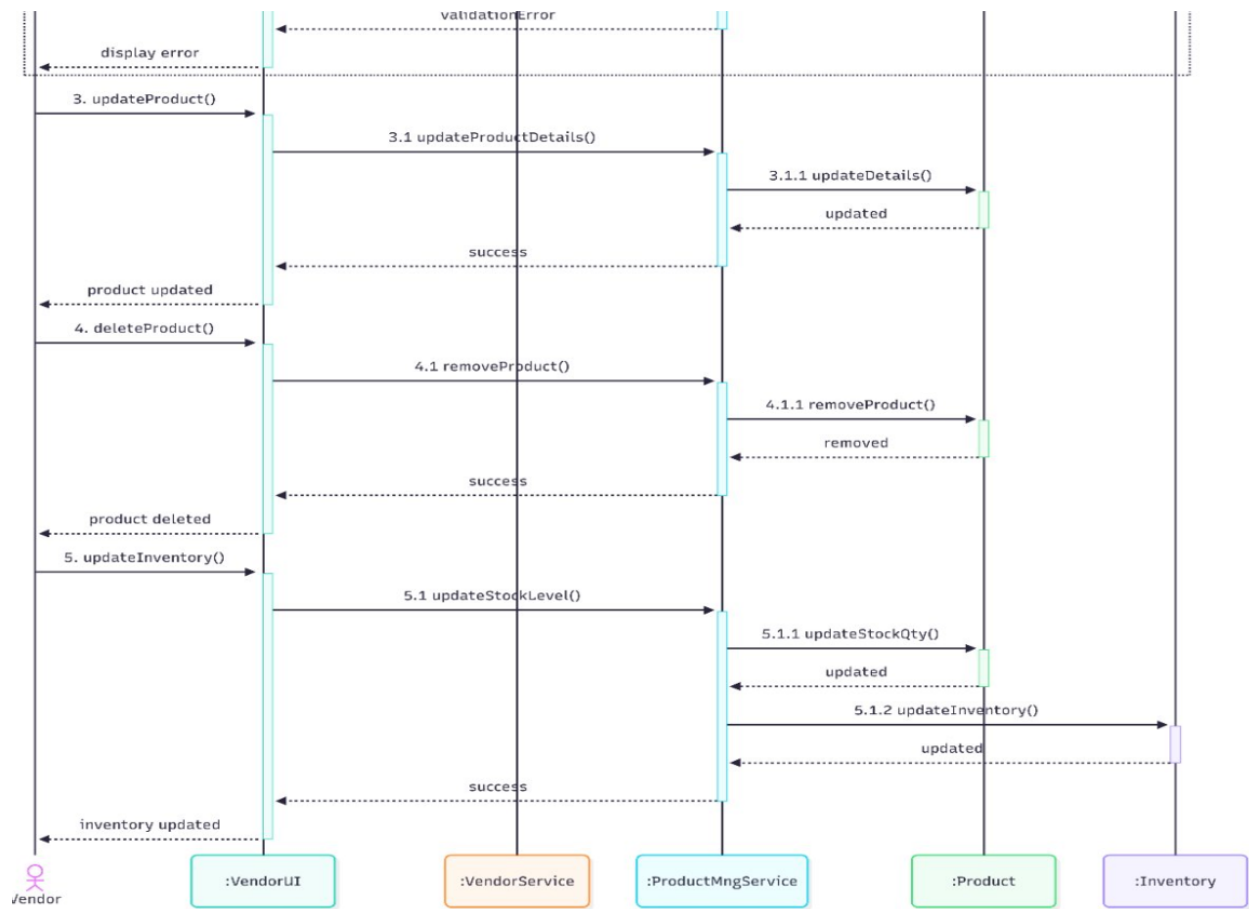


Figure 7.14: Vendor Products Management (Part 2)

7.10 Vendor Orders Management

7.10.1 Vendor Orders Management (Part 1)

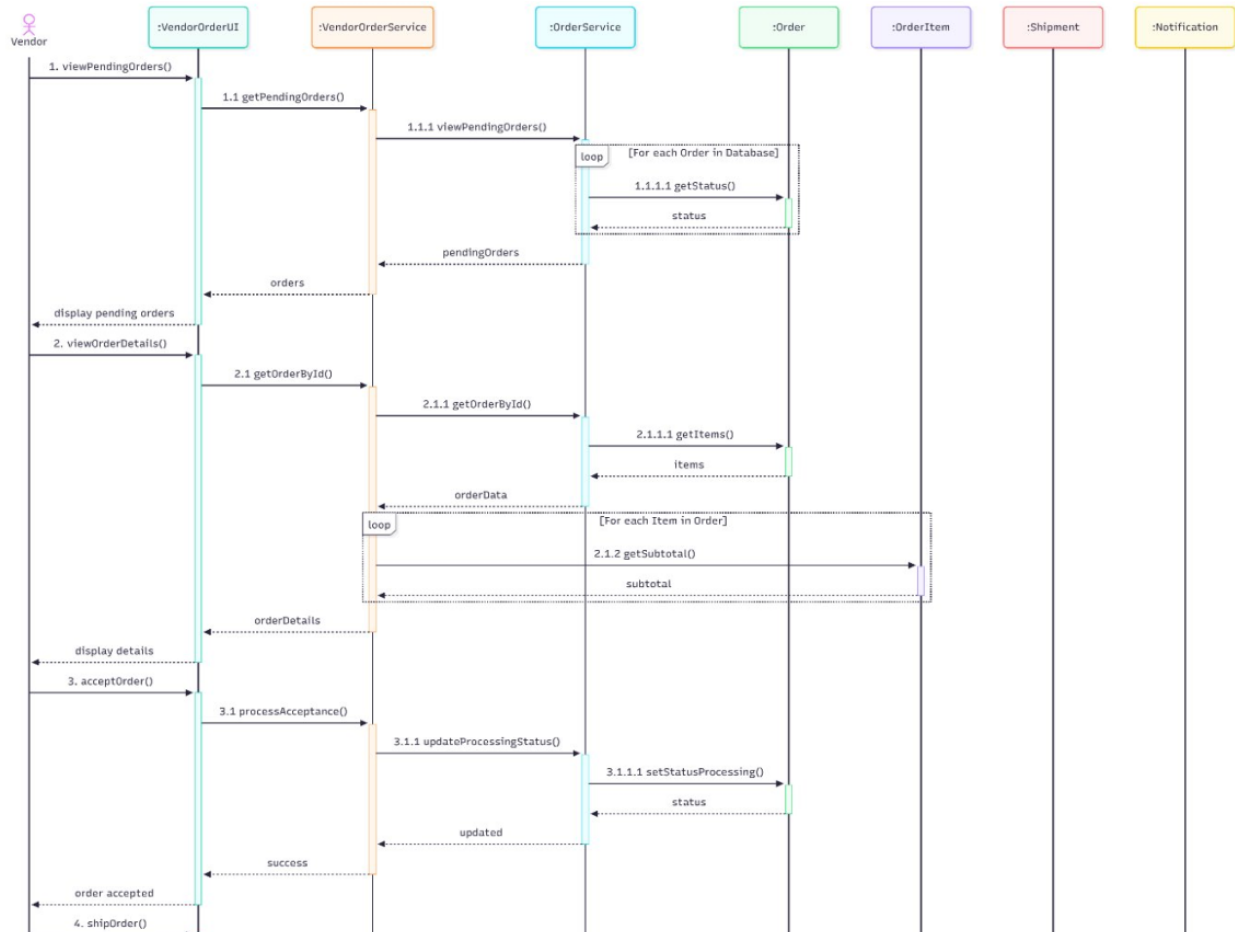


Figure 7.15: Vendor Orders Management (Part 1)

7.10.2 Vendor Orders Management (Part 2)

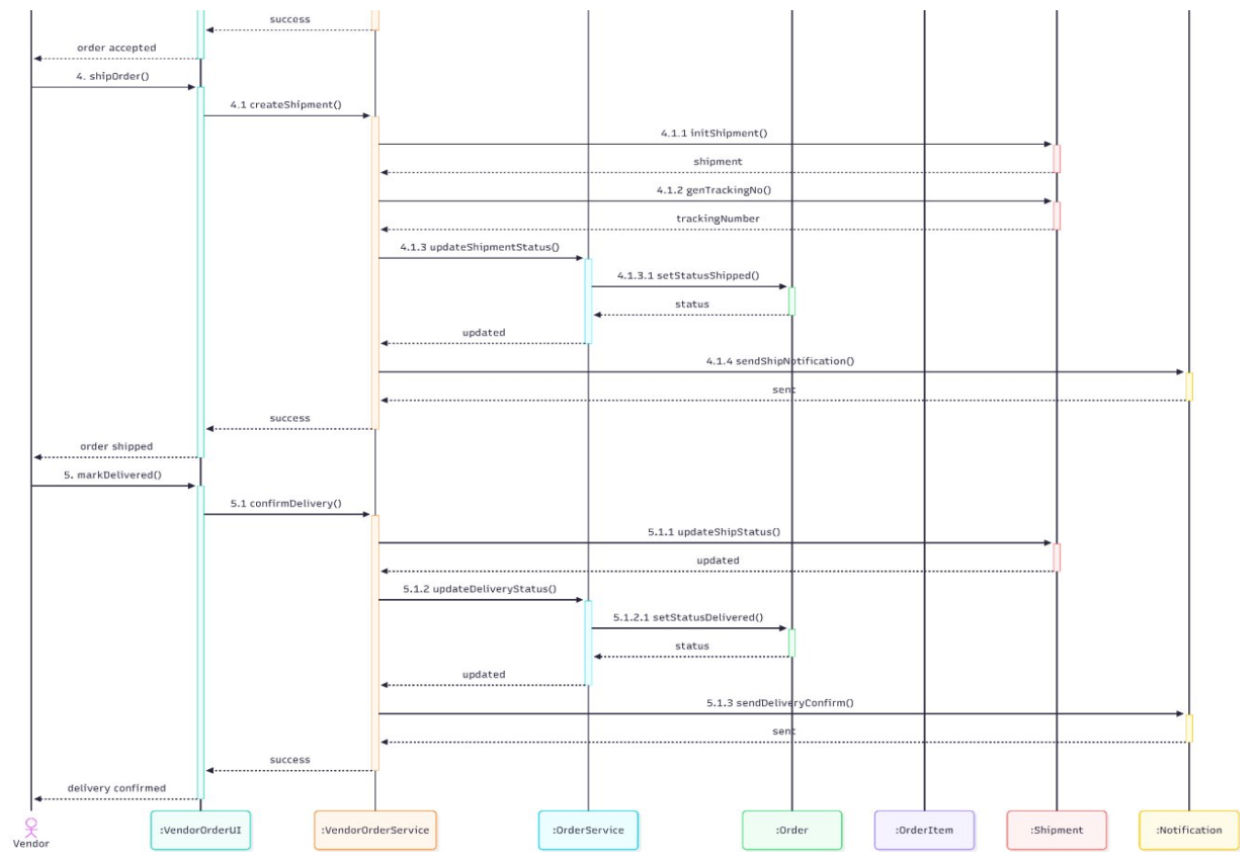


Figure 7.16: Vendor Orders Management (Part 2)

7.11 Review Product

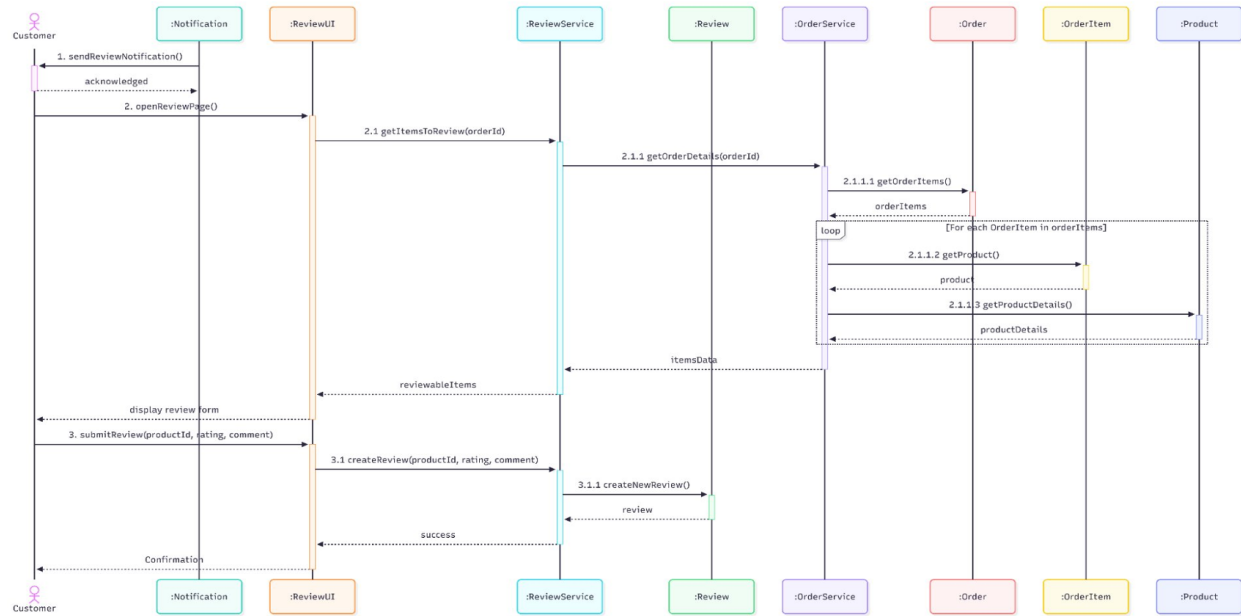


Figure 7.17: Review Product

State

Diagram

The presentation of the state diagrams was delivered by Md. Maruf Hasan Rubab
(2105042)

No modification was suggested for the state diagrams.

8.1 Order State Diagram

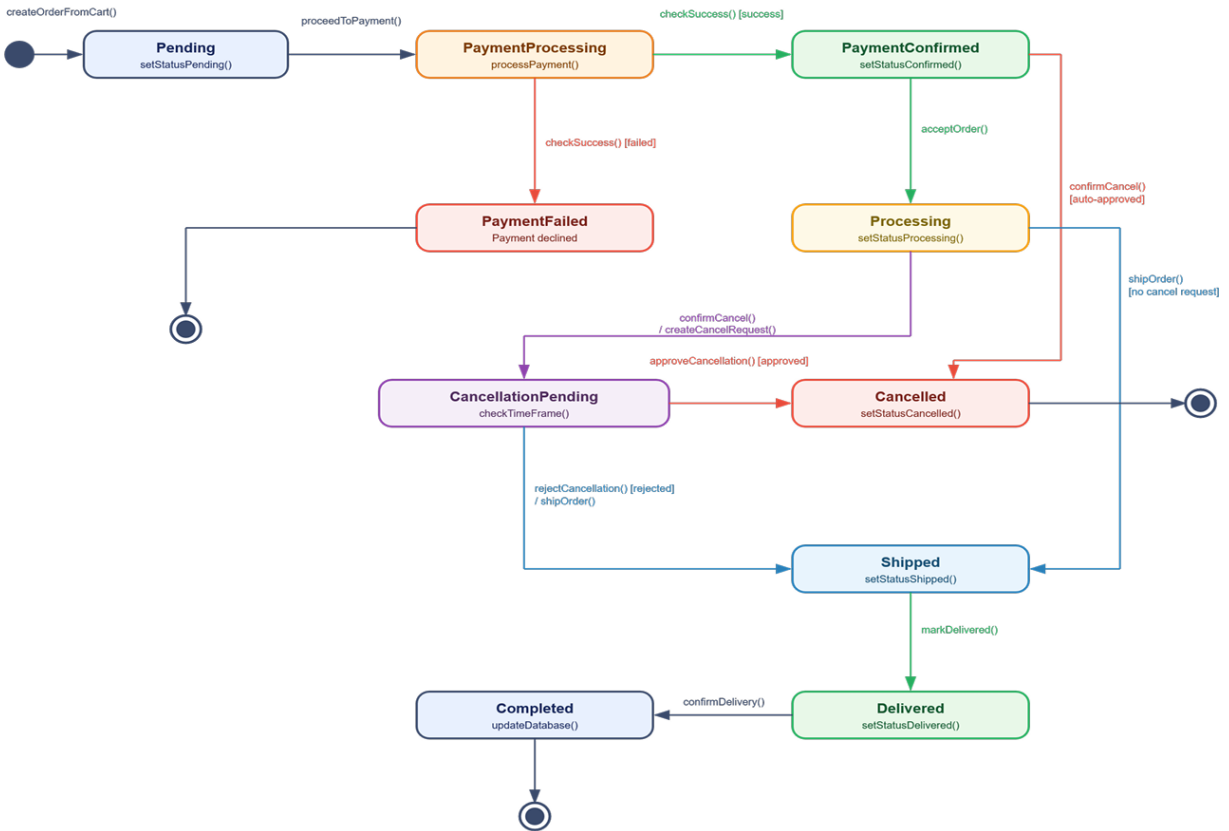


Figure 8.1: Order State Diagram

8.2 Payment State Diagram

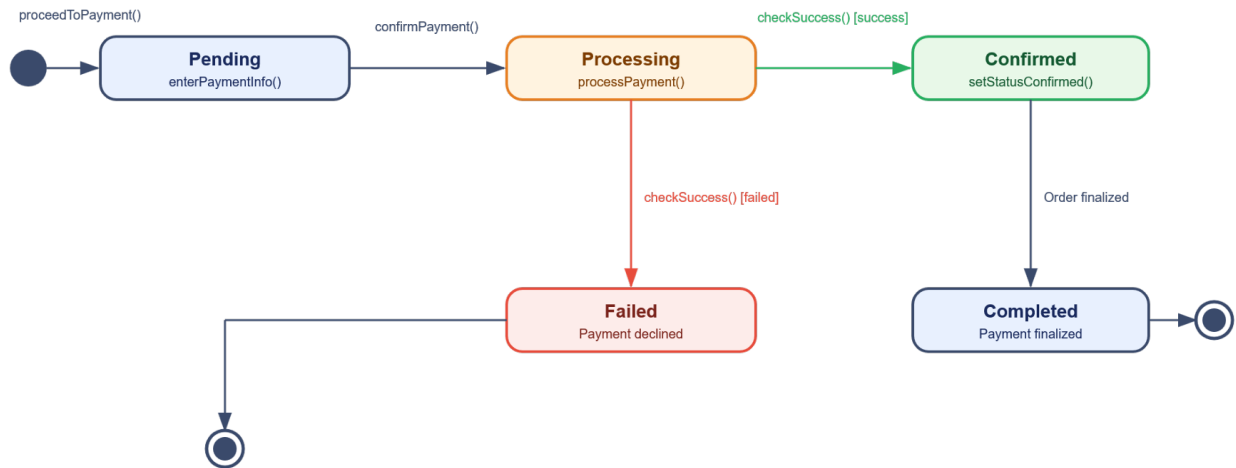


Figure 8.2: Payment State Diagram

8.3 Product Management State Diagram

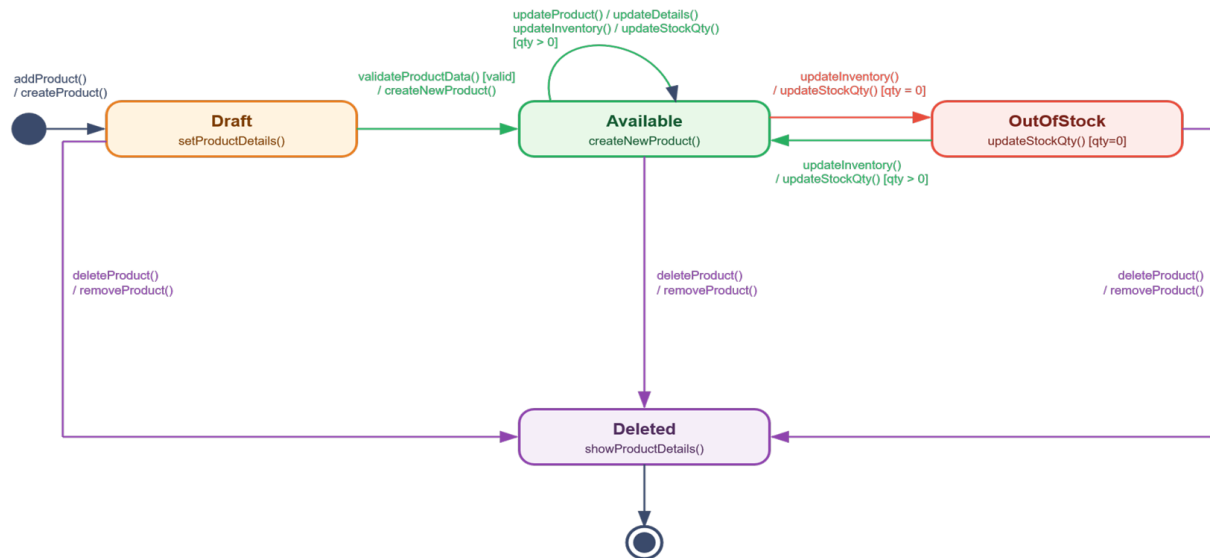


Figure 8.3: Product Management State Diagram

8.4 Cart State Diagram

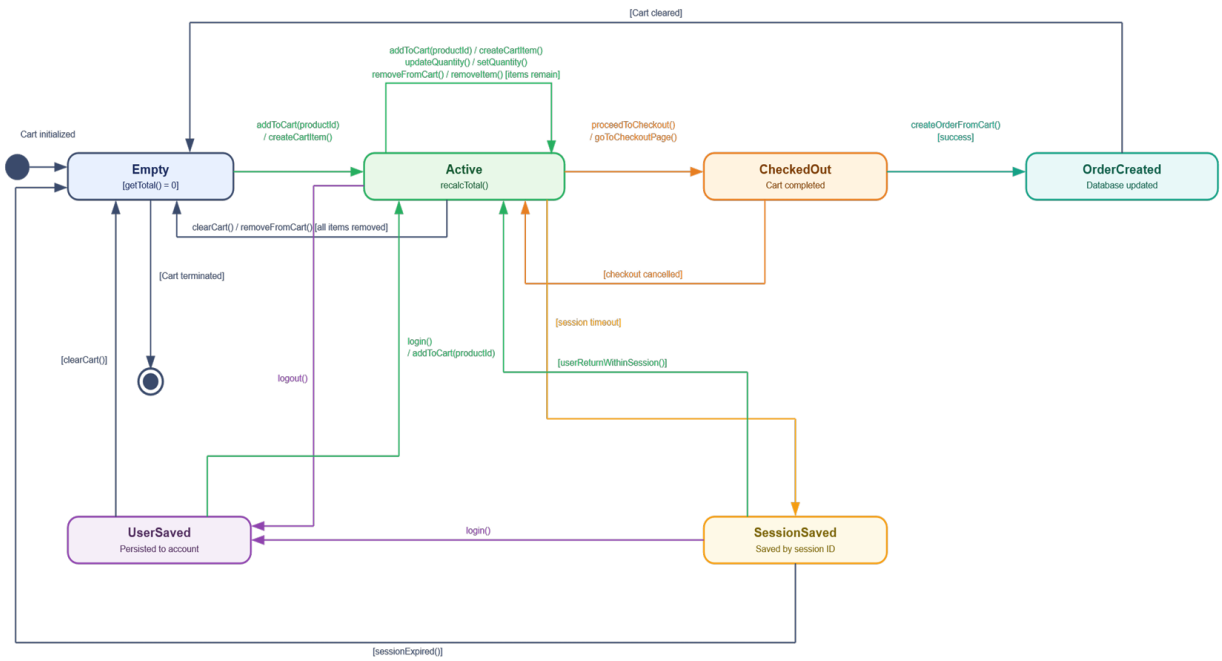


Figure 8.4: Cart State Diagram

8.5 Cancellation Request State Diagram

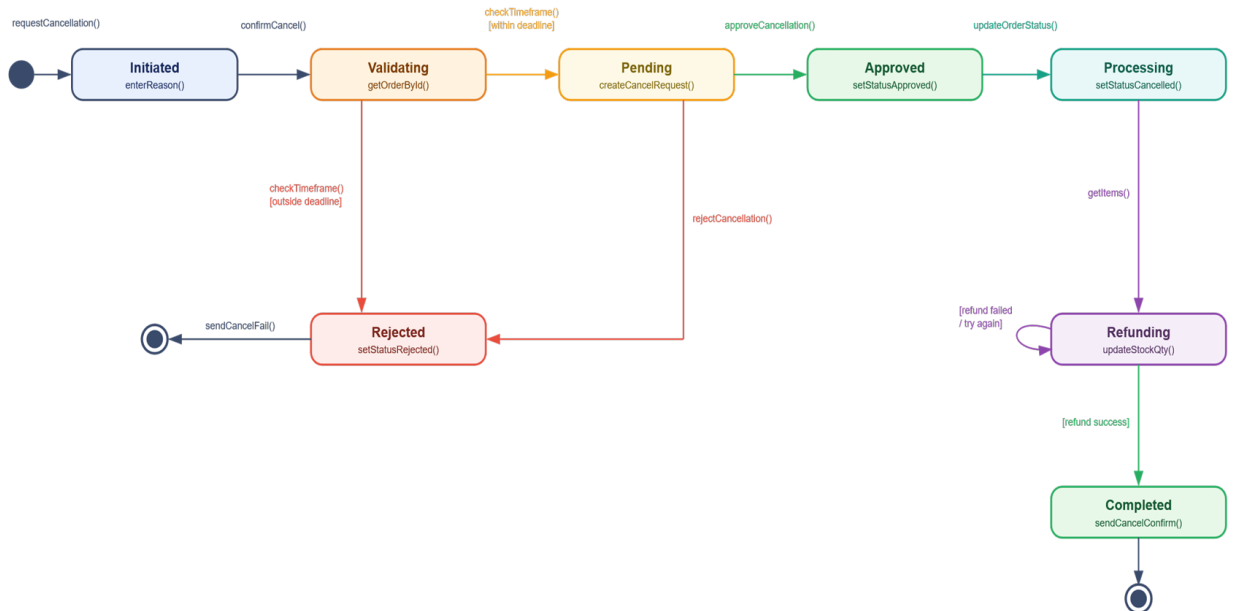


Figure 8.5: Cancellation Request State Diagram

API

Documentation

The presentation of the state diagrams was delivered by Md. Rafat Kabir (2105054)
No modification was suggested for the API Documentation.

9.1 Auth & User Management

9.1.1 POST /api/auth/sync

Access: Public

Tag: Auth

Purpose: Synchronises a Clerk-authenticated user into the local MongoDB database. Called on first sign-in to ensure a local user document exists for relational operations (orders, reviews, etc.). If the user already exists, their name, email, and imageUrl are updated.

Request Body (JSON):

Field	Type	Required	Description
clerkId	string	Yes	Unique Clerk user identifier
name	string	Yes	User display name
email	string	Yes	User email address
imageUrl	string	No	Profile picture URL

Response (200):

```
{
  "success": true,
  "user": {
    "_id": "user_2abc123",
    "name": "John Doe",
    "email": "john@example.com",
    "imageUrl": "https://cdn.clerk.com/image.jpg",
  }
}
```

```

    "cartItems": {},
    "wishlist": [],
    "createdAt": "2025-01-01T00:00:00.000Z",
    "updatedAt": "2025-01-01T00:00:00.000Z"
  }
}

```

9.1.2 GET /api/users/profile

Access: Authenticated

Tag: Auth

Purpose: Returns the profile of the currently authenticated user derived from the bearer token. Returns a computed `cartItemCount` (sum of all quantities in the cart map) and `wishlistCount` (length of the wishlist array) instead of the raw internal fields.

Response (200):

```

{
  "_id": "user_2abc123",
  "name": "John Doe",
  "email": "john@example.com",
  "imageUrl": "https://cdn.clerk.com/image.jpg",
  "cartItemCount": 3,
  "wishlistCount": 5
}

```

9.1.3 PUT /api/users/profile

Access: Authenticated

Tag: Auth

Purpose: Updates the display name and/or profile image of the current user. Returns the full updated User document from MongoDB.

Request Body (JSON):

Field	Type	Required	Description
name	string	No	Updated display name
imageUrl	string	No	Updated profile picture URL

Response (200):

```

{
  "success": true,
  "user": {

```

```

    "_id": "user_2abc123",
    "name": "Jane Doe",
    "email": "john@example.com",
    "imageUrl": "https://cdn.clerk.com/updated.jpg",
    "cartItems": {},
    "wishlist": [],
    "createdAt": "2025-01-01T00:00:00.000Z",
    "updatedAt": "2025-04-01T10:00:00.000Z"
  }
}

```

9.2 Products

9.2.1 GET /api/products

Access: Public

Tag: Products

Purpose: Returns a paginated, filterable, and sortable list of products. Default sort is `boughtInLastMonth` descending (popularity). Used on the homepage and category listing pages.

Query Parameters:

Parameter	Type	Default	Description
<code>page</code>	integer	1	Page number (min: 1)
<code>limit</code>	integer	20	Items per page (max: 100)
<code>sortBy</code>	string	–	<code>price</code> , <code>stars</code> , <code>reviews</code> , <code>boughtInLastMonth</code>
<code>sortOrder</code>	string	<code>desc</code>	<code>asc</code> or <code>desc</code>
<code>category</code>	string	–	Filter by category name (regex, case-insensitive)
<code>minPrice</code>	number	–	Minimum price filter
<code>maxPrice</code>	number	–	Maximum price filter
<code>minRating</code>	number	–	Minimum star rating (0–5)
<code>bestSeller</code>	string	–	Set to <code>true</code> to filter best-sellers only

Response (200):

```

{
  "products": [
    {
      "_id": "64f1a2b3c4d5e6f7a8b9c0d1",
      "asin": "B08N5WRWNW",

```

```

    "title": "Echo Dot (4th Gen)",
    "imgUrl": "https://images-na.ssl-images-amazon.com/images/I/abc.jpg",
    "productURL": "https://www.amazon.com/dp/B08N5WRWNW",
    "stars": 4.7,
    "reviews": 485210,
    "price": 49.99,
    "listPrice": 59.99,
    "category_id": 172282,
    "category_name": "Electronics",
    "isBestSeller": true,
    "boughtInLastMonth": 30000,
    "id": 1
  }
],
"pagination": {
  "page": 1,
  "limit": 20,
  "total": 1450,
  "totalPages": 73,
  "hasNext": true,
  "hasPrev": false
}
}

```

9.2.2 GET /api/products/search

Access: Public

Tag: Products

Purpose: Searches across `title`, `category_name`, and `asin` fields using a case-insensitive regex `$or` query. Powers the search bar on the frontend.

Query Parameters:

Parameter	Type	Required	Description
<code>query</code>	string	Yes	Search term
<code>page</code>	integer	No (default 1)	Page number
<code>limit</code>	integer	No (default 20)	Items per page (max: 100)

Response (200):

```

{
  "products": [
    {
      "_id": "64f1a2b3c4d5e6f7a8b9c0d1",
      "asin": "B08N5WRWNW",

```

```

    "title": "Echo Dot (4th Gen)",
    "imgUrl": "https://images-na.ssl-images-amazon.com/images/I/abc.jpg",
    "productURL": "https://www.amazon.com/dp/B08N5WRWNW",
    "stars": 4.7,
    "reviews": 485210,
    "price": 49.99,
    "listPrice": 59.99,
    "category_id": 172282,
    "category_name": "Electronics",
    "isBestSeller": true,
    "boughtInLastMonth": 30000,
    "id": 1
  }
],
"pagination": {
  "page": 1,
  "limit": 20,
  "total": 12,
  "totalPages": 1,
  "hasNext": false,
  "hasPrev": false
}
}

```

9.2.3 GET /api/products/categories

Access: Public

Tag: Products

Purpose: Aggregates all distinct `category_name` values from the products collection, returning each with its product count and `category_id`. Sorted by count descending. Used to render the category navigation and filter sidebar.

Response (200):

```

[
  { "name": "Electronics",      "count": 320, "categoryId": 172282 },
  { "name": "Books",           "count": 214, "categoryId": 283155 },
  { "name": "Home & Kitchen",   "count": 198, "categoryId": 1055398 },
  { "name": "Sports & Outdoors", "count": 145, "categoryId": 3375251 }
]

```

9.2.4 GET /api/products/best-sellers

Access: Public

Tag: Products

Purpose: Queries products where `isBestSeller` is `true`, sorted by `boughtInLastMonth` descending. Used in the Best Sellers section on the homepage.

Query Parameters:

Parameter	Type	Default	Description
limit	number	10	Number of products to return

Response (200):

```
[
  {
    "_id": "64f1a2b3c4d5e6f7a8b9c0d1",
    "asin": "B08N5WRWNW",
    "title": "Echo Dot (4th Gen)",
    "imgUrl": "https://images-na.ssl-images-amazon.com/images/I/abc.jpg",
    "productURL": "https://www.amazon.com/dp/B08N5WRWNW",
    "stars": 4.7,
    "reviews": 485210,
    "price": 49.99,
    "listPrice": 59.99,
    "category_id": 172282,
    "category_name": "Electronics",
    "isBestSeller": true,
    "boughtInLastMonth": 30000,
    "id": 1
  }
]
```

9.2.5 GET /api/products/:id

Access: Public

Tag: Products

Purpose: Fetches a single product document by its MongoDB ObjectId using a `.lean()` query for performance. Returns 404 if not found. Used on the Product Detail page.

Path Parameters:

Parameter	Type	Description
id	string	MongoDB ObjectId of the product

Response (200):

```
{
  "_id": "64f1a2b3c4d5e6f7a8b9c0d1",
```

```

"asin": "B08N5WRWNW",
"title": "Echo Dot (4th Gen)",
"imgUrl": "https://images-na.ssl-images-amazon.com/images/I/abc.jpg",
"productURL": "https://www.amazon.com/dp/B08N5WRWNW",
"stars": 4.7,
"reviews": 485210,
"price": 49.99,
"listPrice": 59.99,
"category_id": 172282,
"category_name": "Electronics",
"isBestSeller": true,
"boughtInLastMonth": 30000,
"id": 1
}

```

9.2.6 GET /api/products/:id/related

Access: Public

Tag: Products

Purpose: Finds the product by `id`, then queries other products with the same `category_name`, excluding itself, sorted by `boughtInLastMonth` descending. Returns an empty array if the source product is not found. Used in the Related Products section on the Product Detail page.

Path Parameters: `id` – MongoDB ObjectId of the product.

Query Parameters: `limit` (default: 8) – number of related products to return.

Response (200):

```

[
  {
    "_id": "64f1a2b3c4d5e6f7a8b9c0d2",
    "asin": "B07XJ8C8F5",
    "title": "Echo Show 5 (2nd Gen)",
    "imgUrl": "https://images-na.ssl-images-amazon.com/images/I/xyz.jpg",
    "productURL": "https://www.amazon.com/dp/B07XJ8C8F5",
    "stars": 4.6,
    "reviews": 201340,
    "price": 84.99,
    "listPrice": 99.99,
    "category_id": 172282,
    "category_name": "Electronics",
    "isBestSeller": false,
    "boughtInLastMonth": 12000,
  }
]

```

```

    "id": 2
  }
]

```

9.3 Cart

All cart endpoints require authentication. The cart is stored as a `Record<productId, quantity>` map on the User document and is scoped to the authenticated user.

9.3.1 GET /api/cart

Access: Authenticated

Tag: Cart

Purpose: Retrieves the user's cart. Each entry in the `cartItems` map is joined with its full product document. Returns per-item `subtotal`, a grand `total`, and a total `itemCount`.

Response (200):

```

{
  "items": [
    {
      "product": {
        "_id": "64f1a2b3c4d5e6f7a8b9c0d1",
        "asin": "B08N5WRWNW",
        "title": "Echo Dot (4th Gen)",
        "imgUrl": "https://images-na.ssl-images-amazon.com/images/I/abc.jpg",
        "productURL": "https://www.amazon.com/dp/B08N5WRWNW",
        "stars": 4.7,
        "reviews": 485210,
        "price": 49.99,
        "listPrice": 59.99,
        "category_id": 172282,
        "category_name": "Electronics",
        "isBestSeller": true,
        "boughtInLastMonth": 30000,
        "id": 1
      },
      "quantity": 2,
      "subtotal": 99.98
    }
  ],
  "total": 99.98,
  "itemCount": 2
}

```



```
}
```

Response (200) – Empty cart:

```
{ "items": [], "total": 0, "itemCount": 0 }
```

9.3.2 POST /api/cart/update

Access: Authenticated

Tag: Cart

Purpose: Sets the quantity of a product in the cart map. Sending **quantity**: 0 deletes the key from the map entirely. Saves the updated map on the User document, then returns the full recalculated cart. Handles add, increment, decrement, and remove through a single endpoint.

Request Body (JSON):

Field	Type	Required	Description
productId	string	Yes	MongoDB ObjectId of the product
quantity	integer	Yes	Desired quantity (0 = remove; min: 0)

Response (200):

```
{
  "items": [
    {
      "product": {
        "_id": "64f1a2b3c4d5e6f7a8b9c0d1",
        "asin": "B08N5WRWNW",
        "title": "Echo Dot (4th Gen)",
        "imgUrl": "https://images-na.ssl-images-amazon.com/images/I/abc.jpg",
        "productURL": "https://www.amazon.com/dp/B08N5WRWNW",
        "stars": 4.7,
        "reviews": 485210,
        "price": 49.99,
        "listPrice": 59.99,
        "category_id": 172282,
        "category_name": "Electronics",
        "isBestSeller": true,
        "boughtInLastMonth": 30000,
        "id": 1
      },
      "quantity": 3,
      "subtotal": 149.97
    }
  ]
}
```

```

    }
  ],
  "total": 149.97,
  "itemCount": 3
}

```

9.3.3 DELETE /api/cart/clear

Access: Authenticated

Tag: Cart

Purpose: Resets `cartItems` to an empty map on the User document, then returns the empty cart shape. Called automatically after a successful order is placed.

Response (200):

```
{ "items": [], "total": 0, "itemCount": 0 }
```

9.4 Addresses

All address endpoints require authentication. Addresses are strictly scoped to the authenticated user; cross-user access returns 403 Forbidden.

9.4.1 POST /api/addresses

Access: Authenticated

Tag: Addresses

Purpose: Creates a new shipping address for the current user. If it is the user's first address, or `isDefault` is `true`, all other addresses are unset as default before saving the new one.

Request Body (JSON):

Field	Type	Required	Description
<code>fullName</code>	string	Yes	Recipient full name
<code>phoneNumber</code>	string	Yes	Contact phone number
<code>pincode</code>	integer	Yes	PIN/postal code (min: 10000)
<code>area</code>	string	Yes	Street / locality / area
<code>city</code>	string	Yes	City
<code>state</code>	string	Yes	State or province
<code>isDefault</code>	boolean	No	Set as default address

Response (201):

```
{
  "success": true,
  "address": {
    "_id": "64a1b2c3d4e5f6a7b8c9d0e1",
    "userId": "user_2abc123",
    "fullName": "John Doe",
    "phoneNumber": "01712345678",
    "pincode": 12345,
    "area": "123 Main Street, Gulshan",
    "city": "Dhaka",
    "state": "Dhaka Division",
    "isDefault": true,
    "createdAt": "2025-04-01T10:00:00.000Z",
    "updatedAt": "2025-04-01T10:00:00.000Z"
  }
}
```

9.4.2 GET /api/addresses

Access: Authenticated

Tag: Addresses

Purpose: Returns all saved shipping addresses for the current user, sorted with the default address first (`isDefault: -1`), then by newest first. Used to populate the address selection list at checkout.

Response (200):

```
[
  {
    "_id": "64a1b2c3d4e5f6a7b8c9d0e1",
    "userId": "user_2abc123",
    "fullName": "John Doe",
    "phoneNumber": "01712345678",
    "pincode": 12345,
    "area": "123 Main Street, Gulshan",
    "city": "Dhaka",
    "state": "Dhaka Division",
    "isDefault": true,
    "createdAt": "2025-04-01T10:00:00.000Z",
    "updatedAt": "2025-04-01T10:00:00.000Z"
  },
  {
    "_id": "64a1b2c3d4e5f6a7b8c9d0e2",
    "userId": "user_2abc123",
```

```

    "fullName": "John Doe",
    "phoneNumber": "01798765432",
    "pincode": 54321,
    "area": "45 Dhanmondi Road 27",
    "city": "Dhaka",
    "state": "Dhaka Division",
    "isDefault": false,
    "createdAt": "2025-03-15T08:00:00.000Z",
    "updatedAt": "2025-03-15T08:00:00.000Z"
  }
]

```

9.4.3 GET /api/addresses/:id

Access: Authenticated

Tag: Addresses

Purpose: Fetches a single address by its MongoDB ObjectId using a `.lean()` query.
Returns 403 if the address belongs to a different user, and 404 if not found.

Path Parameters: id – Address MongoDB ObjectId.

Response (200):

```

{
  "_id": "64a1b2c3d4e5f6a7b8c9d0e1",
  "userId": "user_2abc123",
  "fullName": "John Doe",
  "phoneNumber": "01712345678",
  "pincode": 12345,
  "area": "123 Main Street, Gulshan",
  "city": "Dhaka",
  "state": "Dhaka Division",
  "isDefault": true,
  "createdAt": "2025-04-01T10:00:00.000Z",
  "updatedAt": "2025-04-01T10:00:00.000Z"
}

```

9.4.4 PUT /api/addresses/:id

Access: Authenticated

Tag: Addresses

Purpose: Updates one or more fields of an existing address using `Object.assign`. If `isDefault: true` is sent, all other addresses of the user are unset as default first.
Returns 403 if the address belongs to a different user.

Path Parameters: id – Address MongoDB ObjectId.

Request Body: Any subset of the fields from POST /api/addresses (all optional).

Response (200):

```
{
  "success": true,
  "address": {
    "_id": "64a1b2c3d4e5f6a7b8c9d0e2",
    "userId": "user_2abc123",
    "fullName": "John Doe",
    "phoneNumber": "01799999999",
    "pincode": 54321,
    "area": "45 Dhanmondi Road 27",
    "city": "Dhaka",
    "state": "Dhaka Division",
    "isDefault": true,
    "createdAt": "2025-03-15T08:00:00.000Z",
    "updatedAt": "2025-04-02T09:30:00.000Z"
  }
}
```

9.4.5 DELETE /api/addresses/:id

Access: Authenticated

Tag: Addresses

Purpose: Permanently deletes an address. If the deleted address was the default, the next most-recently-created address of the user is automatically promoted to default.

Path Parameters: id – Address MongoDB ObjectId.

Response (200):

```
{ "success": true }
```

9.5 Orders

9.5.1 POST /api/orders

Access: Authenticated

Tag: Orders

Purpose: Places a new order. Product prices are snapshotted from the database at order time. A 2% tax is added to the item subtotal to compute the final **amount**. The user's **cartItems** map is reset to {} after a successful save.

Request Body (JSON):

Field	Type	Required	Description
items	OrderItem[]	Yes	Array of {product, quantity} (min 1 item)
addressId	string	Yes	MongoDB ObjectId of the delivery address
paymentMethod	string	Yes	COD or Online

Response (201):

```
{
  "success": true,
  "order": {
    "_id": "64b1c2d3e4f5a6b7c8d9e0f1",
    "userId": "user_2abc123",
    "items": [
      {
        "product": "64f1a2b3c4d5e6f7a8b9c0d1",
        "quantity": 2,
        "price": 49.99
      }
    ],
    "amount": 101.98,
    "address": "64a1b2c3d4e5f6a7b8c9d0e1",
    "status": "Pending",
    "paymentMethod": "Online",
    "paymentStatus": "Pending",
    "paymentId": null,
    "paidAt": null,
    "trackingNumber": null,
    "estimatedDelivery": null,
    "statusHistory": [
      {
        "status": "Pending",
        "timestamp": "2025-04-01T10:00:00.000Z",
        "note": "Order placed"
      }
    ],
    "createdAt": "2025-04-01T10:00:00.000Z",
    "updatedAt": "2025-04-01T10:00:00.000Z"
  }
}
```

9.5.2 GET /api/orders

Access: Authenticated

Tag: Orders

Purpose: Returns the authenticated user's own order history. Both `items.product` and `address` are fully populated. Sorted newest first.

Query Parameters:

Parameter	Type	Default	Description
page	integer	1	Page number
limit	integer	10	Items per page
status	string	–	Filter by order status

Status enum: Pending, Confirmed, Processing, Shipped, Delivered, Cancelled, Returned.

Response (200):

```
{
  "orders": [
    {
      "_id": "64b1c2d3e4f5a6b7c8d9e0f1",
      "userId": "user_2abc123",
      "items": [
        {
          "product": {
            "_id": "64f1a2b3c4d5e6f7a8b9c0d1",
            "asin": "B08N5WRWNW",
            "title": "Echo Dot (4th Gen)",
            "imgUrl": "https://images-na.ssl-images-amazon.com/images/I/abc.jpg",
            "productURL": "https://www.amazon.com/dp/B08N5WRWNW",
            "stars": 4.7,
            "reviews": 485210,
            "price": 49.99,
            "listPrice": 59.99,
            "category_id": 172282,
            "category_name": "Electronics",
            "isBestSeller": true,
            "boughtInLastMonth": 30000,
            "id": 1
          },
          "quantity": 2,
          "price": 49.99
        }
      ]
    }
  ],
}
```

```

    "amount": 101.98,
    "address": {
      "_id": "64a1b2c3d4e5f6a7b8c9d0e1",
      "userId": "user_2abc123",
      "fullName": "John Doe",
      "phoneNumber": "01712345678",
      "pincode": 12345,
      "area": "123 Main Street, Gulshan",
      "city": "Dhaka",
      "state": "Dhaka Division",
      "isDefault": true,
      "createdAt": "2025-04-01T10:00:00.000Z",
      "updatedAt": "2025-04-01T10:00:00.000Z"
    },
    "status": "Pending",
    "paymentMethod": "Online",
    "paymentStatus": "Pending",
    "paymentId": null,
    "paidAt": null,
    "trackingNumber": null,
    "estimatedDelivery": null,
    "statusHistory": [
      {
        "status": "Pending",
        "timestamp": "2025-04-01T10:00:00.000Z",
        "note": "Order placed"
      }
    ],
    "createdAt": "2025-04-01T10:00:00.000Z",
    "updatedAt": "2025-04-01T10:00:00.000Z"
  }
],
"pagination": {
  "page": 1,
  "limit": 10,
  "total": 42,
  "totalPages": 5
}
}

```

9.5.3 GET /api/orders/:id

Access: Authenticated

Tag: Orders

Purpose: Returns a single order by ID with `items.product` and `address` fully populated.
Returns 403 Forbidden if the order belongs to a different user, 404 if not found.

Path Parameters: id – Order MongoDB ObjectId.

Response (200):

```
{
  "_id": "64b1c2d3e4f5a6b7c8d9e0f1",
  "userId": "user_2abc123",
  "items": [
    {
      "product": {
        "_id": "64f1a2b3c4d5e6f7a8b9c0d1",
        "asin": "B08N5WRWNW",
        "title": "Echo Dot (4th Gen)",
        "imgUrl": "https://images-na.ssl-images-amazon.com/images/I/abc.jpg",
        "productURL": "https://www.amazon.com/dp/B08N5WRWNW",
        "stars": 4.7,
        "reviews": 485210,
        "price": 49.99,
        "listPrice": 59.99,
        "category_id": 172282,
        "category_name": "Electronics",
        "isBestSeller": true,
        "boughtInLastMonth": 30000,
        "id": 1
      },
      "quantity": 2,
      "price": 49.99
    }
  ],
  "amount": 101.98,
  "address": {
    "_id": "64a1b2c3d4e5f6a7b8c9d0e1",
    "userId": "user_2abc123",
    "fullName": "John Doe",
    "phoneNumber": "01712345678",
    "pincode": 12345,
    "area": "123 Main Street, Gulshan",
    "city": "Dhaka",
    "state": "Dhaka Division",
    "isDefault": true,
    "createdAt": "2025-04-01T10:00:00.000Z",
    "updatedAt": "2025-04-01T10:00:00.000Z"
  }
}
```

```

},
"status": "Shipped",
"paymentMethod": "Online",
"paymentStatus": "Paid",
"paymentId": "pi_30a1b2C1Kj...",
"paidAt": "2025-04-01T10:05:00.000Z",
"trackingNumber": null,
"estimatedDelivery": "2025-04-06T10:00:00.000Z",
"statusHistory": [
  {
    "status": "Pending",
    "timestamp": "2025-04-01T10:00:00.000Z",
    "note": "Order placed"
  },
  {
    "status": "Confirmed",
    "timestamp": "2025-04-01T10:05:00.000Z",
    "note": "Payment received via Stripe"
  },
  {
    "status": "Processing",
    "timestamp": "2025-04-01T12:00:00.000Z",
    "note": "Status updated to Processing"
  },
  {
    "status": "Shipped",
    "timestamp": "2025-04-02T08:00:00.000Z",
    "note": "Status updated to Shipped"
  }
],
"createdAt": "2025-04-01T10:00:00.000Z",
"updatedAt": "2025-04-02T08:00:00.000Z"
}

```

9.5.4 POST /api/orders/:id/cancel

Access: Authenticated

Tag: Orders

Purpose: Cancels an order. Only **Pending** and **Confirmed** orders can be cancelled; any other status returns 400 Bad Request. If **paymentStatus** was **Paid**, it is automatically flipped to **Refunded**.

Path Parameters: **id** – Order MongoDB ObjectId.

Response (200):

```

{
  "success": true,
  "order": {
    "_id": "64b1c2d3e4f5a6b7c8d9e0f1",
    "userId": "user_2abc123",
    "items": [
      {
        "product": "64f1a2b3c4d5e6f7a8b9c0d1",
        "quantity": 2,
        "price": 49.99
      }
    ],
    "amount": 101.98,
    "address": "64a1b2c3d4e5f6a7b8c9d0e1",
    "status": "Cancelled",
    "paymentMethod": "Online",
    "paymentStatus": "Refunded",
    "paymentId": "pi_30a1b2C1Kj...",
    "paidAt": "2025-04-01T10:05:00.000Z",
    "trackingNumber": null,
    "estimatedDelivery": null,
    "statusHistory": [
      {
        "status": "Pending",
        "timestamp": "2025-04-01T10:00:00.000Z",
        "note": "Order placed"
      },
      {
        "status": "Confirmed",
        "timestamp": "2025-04-01T10:05:00.000Z",
        "note": "Payment received via Stripe"
      },
      {
        "status": "Cancelled",
        "timestamp": "2025-04-01T11:00:00.000Z",
        "note": "Cancelled by customer"
      }
    ],
    "createdAt": "2025-04-01T10:00:00.000Z",
    "updatedAt": "2025-04-01T11:00:00.000Z"
  }
}

```

9.5.5 GET /api/orders/stats

Access: Seller

Tag: Orders

Purpose: Runs two MongoDB aggregations: one `$group` on `status` to produce `statusBreakdown`, and one `$match` on `paymentStatus: Paid` to sum `totalRevenue`.
Used in the seller dashboard.

Response (200):

```
{
  "statusBreakdown": [
    { "_id": "Delivered", "count": 87, "totalAmount": 9230.75 },
    { "_id": "Pending", "count": 12, "totalAmount": 1450.50 },
    { "_id": "Shipped", "count": 9, "totalAmount": 890.00 },
    { "_id": "Cancelled", "count": 8, "totalAmount": 640.00 },
    { "_id": "Processing", "count": 4, "totalAmount": 320.00 }
  ],
  "totalOrders": 120,
  "totalRevenue": 9230.75
}
```

9.5.6 GET /api/orders/all

Access: Seller

Tag: Orders

Purpose: Returns all orders in the system across all users, with `items.product` and `address` fully populated, sorted newest first. Used in the vendor order management dashboard.

Query Parameters: Same as GET /api/orders (`page`, `limit`, `status`).

Response (200):

```
{
  "orders": [
    {
      "_id": "64b1c2d3e4f5a6b7c8d9e0f1",
      "userId": "user_2abc123",
      "items": [
        {
          "product": {
            "_id": "64f1a2b3c4d5e6f7a8b9c0d1",
            "asin": "B08N5WRWNW",
            "title": "Echo Dot (4th Gen)",

```

```

        "imgUrl": "https://images-na.ssl-images-amazon.com/images/I/abc.jpg",
        "productURL": "https://www.amazon.com/dp/B08N5WRWNW",
        "stars": 4.7,
        "reviews": 485210,
        "price": 49.99,
        "listPrice": 59.99,
        "category_id": 172282,
        "category_name": "Electronics",
        "isBestSeller": true,
        "boughtInLastMonth": 30000,
        "id": 1
    },
    "quantity": 2,
    "price": 49.99
}
],
"amount": 101.98,
"address": {
    "_id": "64a1b2c3d4e5f6a7b8c9d0e1",
    "userId": "user_2abc123",
    "fullName": "John Doe",
    "phoneNumber": "01712345678",
    "pincode": 12345,
    "area": "123 Main Street, Gulshan",
    "city": "Dhaka",
    "state": "Dhaka Division",
    "isDefault": true,
    "createdAt": "2025-04-01T10:00:00.000Z",
    "updatedAt": "2025-04-01T10:00:00.000Z"
},
"status": "Pending",
"paymentMethod": "Online",
"paymentStatus": "Pending",
"paymentId": null,
"paidAt": null,
"trackingNumber": null,
"estimatedDelivery": null,
"statusHistory": [
    {
        "status": "Pending",
        "timestamp": "2025-04-01T10:00:00.000Z",
        "note": "Order placed"
    }
],
"createdAt": "2025-04-01T10:00:00.000Z",

```

```

      "updatedAt": "2025-04-01T10:00:00.000Z"
    }
  ],
  "pagination": {
    "page": 1,
    "limit": 10,
    "total": 120,
    "totalPages": 12
  }
}

```

9.5.7 POST /api/orders/update-status

Access: Seller

Tag: Orders

Purpose: Updates the lifecycle status of any order following enforced transitions. Auto-sets `estimatedDelivery` to 5 days ahead when status becomes **Shipped**. Auto-sets `paymentStatus`: `Paid` and `paidAt` when a COD order reaches **Delivered**.

Valid Status Transitions:

Current Status	Allowed Next Status
Pending	Confirmed, Cancelled
Confirmed	Processing, Cancelled
Processing	Shipped, Cancelled
Shipped	Delivered, Returned
Delivered	Returned
Cancelled	(none)
Returned	(none)

Request Body (JSON):

Field	Type	Required	Description
<code>orderId</code>	string	Yes	MongoDB ObjectId of the order
<code>status</code>	string	Yes	New status (see transitions above)
<code>note</code>	string	No	Optional note appended to status history

Response (200):

```

{
  "success": true,
  "order": {
    "_id": "64b1c2d3e4f5a6b7c8d9e0f1",

```

```

    "userId": "user_2abc123",
    "items": [
      {
        "product": "64f1a2b3c4d5e6f7a8b9c0d1",
        "quantity": 2,
        "price": 49.99
      }
    ],
    "amount": 101.98,
    "address": "64a1b2c3d4e5f6a7b8c9d0e1",
    "status": "Shipped",
    "paymentMethod": "Online",
    "paymentStatus": "Paid",
    "paymentId": "pi_30a1b2C1Kj...",
    "paidAt": "2025-04-01T10:05:00.000Z",
    "trackingNumber": null,
    "estimatedDelivery": "2025-04-06T10:00:00.000Z",
    "statusHistory": [
      {
        "status": "Pending",
        "timestamp": "2025-04-01T10:00:00.000Z",
        "note": "Order placed"
      },
      {
        "status": "Confirmed",
        "timestamp": "2025-04-01T10:05:00.000Z",
        "note": "Payment received via Stripe"
      },
      {
        "status": "Processing",
        "timestamp": "2025-04-01T12:00:00.000Z",
        "note": "Status updated to Processing"
      },
      {
        "status": "Shipped",
        "timestamp": "2025-04-02T08:00:00.000Z",
        "note": "Status updated to Shipped"
      }
    ],
    "createdAt": "2025-04-01T10:00:00.000Z",
    "updatedAt": "2025-04-02T08:00:00.000Z"
  }
}

```

9.6 Payments

Payment endpoints integrate with the Stripe API. Online payment flow: create checkout session → redirect user to Stripe-hosted checkout page → verify session on return.

9.6.1 POST /api/payments/checkout/:orderId

Access: Authenticated

Tag: Payments

Purpose: Creates a Stripe Checkout Session for the specified order. Each order item becomes a Stripe line item at its snapshot price. A 2% tax line item is appended. Returns the Stripe session ID and hosted checkout URL. Rejects if the order is already paid, uses COD, or belongs to a different user.

Path Parameters: `orderId` – Order MongoDB ObjectId.

Response (201):

```
{
  "sessionId": "cs_test_a1b2c3d4e5f6g7h8i9j0k1l2m3n4o5p6",
  "url": "https://checkout.stripe.com/c/pay/cs_test_a1b2c3d4..."
}
```

9.6.2 POST /api/payments/verify

Access: Authenticated

Tag: Payments

Purpose: Retrieves the Stripe session by `sessionId` and checks `payment_status`. If paid, the order's `paymentStatus` is set to Paid, `paymentId` is stored from `session.payment_intent`, `paidAt` is recorded, and `status` is advanced to Confirmed with a history entry. Idempotent on already-paid orders.

Request Body (JSON):

Field	Type	Required	Description
<code>sessionId</code>	string	Yes	Stripe Checkout Session ID
<code>orderId</code>	string	Yes	MongoDB ObjectId of the order

Response (200) – Payment confirmed:

```
{
  "success": true,
  "paymentStatus": "Paid",
  "order": {
```



```

    "_id": "64b1c2d3e4f5a6b7c8d9e0f1",
    "userId": "user_2abc123",
    "items": [
      {
        "product": "64f1a2b3c4d5e6f7a8b9c0d1",
        "quantity": 2,
        "price": 49.99
      }
    ],
    "amount": 101.98,
    "address": "64a1b2c3d4e5f6a7b8c9d0e1",
    "status": "Confirmed",
    "paymentMethod": "Online",
    "paymentStatus": "Paid",
    "paymentId": "pi_30a1b2C1Kj4MT4vL1A2B3C4",
    "paidAt": "2025-04-01T10:05:00.000Z",
    "trackingNumber": null,
    "estimatedDelivery": null,
    "statusHistory": [
      {
        "status": "Pending",
        "timestamp": "2025-04-01T10:00:00.000Z",
        "note": "Order placed"
      },
      {
        "status": "Confirmed",
        "timestamp": "2025-04-01T10:05:00.000Z",
        "note": "Payment received via Stripe"
      }
    ],
    "createdAt": "2025-04-01T10:00:00.000Z",
    "updatedAt": "2025-04-01T10:05:00.000Z"
  }
}

```

Response (200) – Payment not completed:

```
{ "success": false, "paymentStatus": "unpaid" }
```

9.7 Reviews

9.7.1 POST /api/reviews

Access: Authenticated

Tag: Reviews

Purpose: Submits a new product review. Enforces one review per user per product via a unique compound index on `userId` + `productId`. If `orderId` is provided and that order is `Delivered` and contains the product, `isVerifiedPurchase` is set to `true`.

Request Body (JSON):

Field	Type	Required	Description
<code>productId</code>	string	Yes	MongoDB ObjectId of the product
<code>orderId</code>	string	No	Order ID for verified purchase badge
<code>rating</code>	integer	Yes	Star rating, 1 to 5
<code>title</code>	string	No	Review headline
<code>comment</code>	string	No	Full review body text

Response (201):

```
{
  "success": true,
  "review": {
    "_id": "64c1d2e3f4a5b6c7d8e9f0a1",
    "userId": "user_2abc123",
    "productId": "64f1a2b3c4d5e6f7a8b9c0d1",
    "orderId": "64b1c2d3e4f5a6b7c8d9e0f1",
    "rating": 5,
    "title": "Great product!",
    "comment": "Works exactly as described. Very happy with this purchase.",
    "isVerifiedPurchase": true,
    "createdAt": "2025-04-01T11:00:00.000Z",
    "updatedAt": "2025-04-01T11:00:00.000Z"
  }
}
```

9.7.2 GET /api/reviews/product/:productId

Access: Public

Tag: Reviews

Purpose: Returns paginated reviews for a product. Each review is enriched with `userName` and `userImage` fetched from the User collection. The `summary` block is produced

by two separate aggregation pipelines: one for average rating and total, one for the per-star distribution.

Path Parameters: productId – Product MongoDB ObjectId.

Query Parameters:

Parameter	Type	Default	Description
page	integer	1	Page number
limit	integer	10	Items per page
sortBy	string	newest	newest, highest, lowest

Response (200):

```
{
  "reviews": [
    {
      "_id": "64c1d2e3f4a5b6c7d8e9f0a1",
      "userId": "user_2abc123",
      "productId": "64f1a2b3c4d5e6f7a8b9c0d1",
      "orderId": "64b1c2d3e4f5a6b7c8d9e0f1",
      "rating": 5,
      "title": "Great product!",
      "comment": "Works exactly as described. Very happy with this purchase.",
      "isVerifiedPurchase": true,
      "createdAt": "2025-04-01T11:00:00.000Z",
      "updatedAt": "2025-04-01T11:00:00.000Z",
      "userName": "John Doe",
      "userImage": "https://cdn.clerk.com/image.jpg"
    }
  ],
  "pagination": {
    "page": 1,
    "limit": 10,
    "total": 38,
    "totalPages": 4
  },
  "summary": {
    "averageRating": 4.6,
    "totalReviews": 38,
    "ratingDistribution": {
      "5": 22,
      "4": 10,
      "3": 4,
      "2": 1,
      "1": 1
    }
  }
}
```

```

    }
  }
}

```

9.7.3 GET /api/reviews/my-reviews

Access: Authenticated

Tag: Reviews

Purpose: Returns all reviews written by the authenticated user, sorted newest first. The `productId` field is fully populated as a complete Product object via Mongoose `.populate('productId')`.

Response (200):

```

[
  {
    "_id": "64c1d2e3f4a5b6c7d8e9f0a1",
    "userId": "user_2abc123",
    "productId": {
      "_id": "64f1a2b3c4d5e6f7a8b9c0d1",
      "asin": "B08N5WRWNW",
      "title": "Echo Dot (4th Gen)",
      "imgUrl": "https://images-na.ssl-images-amazon.com/images/I/abc.jpg",
      "productURL": "https://www.amazon.com/dp/B08N5WRWNW",
      "stars": 4.7,
      "reviews": 485210,
      "price": 49.99,
      "listPrice": 59.99,
      "category_id": 172282,
      "category_name": "Electronics",
      "isBestSeller": true,
      "boughtInLastMonth": 30000,
      "id": 1
    },
    "orderId": "64b1c2d3e4f5a6b7c8d9e0f1",
    "rating": 5,
    "title": "Great product!",
    "comment": "Works exactly as described. Very happy with this purchase.",
    "isVerifiedPurchase": true,
    "createdAt": "2025-04-01T11:00:00.000Z",
    "updatedAt": "2025-04-01T11:00:00.000Z"
  }
]

```

9.7.4 PUT /api/reviews/:id

Access: Authenticated

Tag: Reviews

Purpose: Updates an existing review using field-by-field assignment. Only the original author can update it (400 otherwise). Only provided fields are overwritten; omitted fields remain unchanged.

Path Parameters: id – Review MongoDB ObjectId.

Request Body: Any subset of `rating` (1–5), `title`, `comment` (all optional).

Response (200):

```
{
  "success": true,
  "review": {
    "_id": "64c1d2e3f4a5b6c7d8e9f0a1",
    "userId": "user_2abc123",
    "productId": "64f1a2b3c4d5e6f7a8b9c0d1",
    "orderId": "64b1c2d3e4f5a6b7c8d9e0f1",
    "rating": 4,
    "title": "Updated title",
    "comment": "On reflection, a solid product with minor flaws.",
    "isVerifiedPurchase": true,
    "createdAt": "2025-04-01T11:00:00.000Z",
    "updatedAt": "2025-04-02T09:00:00.000Z"
  }
}
```

9.7.5 DELETE /api/reviews/:id

Access: Authenticated

Tag: Reviews

Purpose: Permanently deletes a review using `.deleteOne()`. Only the original author can delete it (400 otherwise).

Path Parameters: id – Review MongoDB ObjectId.

Response (200):

```
{ "success": true }
```

9.8 Wishlist

All wishlist endpoints require authentication. The wishlist is stored as a `string[]` array of product ObjectIds on the User document.

9.8.1 GET /api/wishlist

Access: Authenticated

Tag: Wishlist

Purpose: Reads the `wishlist` array from the User document, converts each string ID to a MongoDB ObjectId, and fetches all matching product documents. Returns them under an `items` key.

Response (200):

```
{
  "items": [
    {
      "_id": "64f1a2b3c4d5e6f7a8b9c0d1",
      "asin": "B08N5WRWNW",
      "title": "Echo Dot (4th Gen)",
      "imageUrl": "https://images-na.ssl-images-amazon.com/images/I/abc.jpg",
      "productURL": "https://www.amazon.com/dp/B08N5WRWNW",
      "stars": 4.7,
      "reviews": 485210,
      "price": 49.99,
      "listPrice": 59.99,
      "category_id": 172282,
      "category_name": "Electronics",
      "isBestSeller": true,
      "boughtInLastMonth": 30000,
      "id": 1
    }
  ]
}
```

Response (200) – Empty wishlist:

```
{ "items": [] }
```

9.8.2 POST /api/wishlist/:productId

Access: Authenticated

Tag: Wishlist

Purpose: Pushes the `productId` string into the user's `wishlist` array only if it is not already present (idempotent). Returns the new count.

Path Parameters: `productId` – Product MongoDB ObjectId.

Response (201):

```
{ "success": true, "wishlistCount": 3 }
```

9.8.3 DELETE /api/wishlist/:productId

Access: Authenticated

Tag: Wishlist

Purpose: Filters out `productId` from the user's `wishlist` array and saves the document. Returns the updated count.

Path Parameters: `productId` – Product MongoDB ObjectId.

Response (200):

```
{ "success": true, "wishlistCount": 2 }
```

9.8.4 GET /api/wishlist/:productId/check

Access: Authenticated

Tag: Wishlist

Purpose: Checks whether `productId` is present in the user's `wishlist` string array using `Array.includes()`. Used to toggle the wishlist heart icon state on product cards and the detail page without fetching the full wishlist.

Path Parameters: `productId` – Product MongoDB ObjectId.

Response (200):

```
{ "inWishlist": true }
```

Feature

Implementation

The Feature Implementation chapter demonstrates the successful realization of the Amazon platform's order management system. This chapter covers three core features (Place Order, Checkout, and Order Review), along with the complete system architecture, technology stack, deployment infrastructure, and team contributions.

10.1 Introduction

The Feature Implementation phase represents the practical development and deployment of the designed e-commerce system. Our team successfully implemented a full-stack Amazon-like platform with integrated order management, checkout flow, and user review system. The implementation showcases modern web technologies, cloud deployment, and secure payment integration using industry-standard tools. All features have been deployed on production-ready infrastructure with comprehensive CI/CD pipelines for continuous integration and deployment.

10.2 Features Overview

10.2.1 Feature A: Place Order

This feature allows authenticated users to convert their cart items into confirmed orders.

Main Capabilities:

- Select delivery address from saved addresses or add new address
- Choose payment method (Cash on Delivery or Online Payment)
- Create order with item snapshot capturing price and quantity at order time
- Store order status and payment status for tracking

- Track order lifecycle (Pending, Processing, Shipped, Delivered, Cancelled, etc.)
- Order confirmation with order ID and tracking information

10.2.2 Feature B: Checkout

This feature handles the final purchase flow before order confirmation.

Main Capabilities:

- Calculate and display order summary (items, subtotal, taxes, total)
- Address selection and validation
- Payment flow integration with Stripe for online payments
- Payment verification and Stripe webhook integration
- Prevent duplicate payment confirmation behavior
- Automatic redirect to order detail page after successful checkout

10.2.3 Feature C: Order Review

This feature enables post-purchase user feedback on products.

Main Capabilities:

- Authenticated users can submit, update, or delete ratings and reviews
- One-review-per-user-per-product enforcement
- Verified purchase context through order linkage
- Product-level aggregates (stars, reviews) automatically updated after review operations
- Review summary reflected across product detail and listing pages
- Star ratings calculation and display

10.3 Repository Links

GitHub Repositories:

- **Frontend Repository:** <https://github.com/mehedi-107/amazon-frontend>
- **Backend Repository:** <https://github.com/mehedi-107/amazon-backend>

All source code is version-controlled and follows Git best practices with multiple branches for development, staging, and production.

10.4 Hosted Website Links

Live Deployment URLs:

- **Frontend (Vercel):** `https://amazon-frontend-rosy.vercel.app`
- **Backend API (Vercel):** `https://amazon-backend-swart.vercel.app`
- **Stripe Payment Gateway:** `https://checkout.stripe.com`

Frontend environment variable configuration:

`VITE_API_URL=https://amazon-backend-swart.vercel.app`

10.5 Screenshots of Different Pages

The following section showcases the user interface of the implemented features across different application pages.

10.5.1 Home Page

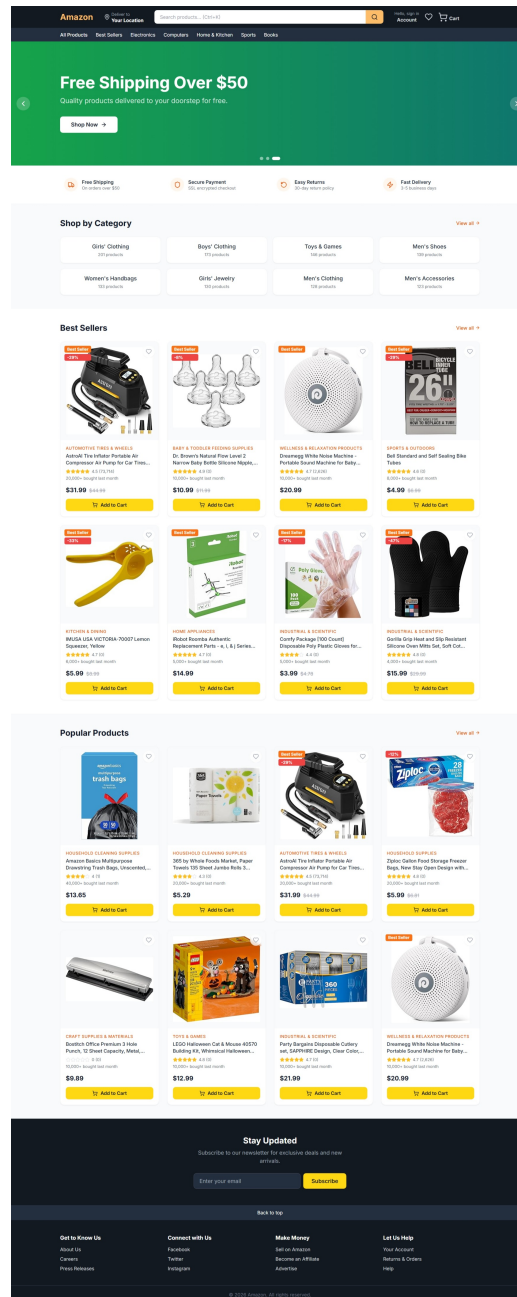


Figure 10.1: Home Page - Product Listing and Browsing

10.5.2 Product Detail Page

The screenshot displays the Amazon product detail page for 'Amazon Basics Multipurpose Drawstring Trash Bags, Unscented, 30 Gallon, 50 Count (Previously Solimo)'. The page features a product image of a blue trash bag, a star rating of 4.0 with 1 review, and a price of \$13.65. Below the product information, there is a section for customer reviews, showing a 4.0 star rating and a single review from 'Mid Maruf Hasan' dated March 10, 2020. The 'Related Products' section at the bottom displays eight other household cleaning supplies, each with its own image, title, price, and 'Add to Cart' button. The footer contains links for 'Get to Know Us', 'Connect with Us', 'Make Money', and 'Let Us Help'.

Amazon Deliver to Your Location Search products... (2019-03) Hello, sign in Account Cart

All Products Best Sellers Electronics Computers Home & Kitchen Sports Books

Home > Products > Household Cleaning Supplies > Amazon Basics Multipurpose Drawstring Tra...

HOUSEHOLD CLEANING SUPPLIES

Amazon Basics Multipurpose Drawstring Trash Bags, Unscented, 30 Gallon, 50 Count (Previously Solimo)

★★★★ 4.0 (1 reviews)

40,000+ bought in the last month

\$13.65

Quantity: - 1 +

Add to Cart

Free Shipping Secure Payment 30-Day Returns

ASIN: B08SPL2MPZ Category: Household Cleaning Supplies

Customer Reviews

4.0 5 star 4 star 3 star 2 star 1 star

1 reviews

Mid Maruf Hasan Verified Purchase

very good product its perfect

March 10, 2020

Related Products

HOUSEHOLD CLEANING SUPPLIES 365 by Whole Foods Market, Paper Towels 125 Sheet Jumbo Rolls 2... **\$5.29** **Add to Cart**

HOUSEHOLD CLEANING SUPPLIES Lysol Disinfectant Wipes Multi-Surface Antibacterial Cleaning Wipe... **\$11.94** **Add to Cart**

HOUSEHOLD CLEANING SUPPLIES Small Trash Bags 4-6 Gallon, Heavyduty 200 Count Biodegradable Trash Ba... **\$14.99** **Add to Cart**

HOUSEHOLD CLEANING SUPPLIES Ref. ProGrade Contractor Trash Bags 55 Gallon | 20 Bags w/ Tie | Heavy... **\$23.99** **Add to Cart**

HOUSEHOLD CLEANING SUPPLIES The Honest Company Clean Conscious Wipes | 100% Water... **\$5.58** **Add to Cart**

HOUSEHOLD CLEANING SUPPLIES Wall Map with 82 Inches Long Handle, 360 Degree Rotating Microfiber Tr... **\$24.99** **Add to Cart**

HOUSEHOLD CLEANING SUPPLIES ADEA Microfiber Cleaning Cloths- 100% poly, Super Highly Absorbent... **\$30.95** **Add to Cart**

HOUSEHOLD CLEANING SUPPLIES 3 in 1 Tub Shower Scrubbing Cleaning Brush with 58"... **\$15.99** **Add to Cart**

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Figure 10.2: Product Detail Page with Reviews and Ratings

10.5.3 Cart Page

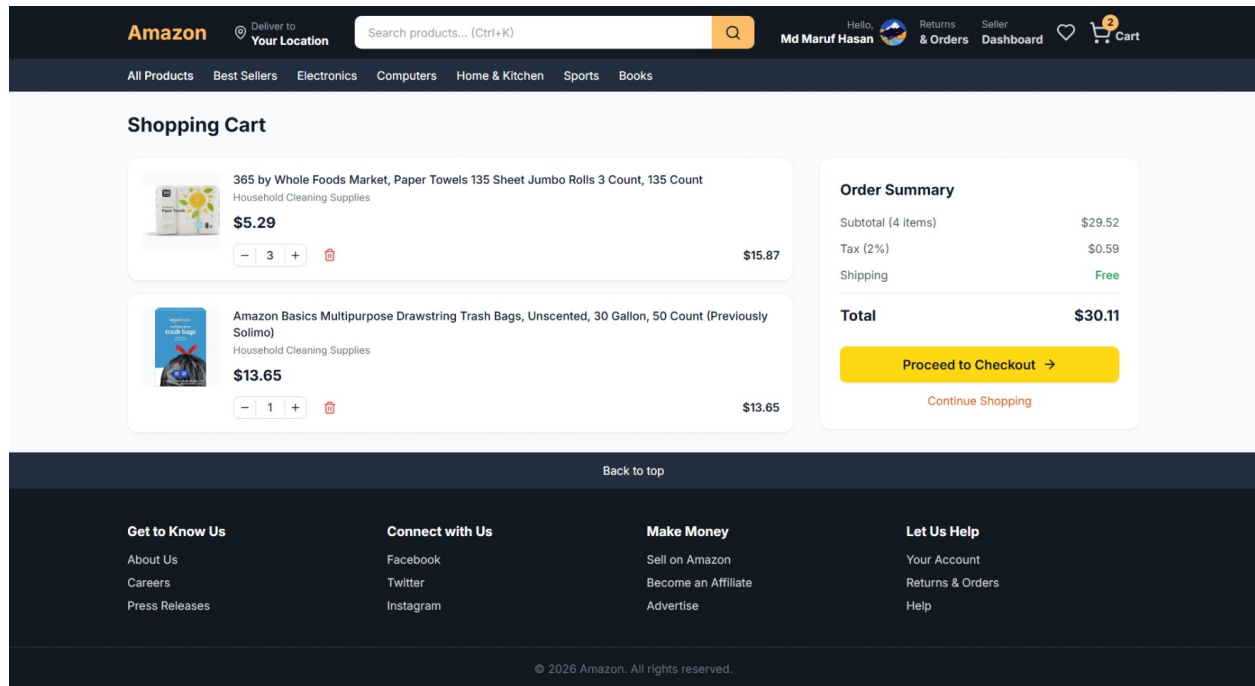


Figure 10.3: Shopping Cart Page with Item Management

10.5.4 Checkout Page - Address Selection

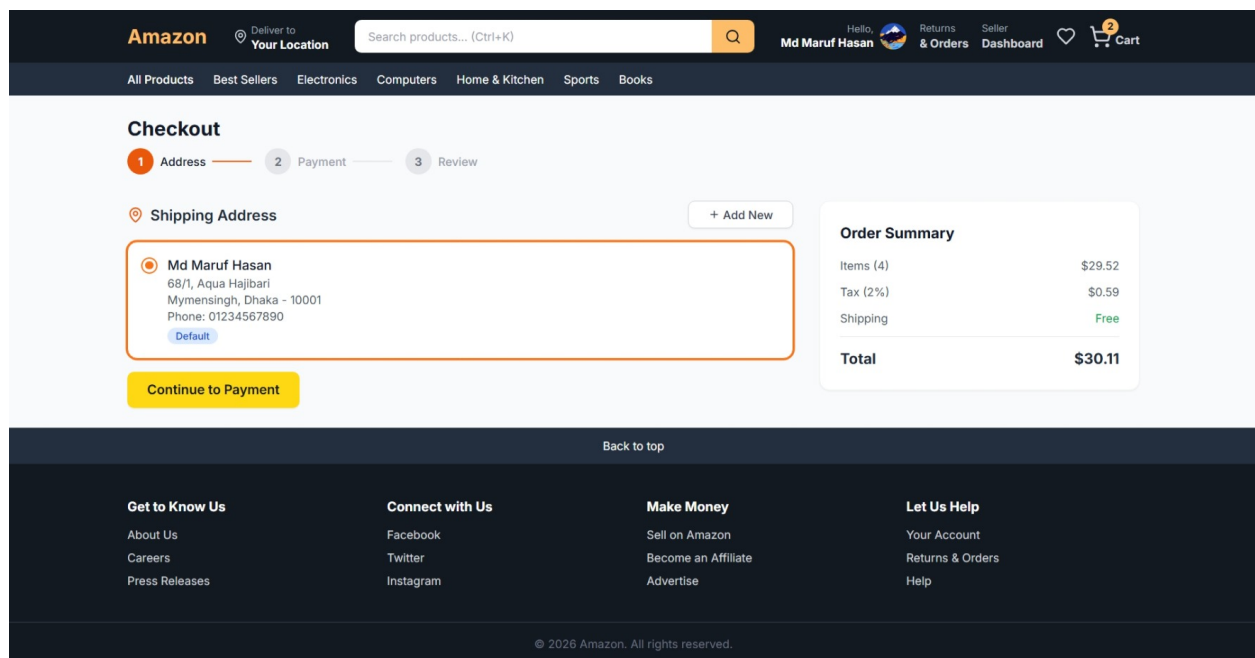


Figure 10.4: Checkout Page - Select Delivery Address

10.5.5 Checkout Page - Order Summary

The screenshot displays the Amazon checkout page. At the top, the Amazon logo is on the left, followed by a location selector showing 'Deliver to Your Location'. A search bar with the placeholder 'Search products... (Ctrl+K)' is in the center. On the right, the user's name 'Md Maruf Hasan' is shown next to a globe icon, with links for 'Returns & Orders' and 'Seller Dashboard'. A shopping cart icon with a '2' badge is also present. Below the header, a navigation bar lists categories: All Products, Best Sellers, Electronics, Computers, Home & Kitchen, Sports, and Books.

The main content area is titled 'Review Your Order'. It is divided into three sections:

- Shipping Address:** Shows 'Md Maruf Hasan, Palashi, Buet, Dhaka - 10002' with a 'Change' link.
- Payment Method:** Shows 'Pay Online (Stripe)' with a 'Change' link.
- Order Items:** Lists two items:
 - 365 by Whole Foods Market, Paper Towels 135 Sheet Jumbo Rolls 3 Count, 135 Count. Qty: 3. Price: \$15.87.
 - Amazon Basics Multipurpose Drawstring Trash Bags, Unscented, 30 Gallon, 50 Count (Previously ...). Qty: 1. Price: \$13.65.

At the bottom of the 'Review Your Order' section, there are two buttons: 'Back' and 'Place Order — \$30.11'.

To the right of the 'Review Your Order' section is an 'Order Summary' box:

Order Summary	
Items (4)	\$29.52
Tax (2%)	\$0.59
Shipping	Free
Total	\$30.11

Below the 'Review Your Order' section, there is a 'Back to top' link. The footer contains four columns of links: 'Get to Know Us' (About Us, Careers, Press Releases), 'Connect with Us' (Facebook, Twitter, Instagram), 'Make Money' (Sell on Amazon, Become an Affiliate, Advertise), and 'Let Us Help' (Your Account, Returns & Orders, Help). The footer also includes the copyright notice '© 2026 Amazon. All rights reserved.'

Figure 10.5: Checkout Page - Order Summary and Details

10.5.6 Payment Method Selection

The screenshot displays the Amazon checkout interface. At the top, the Amazon logo is on the left, followed by a location pin icon and 'Your Location'. A search bar with the placeholder 'Search products... (Ctrl+K)' and a magnifying glass icon is in the center. On the right, the user's name 'Md Maruf Hasan' is shown next to a globe icon, followed by links for 'Returns & Orders', 'Seller Dashboard', a heart icon, and a shopping cart icon with a '2' badge.

Below the header, a navigation bar lists categories: 'All Products', 'Best Sellers', 'Electronics', 'Computers', 'Home & Kitchen', 'Sports', and 'Books'.

The main content area is titled 'Checkout' and features a progress indicator with three steps: '1 Address', '2 Payment' (the current step, highlighted with an orange circle), and '3 Review'.

Under the 'Payment Method' section, there are two options:

- Pay Online (Stripe)**: This option is selected, indicated by a radio button and a blue border. Below it, the text reads 'Credit/Debit card via Stripe secure checkout'.
- Cash on Delivery**: This option is unselected, indicated by a radio button. Below it, the text reads 'Pay when your order is delivered'.

At the bottom of the payment section are two buttons: 'Back' and 'Review Order' (highlighted in yellow).

To the right of the payment options is an 'Order Summary' box. It contains the following items and prices:

Order Summary	
Items (4)	\$29.52
Tax (2%)	\$0.59
Shipping	Free
Total	\$30.11

Below the checkout area, a dark blue footer contains a 'Back to top' link. The footer is divided into four columns of links:

- Get to Know Us**: About Us, Careers, Press Releases
- Connect with Us**: Facebook, Twitter, Instagram
- Make Money**: Sell on Amazon, Become an Affiliate, Advertise
- Let Us Help**: Your Account, Returns & Orders, Help

At the very bottom, a small copyright notice reads: '© 2026 Amazon. All rights reserved.'

Figure 10.6: Payment Method Selection (COD or Online Payment)

10.5.7 Order Confirmation Page

The screenshot displays the Amazon Order Confirmation page for Order #64AF935A, placed on April 1, 2026, at 09:16 PM. The page is titled "Order #64AF935A" and shows a "Confirmed" status with a "Cancel Order" link. The order progress is shown as a sequence of steps: Pending, Confirmed, Processing, Shipped, and Delivered. The items section lists two items: 365 by Whole Foods Market, Paper Towels 135 Sheet Jumbo Rolls 3 Count, 135 Count (Qty: 3) for \$15.87, and Amazon Basics Multipurpose Drawstring Trash Bags, Unscented, 30 Gallon, 50 Count (Previously Solimo) (Qty: 1) for \$13.65. The shipping address is Md Maruf Hasan, Palashi, Buet, Dhaka 10002, 01234567890. The payment summary shows the method as Online, status as Paid, subtotal as \$30.11, and total as \$30.11. The status history shows the order was placed on April 1, 2026, at 09:16 PM and payment was received via Stripe on April 1, 2026, at 09:17 PM. The footer includes links for "Get to Know Us", "Connect with Us", "Make Money", and "Let Us Help", along with a copyright notice for 2026 Amazon.

Amazon Deliver to Your Location Search products... (Ctrl+K) Hello, Md Maruf Hasan Returns & Orders Seller Dashboard Cart

All Products Best Sellers Electronics Computers Home & Kitchen Sports Books

← Back to Orders

Order #64AF935A Confirmed Cancel Order
Placed on Apr 1, 2026, 09:16 PM

Order Progress

Pending Confirmed Processing Shipped Delivered

Items (2)

365 by Whole Foods Market, Paper Towels 135 Sheet Jumbo Rolls 3 Count, 135 Count
Qty: 3 \$15.87

Amazon Basics Multipurpose Drawstring Trash Bags, Unscented, 30 Gallon, 50 Count (Previously Solimo)
Qty: 1 \$13.65

Shipping Address
Md Maruf Hasan
Palashi
Buet, Dhaka 10002
01234567890

Payment Summary

Method	Online
Status	Paid
Subtotal	\$30.11
Total	\$30.11

Status History

- Pending**
Apr 1, 2026, 09:16 PM
Order placed
- Confirmed**
Apr 1, 2026, 09:17 PM
Payment received via Stripe

Back to top

Get to Know Us
About Us
Careers
Press Releases

Connect with Us
Facebook
Twitter
Instagram

Make Money
Sell on Amazon
Become an Affiliate
Advertise

Let Us Help
Your Account
Returns & Orders
Help

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Figure 10.7: Order Confirmation Page with Order ID and Details

10.5.8 My Orders Page

The screenshot displays the Amazon 'My Orders' page. At the top, the Amazon logo is on the left, and navigation links for 'Hello, Md Maruf Hasan', 'Returns & Orders', 'Seller Dashboard', and 'Cart' are on the right. A search bar is also present. Below the header, the 'My Orders' section is titled, with a filter for 'All Statuses'. The page lists six orders, each with a header row showing 'Order ID', 'Date', and 'Total'. The status of each order is indicated by a colored pill: 'Confirmed' (blue), 'Delivered' (green), and 'Cancelled' (red). Each order entry includes a product image, the product name, quantity, and price. Below each order, there is a 'View Details' button. The footer contains a 'Back to top' link and four columns of links: 'Get to Know Us' (About Us, Careers, Press Releases), 'Connect with Us' (Facebook, Twitter, Instagram), 'Make Money' (Sell on Amazon, Become an Affiliate, Advertise), and 'Let Us Help' (Your Account, Returns & Orders, Help). The copyright notice '© 2026 Amazon. All rights reserved.' is at the bottom.

Order ID	Date	Total	Status
64AF935A	April 1, 2026	\$30.11	Confirmed
365 by Whole Foods Market, Paper Towels 135 Sheet Jumbo Rolls 3 Count, 135 Count Qty: 3 × \$5.29			
Amazon Basics Multipurpose Drawstring Trash Bags, Unscented, 30 Gallon, 50 Count (Previously Solimo) Qty: 1 × \$13.65			
View Details Cancel			
84C09DFD	March 19, 2026	\$13.92	Delivered
Amazon Basics Multipurpose Drawstring Trash Bags, Unscented, 30 Gallon, 50 Count (Previously Solimo) Qty: 1 × \$13.65			
View Details			
84C09EC4	March 19, 2026	\$5.40	Cancelled
365 by Whole Foods Market, Paper Towels 135 Sheet Jumbo Rolls 3 Count, 135 Count Qty: 1 × \$5.29			
View Details			
CB6EFB21	March 17, 2026	\$14.38	Cancelled
Ddrops Organic Baby 400 IU 90 Drops - Daily Vitamin D Liquid for Infants. Supports Teeth & Bone Health. No Preservatives, No Sugar, Non-GMO, Allergy-Friendly Qty: 1 × \$14.30			
View Details			
CB6EFA98	March 17, 2026	\$97.89	Cancelled
AstroAI Tire Inflator Portable Air Compressor Air Pump for Car Tires - Car Accessories, 12V DC Auto Pump with Digital Pressure Gauge, 100PSI with Emergency LED Li... Qty: 3 × \$31.99			
View Details			
B92D77A0	March 16, 2026	\$55.58	Delivered
Super Mega Baseball 4 - Nintendo Switch Qty: 1 × \$49.99			
Baseball bat Qty: 1 × \$4.50			
View Details			

Figure 10.8: My Orders Page - View All Customer Orders

10.5.9 Order Detail Page(Seller View)

The screenshot displays the Amazon Seller Center interface. On the left is a dark sidebar with the Amazon Seller Center logo and a menu for Md Maruf Hasan (marufhasanrubab@gmail.com). The menu includes Dashboard, Orders, Products, and Add Product. The main content area is titled 'Dashboard' and shows the order details for Order #2D6E867A, placed on Mar 19, 2026, at 12:41 PM. The order status is 'Shipped'. The items list includes Nike Boys' Ankle Socks (6 Pairs) for \$18.62 and Boys Chino Pants for \$23.99. The shipping address is Anuron Maitro, Ahsanullah hall, Buet, Azimpur, Dhaka, Texas 65000, with zip code 01774496727. The payment summary shows a total of \$43.46, with a pending status. The status history shows the order progressing from Pending to Confirmed, Processing, and finally Shipped.

Amazon Seller Center

Md Maruf Hasan
marufhasanrubab@gmail.com

Dashboard
Manage your store

← Back to Store

← Back to Seller Orders

Order #2D6E867A
Placed on Mar 19, 2026, 12:41 PM

Items (2)

	Nike Boys' Ankle Socks (6 Pairs) Qty: 1	\$18.62
	Boys Chino Pants Qty: 1	\$23.99

Shipping Address
Anuron Maitro
Ahsanullah hall, Buet, Azimpur
Dhaka, Texas 65000
01774496727

Payment Summary

Method	COD
Status	Pending
Total	\$43.46

Status History

- Pending**
Mar 19, 2026, 12:41 PM
Order placed
- Confirmed**
Mar 19, 2026, 12:55 PM
Status updated to Confirmed
- Processing**
Mar 19, 2026, 02:56 PM
Status updated to Processing
- Shipped**
Mar 19, 2026, 03:31 PM
Status updated to Shipped

Figure 10.9: Order Detail Page with Tracking Information

10.5.10 View Product Review(seller View)

The screenshot displays the Amazon Seller Center dashboard for a user named Md Maruf Hasan. The left sidebar contains a menu with options: Dashboard, Orders, Products, and Add Product. The main content area shows the product details for 'Amazon Basics Multipurpose Drawstring Trash Bags, Unscented, 30 Gallon, 50 Count (Previously Solimo)'. The product has a 4.0 star rating from 1 review and is priced at \$13.65. The dashboard also shows the stock level (235) and the number of units bought last month (40001). Below the product details, there is a section for 'Customer Reviews & Ratings' showing the 4.0 star rating and a bar chart for the number of reviews per star. A verified purchase review is also displayed, stating 'very good product its perfect' with a date of March 19, 2026.

Amazon Seller Center

Md Maruf Hasan
marufhasanrubab@gmail.com

Dashboard
Manage your store

← Back to Store

← Back to Seller Products

HOUSEHOLD CLEANING SUPPLIES

Amazon Basics Multipurpose Drawstring Trash Bags, Unscented, 30 Gallon, 50 Count (Previously Solimo)

★★★★☆ 4.0 (1 reviews)

\$13.65

Stock: 235

Bought last month: 40001

Customer Reviews & Ratings

4.0
★★★★☆ 4.0 (1 reviews)

5 star: 0
4 star: 1
3 star: 0
2 star: 0
1 star: 0

Md Maruf Hasan Verified Purchase

★★★★☆
very good product
its perfect
March 19, 2026

Figure 10.10: Review Update - Edit Existing Review

10.5.11 Stripe Payment Gateway

The screenshot displays the Stripe payment gateway interface. On the left, a shopping cart is visible with the following items:

Item	Quantity	Price (BDT)
365 by Whole Foods Market, Paper Towels 135 Sheet Jumbo...	Qty 3	BDT 2,029.91
Amazon Basics Multipurpose Drawstring Trash Bags,...	Qty 1	BDT 1,745.95
Tax (2%)	Qty 1	BDT 75.47

Below the cart, a currency selection dropdown is set to BDT 3,851.33, with a conversion rate of 1 USD = 127.9087 BDT. A \$30.11 option is also visible.

On the right, the payment form is titled "Pay with link". It includes a "Contact information" section with an email field containing "marufhasanrubab@gmail.com". The "Payment method" section is set to "Card" and includes fields for card information (4242 4242 4242 4242, 12 / 28, 123), cardholder name (Rubab), and country or region (Bangladesh). A checkbox for "Save my information for faster checkout" is present. A blue "Pay" button is at the bottom. The footer indicates "Powered by stripe" with links to "Terms" and "Privacy".

Figure 10.11: Stripe Payment Gateway - Secure Payment Processing

10.6 System Architecture Overview

10.6.1 Overall Architecture

Architecture Style: Layered Client-Server Architecture (3-Tier)

The system follows a modern three-tier architecture pattern:

- **Frontend Tier (Client):** Single Page Application (SPA) built with React
- **Backend Tier (API):** RESTful API built with NestJS with modular feature architecture
- **Database Tier:** MongoDB document store hosted on MongoDB Atlas

10.6.2 High-Level Request Flow

1. User interacts with the React frontend application
2. Frontend sends HTTP requests to backend REST endpoints via Axios
3. Backend applies authentication/authorization, validates input, and executes business logic
4. Backend communicates with Mongoose-based database persistence layer
5. For payments, backend integrates with Stripe API for verification
6. Backend returns JSON responses to frontend
7. Frontend renders updated UI based on response

10.6.3 System Components

- **Authentication Service:** Clerk for client and server-side authentication
- **Payment Service:** Stripe integration for online payments and verification
- **Product Service:** Product catalog, inventory, and search
- **Order Service:** Order creation, tracking, and management
- **Review Service:** Product reviews, ratings, and aggregations
- **Cart Service:** Shopping cart management and calculations
- **Address Service:** Address management with suggestions and validation
- **User Service:** User profile and preference management

10.7 Technologies and Frameworks Used

10.7.1 Frontend Stack

- **Framework:** React 18 with functional components and hooks
- **Build Tool:** Vite for fast development and optimized builds
- **Routing:** React Router for client-side routing
- **HTTP Client:** Axios for API communication
- **Styling:** Tailwind CSS for utility-first responsive design
- **Authentication:** Clerk for user authentication and session management
- **Payment Integration:** Stripe.js for payment flow
- **Notifications:** React Hot Toast for user notifications
- **State Management:** React Context API and hooks

10.7.2 Backend Stack

- **Framework:** NestJS with TypeScript for structured backend development
- **Language:** TypeScript for type safety and better development experience
- **ORM/ODM:** Mongoose for MongoDB object mapping and validation
- **Authentication:** Clerk backend SDK for secure token verification
- **Payments:** Stripe SDK for payment processing and webhook handling
- **API Documentation:** Swagger/OpenAPI for interactive API documentation
- **Validation:** Class validators and transformers for input validation
- **Middleware:** Custom middleware for logging, error handling, and CORS

10.7.3 Database

- **Database:** MongoDB - NoSQL document database
- **Hosting:** MongoDB Atlas - Managed cloud service on AWS
- **Collections:** Users, Products, Orders, OrderItems, Reviews, Carts, Addresses, Payments
- **Indexing:** Optimized indexes on frequently queried fields for performance

10.8 Deployment Details

10.8.1 Hosting Locations

- **Frontend:** Vercel - Static SPA with automatic routing rewrites
- **Backend:** Vercel - Node.js serverless functions
- **Database:** MongoDB Atlas - AWS-backed managed service (us-east-1 region)

10.8.2 Cloud Platform Used

Vercel Infrastructure:

- Automatic deployments from GitHub push to main branch
- Global CDN for fast content delivery
- Automatic HTTPS with Let's Encrypt certificates
- Serverless function scaling
- Environment variable management for secrets

MongoDB Atlas:

- Shared cluster for development/production
- Automated backups and snapshots
- IP whitelist security
- Connection pooling for optimal performance

10.8.3 Environment Configuration

Frontend environment variables (`.env.local`):

- `VITE_API_URL` - Backend API base URL
- `VITE_CLERK_PUBLISHABLE_KEY` - Clerk authentication key
- `VITE_STRIPE_PUBLIC_KEY` - Stripe public key for payment UI

Backend environment variables (`.env`):

- `MONGODB_URI` - MongoDB connection string
- `CLERK_SECRET_KEY` - Clerk server secret
- `STRIPE_SECRET_KEY` - Stripe server secret for payment processing
- `DATABASE_NAME` - MongoDB database name

10.9 DevOps and Deployment Pipeline

10.9.1 Git-Based Deployment

- Source code hosted on GitHub with multiple branches (main, develop, feature/*)
- Both frontend and backend repositories configured for automatic Vercel deployment
- Push to 'main' branch triggers automatic deployment

10.9.2 Continuous Integration/Continuous Deployment (CI/CD)

Backend GitHub Actions Workflow:

- Automated trigger on push to main branch
- Runs linting and type checking (if configured)
- Triggers Vercel deployment hook
- Automatic redeployment on successful build

Frontend Deployment:

- Automatic Vercel project setup triggered by GitHub connection
- Build command: `vite build`
- Output directory: `dist/`
- Automatic routing configuration for React Router (rewrite routes to `index.html`)

10.9.3 Containerization

Current Implementation: No Docker-based containerization is used. The system relies on Vercel's native Node.js runtime for backend deployment and Vercel's static hosting for frontend.

Future Considerations: Docker containerization can be added for local development consistency and custom CI/CD pipelines if needed.

10.9.4 Build and Deployment Process

For Frontend:

1. Developer pushes code to GitHub main branch
2. Vercel detects push event
3. Vercel runs build command: `npm run build`
4. Vite creates optimized production bundle
5. Static files deployed to Vercel CDN
6. Live URL updated automatically

For Backend:

1. Developer pushes code to GitHub main branch
2. GitHub Actions workflow triggered (if configured)
3. Workflow triggers Vercel deploy hook via webhook
4. Vercel detects changes and redeploys
5. NestJS serverless function bundle created
6. New API endpoints live at updated deployment URL

10.10 Team Contributions

Team Member Roles and Feature Implementation Contributions:

- **Md. Mehedi Hasan (2105052)**
 - Set up frontend and backend repositories on GitHub.
 - Pushed the initial skeleton project structure.
 - Designed and implemented the schema for products.
 - Implemented the end-to-end product browsing module across both frontend and backend.
 - Built the frontend experience including Home, Product List, filtering/sorting, Search, and Product Details pages.

- Developed the corresponding backend Products APIs to support listing, pagination, filtering, search, categories/best-sellers, product details, and related products.
 - Contributed to deployment by publishing the application on Vercel.
- **Anuron Maitro(2105037)**
 - Designed and implemented Order Schema
 - Implemented order-related frontend pages, including Checkout, Place Order, Order Details.
 - Developed the frontend order workflow for cart review, order placement, and detailed order presentation.
 - Built backend order-management components, including Order Service, Add to Order functionality, and Order Controller.
 - Implemented backend logic for order creation and processing to persist and retrieve order data reliably.
 - Integrated a dummy payment method for testing and ensured end-to-end integration between frontend order pages and backend APIs.
 - Implemented Wishlist System Frontend Pages
 - Implemented Wishlist backend logic like wishlist controller,module and service to manage user wishlists effectively.
 - **Md Asif Ali(2105038)**
 - Added implementation of the User Profile module across frontend and backend layers.
 - Integrated Clerk for authentication and user identity management, and configured it as the primary profile/auth system.
 - Developed backend user-management components, including User Service and User Controller, to handle user operations and API requests.
 - Built the frontend User Profile page to enable profile viewing and management.
 - Implemented product stock management and added backend logic to automatically update inventory quantities after purchases.
 - **Md Maruf Hasan Rubab(2105042)**
 - Implemented the seller-side actions across both frontend and backend modules.
 - Developed seller features on the frontend, including seller orders, seller order details, and seller product management/detail flows, with improvements to filtering, navigation, and data presentation.
 - Implemented seller-focused backend functionality, including seller order-detail API endpoints and seller product management support (add/update/filter operations).

- Contributed additional refinements to customer-facing pages, including wishlist behavior and review/rating display consistency.
- Imported the Amazon products Kaggle dataset into MongoDB and integrated it as the project's product data source.

- **Himadri Gobinda Biswas(2105047)**

- Set up the initial MongoDB Atlas database and prepared the required environment/API key configuration for data access and authentication.
- Configured Clerk-related keys in the environment setup to support secure user authentication.
- Designed and implemented the address schema.
- Developed backend cart and address service/controller functionalities.
- Built the Cart frontend page with full functionalities, including product quantity update features and with a dynamic order summary with total price.
- Developed the fronted pages for address with complete functional flow, including features for adding new address and selecting one from previously saved addresses.
- Prepared Vercel deployment configuration files for both frontend and backend.
- Contributed minor post-merge bug fixes.

- **Md Rafat Kabir(2105054)**

- Added the end-to-end payments implementation by integrating Stripe into the checkout and order-processing workflow.
- Replaced the dummy checkout flow with a production payment flow that redirects customers to Stripe and handles return outcomes.
- Enhanced order pages (My Orders and Order Details) to support real payment sessions, payment verification on redirect, and clear success/failure handling.
- Implemented backend payment verification and order payment-state updates, including payment tracking within order lifecycle management.
- Designed and implemented the review data model with proper constraints and structured request validation support.
- Built the backend review module (service/controller) with pagination and rating analytics to improve product feedback visibility.

Key Responsibilities Across Features:

- **Place Order Feature:** Complete implementation including order creation, status tracking, and order history management
- **Checkout Feature:** Full checkout flow from cart to payment processing with address validation

- **Order Review Feature:** Review submission, aggregation, and display across product pages
- **Deployment:** Infrastructure setup, CI/CD pipeline configuration, and production monitoring
- **Payment Integration:** Stripe integration with secure webhook handling for payment confirmation
- **Database:** Schema design, indexing, and query optimization for performance